Administration console

Wings Notification Hub

User manual

For administrators, template designers and approvers, campaign managers, operators and auditors

Email · SMS · WhatsApp · In-app · Push · Web push · Webhooks

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How to use this manual

This manual is for the people who run Wings Notification Hub from day to day through its administration console: administrators, template designers and approvers, campaign managers, operators and auditors. You do not need to be a developer.

It is written around tasks. Each task has a heading that starts with what you want to do, followed by numbered steps, what you will see, and what to do when something goes wrong.

- **New to the hub?** Read Chapter 1 on page 5 to Chapter 4 on page 11 first.
- **Looking after messages today?** Go to Part 2: Chapter 6 on page 16 covers finding, retrying and resolving deliveries.
- **Writing messages or automation?** Go to Part 3, starting at Chapter 9 on page 26.
- **Setting up a customer or provider?** Go to Part 4, starting at Chapter 14 on page 53.
- **Something went wrong?** Go straight to Appendix D on page 81.

Conventions

You will see	It means
Save draft	A label on the screen: a button, tab, menu item or field.
`INVOICE_CREATED_V1`	Something you type or read exactly as shown: a code, a value, a variable.
Delivered Failed	A status, shown as in the console. Status is always an icon plus text, never colour alone.
Who can do this	The roles that can perform a task. If you cannot see a button that a task mentions, your role does not include it. Appendix C lists every role.

Tip Tips save you time.

Note Notes explain behaviour that may surprise you.

Warning Warnings mark actions that cannot be undone or that affect live messages.

Note The screens in this manual show sample data for fictional businesses such as ABC Traders Pvt Ltd. Your own console shows your data. Providers with “Mock” in their name exist only in demo data: they simulate sending and never reach a real person. Your provider list will show your real accounts.

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PART 1

Getting started

What the hub does, how to sign in, how to find your way around, and what your role lets you do.

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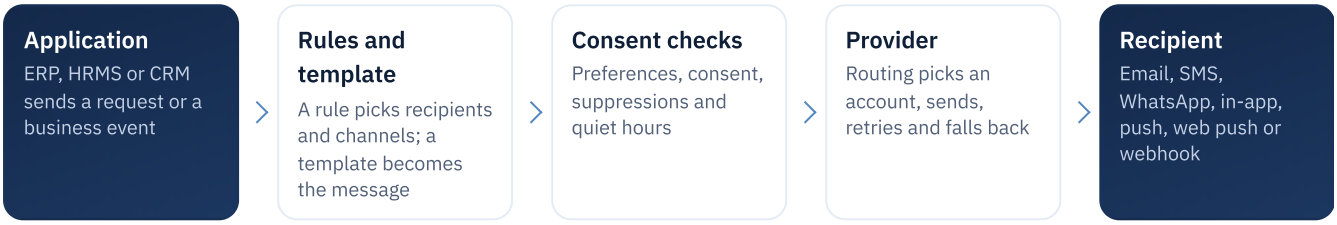
CHAPTER 1

What the hub does

Wings Notification Hub is the one place your business applications go to send messages. An ERP, HRMS, CRM or other system hands the hub a request or a business event. The hub picks the wording, checks consent, chooses a provider, sends, retries and records exactly what happened.

It carries seven **channels**: email, SMS, WhatsApp, in-app, push, web push and webhook. Customers can also reply on WhatsApp, SMS and email, and those replies arrive in the console's Inbox.

1.1 A notification's journey



Every step is recorded. Receipts from providers update the delivery's status, which you read in the Notification explorer and your application can receive by webhook.

1.2 Key ideas

Term	What it means
Tenant	A customer organisation on the hub. It has its own retention periods, consent setting and branding. Data never crosses between tenants.
Company	A business unit or legal entity inside a tenant, with its own time zone, quiet hours and branding.
Application	A system that sends through the hub, such as Wings ERP Auto. It can send only for the companies it is enabled for.
Notification	One request to tell someone something, for example “invoice INV-10045 was created”.
Delivery	One message to one address on one channel. A single notification sent by email and WhatsApp has two deliveries, each with its own status.
Channel	The route a message takes: email, SMS, WhatsApp, in-app, push, web push or webhook.
Template	The wording of a message for one channel and language, with variables such as `{{CustomerName}}`. Templates are versioned and approved by a second person.
Notification type	A named kind of message, such as `INVOICE_CREATED`, with its message class, default channels and default template.
Rule	Turns a business event, such as `payment.overdue`, into notifications: who gets what, on which channel, under which conditions.
Provider	The outside service that carries a message, such as a mail server or SMS gateway. A **provider account** holds its settings and credentials.
Sender identity	The from address, SMS sender id or WhatsApp number a message goes out under.
Message class	Whether a message is transactional, promotional or critical. See below.

1.3 Message classes

Every message has a class. The class decides which consent checks apply and whether quiet hours can delay it.

Class	Use it for	How it behaves
Transactional	Messages about something the recipient is party to: invoices, approvals, service updates.	Honours the recipient's preferences, unless the notification type is mandatory. Ignores unsubscribes.
Promotional	Marketing and offers.	Needs marketing consent when the tenant requires it. Honours preferences and unsubscribes. Can be delayed by quiet hours.
Critical	Security and regulatory notices.	Ignores preferences and opt-outs. Never delayed by quiet hours. Still cannot reach a hard-bounced or invalid address, and WhatsApp still needs opt-in.

The full rules are in section 12.4 on page 46.

CHAPTER 2

Signing in

You sign in with an email address and password from your administrator. Your session is kept only in the current browser tab, so a new tab asks you to sign in again.

To sign in

1. Open your hub's web address in a browser.
2. Enter your **Email** and **Password**. Use the eye icon to show what you typed.
3. Select **Sign in**. The console opens on the first screen your role allows, which is usually the **Dashboard**.

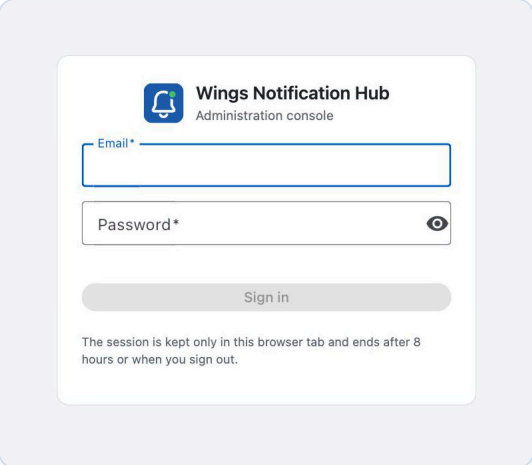


Figure 1: The sign-in page.

To sign out

1. Select your name at the top right of the console.
2. Select **Sign out**.

What you see	What it means and what to do
`Sign-in failed: Invalid email or password.`	Check both fields. If you still cannot get in, ask your administrator. The console has no “forgot password” link.
`Your session expired or was rejected. Sign in again.`	A session lasts 8 hours. It also ends when you sign out. Sign in again.
`No screens available`	Your account has no roles. Ask an administrator to grant one (Chapter 17 on page 64).

CHAPTER 3

Finding your way around

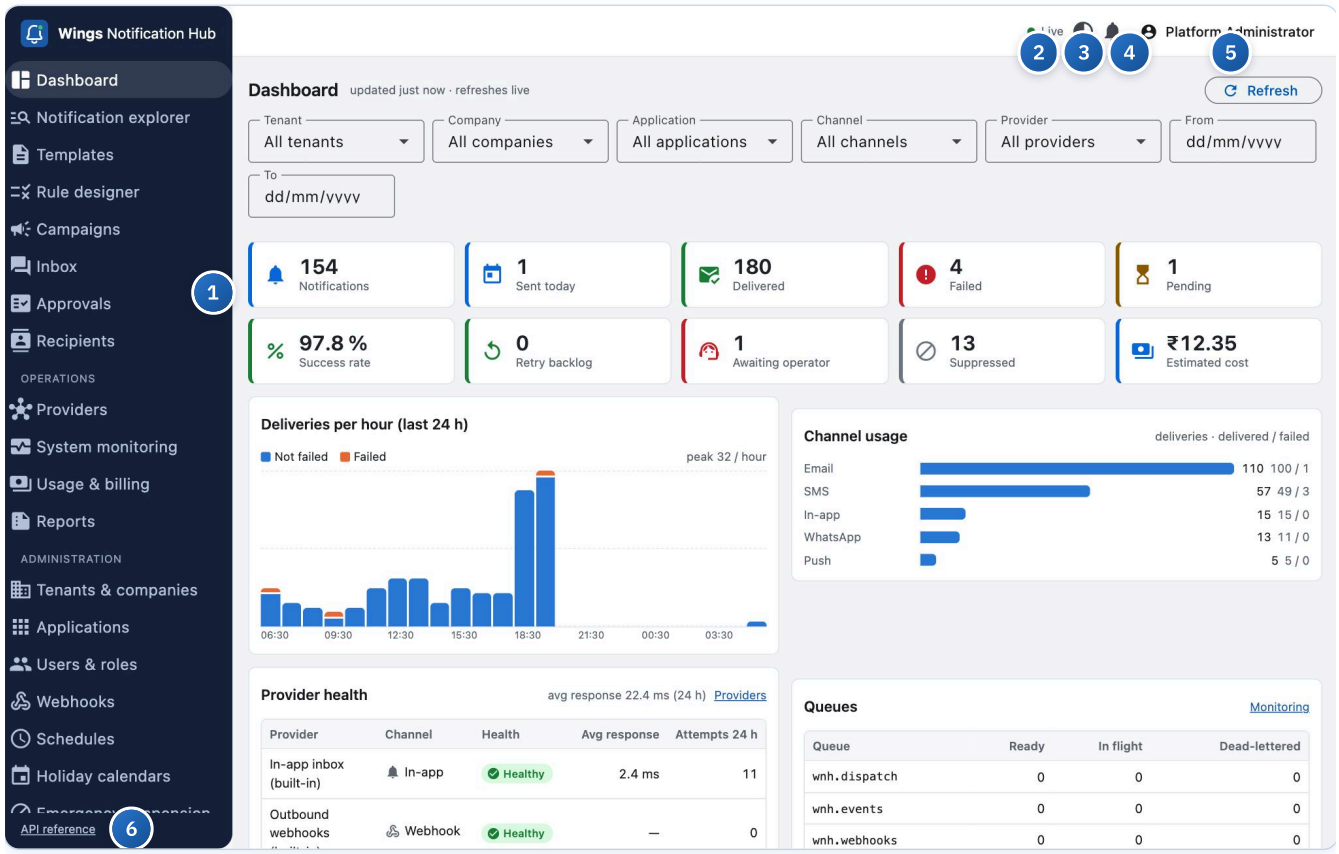


Figure 2: The console window. Numbered items are explained below.

1. **Menu.** Grouped into main screens, **Operations** and **Administration**. You see only what your role allows.
2. **Live indicator. Live** with a green dot means updates arrive by themselves. **Offline** means the connection dropped; the console keeps retrying.
3. **Theme.** Each click moves between system, light and dark.
4. **Live events bell.** Failed deliveries and customer replies, as they happen.
5. **Your name.** Your roles, theme choice and **Sign out**.
6. **API reference.** The technical reference for developers.

3.1 Using the live events bell

A badge on the bell counts events you have not seen. Select the bell to list them, then select an entry to open that delivery or conversation. Only failures and outcomes that need a decision are listed, so routine progress does not crowd it out.

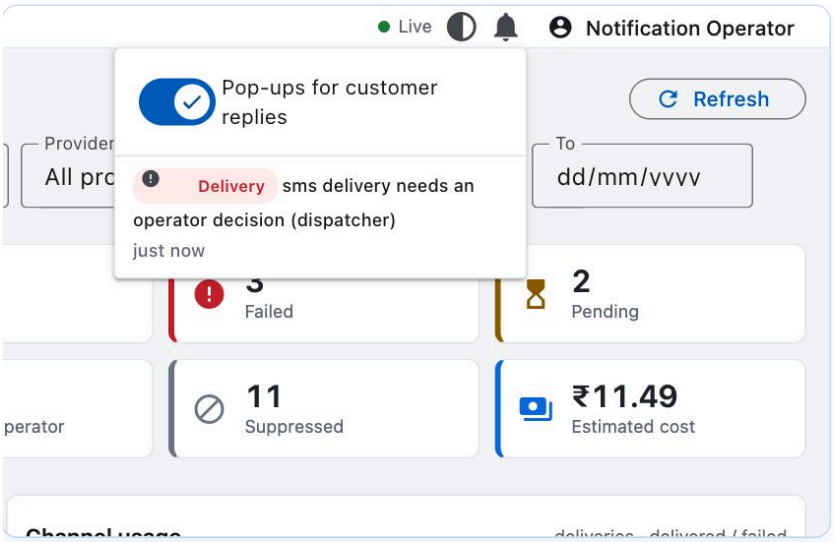


Figure 3: The live events list, with the switch for pop-ups.

- The list starts when you open the console and is kept only in that tab. For history, use the **Notification explorer**.
- Turn **Pop-ups for customer replies** on or off at the top of the list. When it is on, a pop-up with an **Open** link appears at the bottom of the screen for each reply.

3.2 What every screen has in common

- **Filters** sit above the data. The screen updates as you change a select list. In a text box, press Enter.
- **Rows open things.** Select a row in a table to see its detail, usually in a panel at the side or below.
- **Refresh** reloads the data. Screens that show live data also update by themselves.
- **Times** are shown in your browser's time zone. Exports from the explorer use UTC.
- **Errors** appear in a red banner with a short message, the error code and a correlation id. Quote the correlation id when you ask for help.

3.3 Choosing a light or dark theme

Select the theme button in the top bar, or select your name and choose **System theme**, **Light theme** or **Dark theme**. **System theme** follows your computer's setting. The same menu shows your roles and scope.

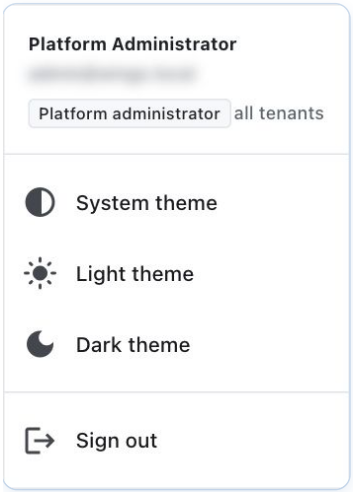


Figure 4: The menu under your name.

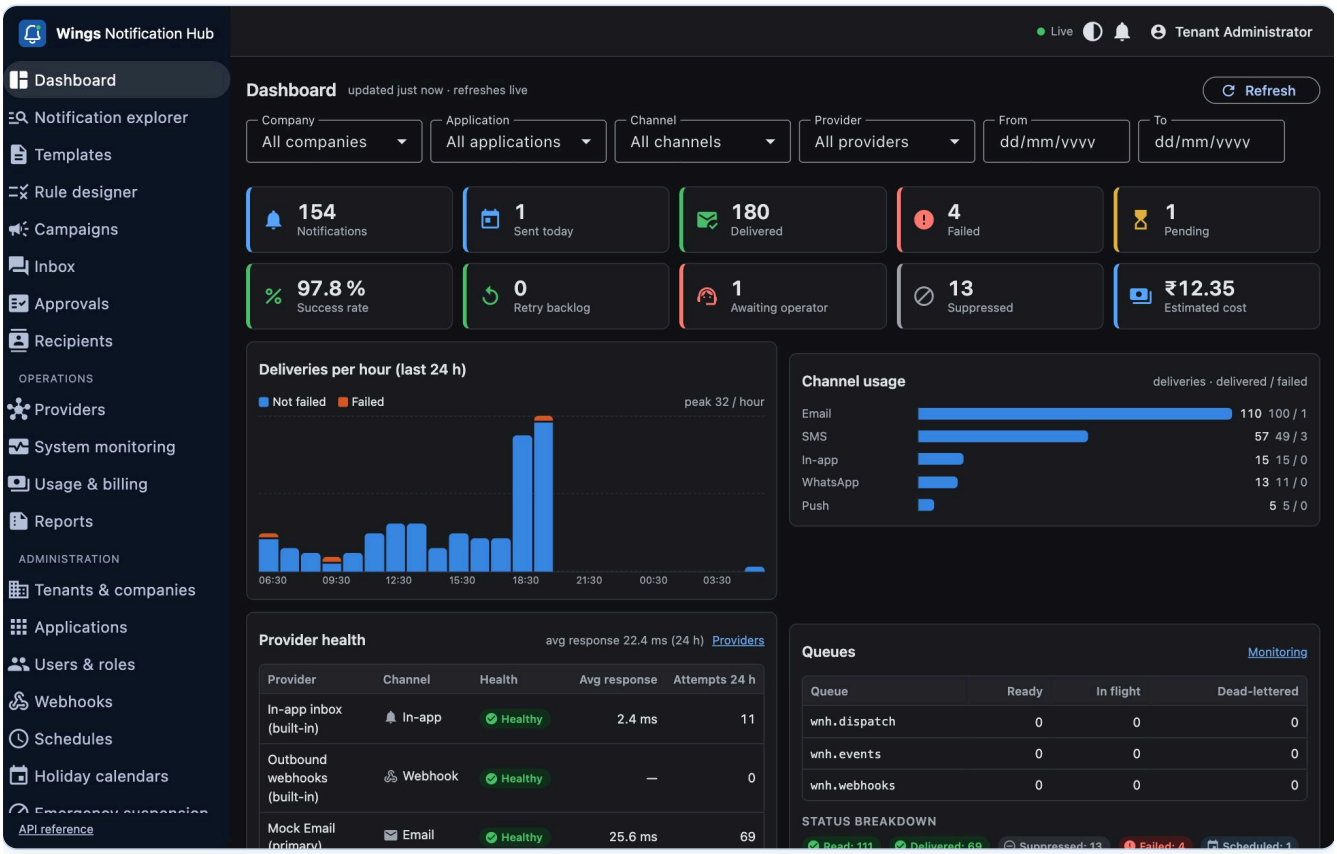


Figure 5: The dashboard in the dark theme.

3.4 Using the console on a phone

On a narrow screen the menu folds away behind the **Menu** button at the top left, cards stack, and wide tables scroll sideways inside their own area. Everything still works, but templates, rules and reports are easier on a larger screen.

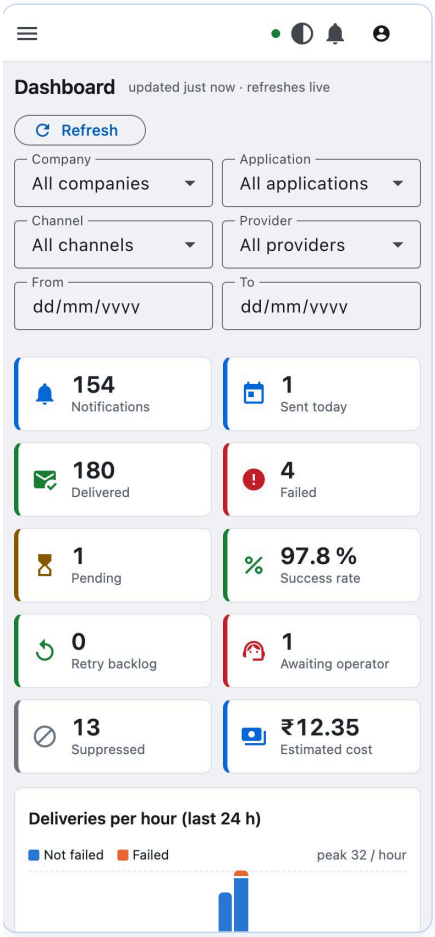


Figure 6: Dashboard on a phone.

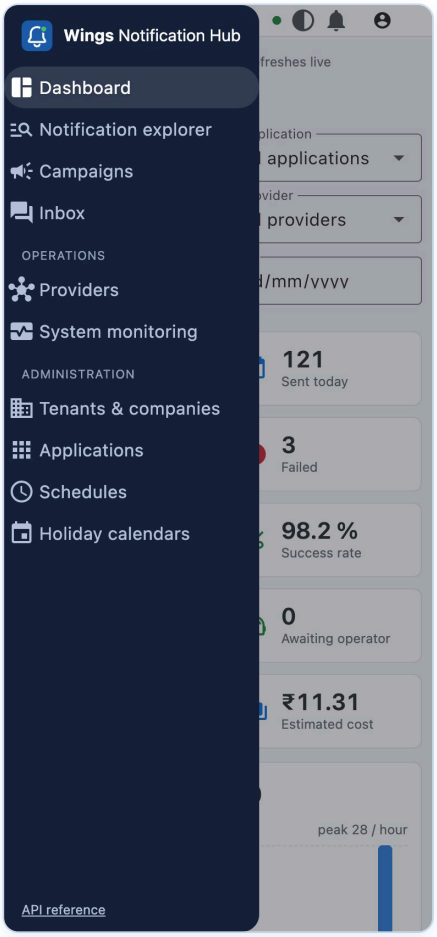


Figure 7: The menu opened.

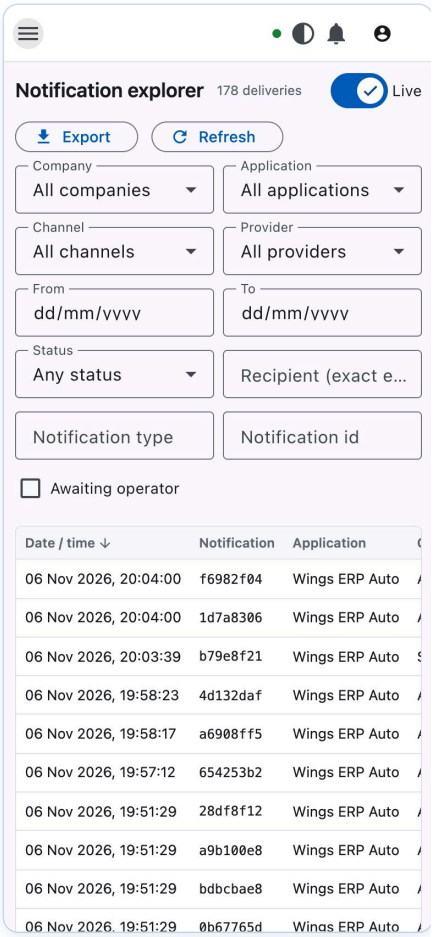


Figure 8: Notification explorer.

CHAPTER 4

What each role sees

The console shows each person only the screens and buttons their role allows. Roles are granted at a **scope**: a whole tenant, one company, one application or one channel.

Role	Typical work	Notable limits
Platform administrator	Runs the hub for all tenants: tenants, shared providers, credits and rates.	None. The only role that can create tenants and change credits or rates.
Tenant administrator	Runs one tenant: companies, applications, users, providers, approvals.	Cannot suspend its own tenant, add credits or change rates.
Company administrator	Runs one company: templates, rules, campaigns, providers, approvals.	No **Users & roles** screen. Cannot edit tenants or companies, or manage credentials.
Application administrator	Runs one application: templates, rules, contacts, webhooks, credentials.	No approvals, audit log or emergency suspension.
Template designer	Writes templates and notification types.	Sees rules but cannot change them. Cannot approve. Cannot deliver a test of an active template.
Template approver	Reviews, approves, rejects, rolls back and retires templates.	Cannot edit template content. Sees only template requests in **Approvals**.
Campaign manager	Builds campaigns, imports recipients, manages contacts.	Cannot approve campaigns.
Notification operator	Watches deliveries, retries, resolves, answers the Inbox, requeues dead letters.	Sees providers and campaigns but cannot change them.
Auditor	Reads everything, including the audit log and reports.	Cannot change anything except acknowledging alerts. Never sees message bodies. Addresses are masked.

The full list is in Appendix C on page 80. Two examples of the difference a role makes:

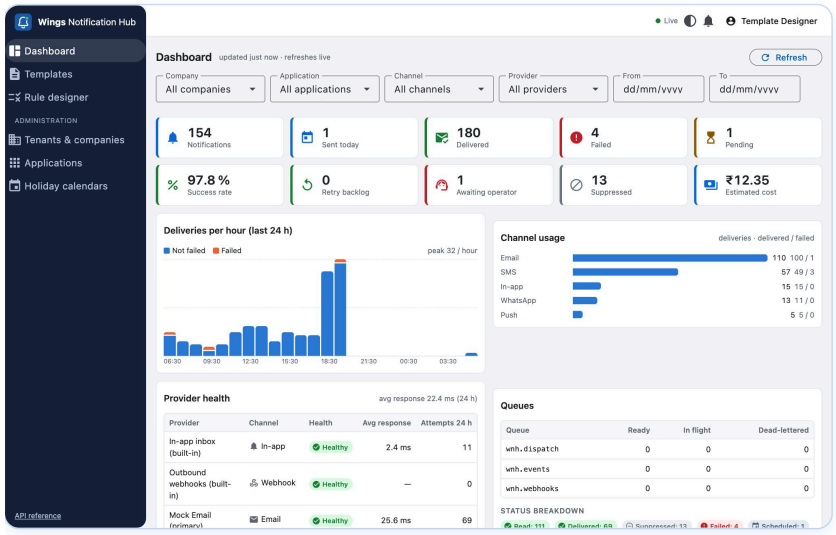


Figure 9: A template designer's console: the menu shows only the screens the role can use.

06 Nov 2026, 19:24:24	4926647e	Wings ERP Auto	ABC Traders Pvt Ltd	u-1002	STOCK_BELOW_REORDER	In-app	In-app inbox (built-in)	Delivered	1	₹0.00
06 Nov 2026, 19:24:24	84c0f24e	Wings ERP Auto	ABC Traders Pvt Ltd	69905c85-9498-4d91-8ec…	ORDER_APPROVED	Push	Mock Push (FCM/AP…	Delivered	1	₹0.00
06 Nov 2026, 19:24:24	84c0f24e	Wings ERP Auto	ABC Traders Pvt Ltd	u-1001	ORDER_APPROVED	In-app	In-app inbox (built-in)	Delivered	1	₹0.00
06 Nov 2026, 19:24:23	25872884	Wings ERP Auto	ABC Traders Pvt Ltd	r*****7@example.com	PROMO_FESTIVE	Email	Mock Email (primary)	Read	1	₹0.01
06 Nov 2026, 19:24:23	eeb43060	Wings ERP Auto	ABC Traders Pvt Ltd	r*****8@example.com	PROMO_FESTIVE	Email	Mock Email (primary)	Read	1	₹0.01
06 Nov 2026, 19:24:23	c856b549	Wings ERP Auto	ABC Traders Pvt Ltd	r*****0@example.com	PROMO_FESTIVE	Email	Mock Email (primary)	Read	1	₹0.01
06 Nov 2026, 19:24:23	9ebabed7	Wings ERP Auto	Sunrise Retail LLP	+91******2222	INVOICE_CREATED	SMS	Sunrise SMS gateway	Provider accepted	1	—
06 Nov 2026, 19:24:23	623941a3	Wings ERP Auto	ABC Traders Pvt Ltd	c*****r@example.com	INVOICE_CREATED	Email	—	Scheduled	0	—

Figure 10: An auditor sees masked addresses, such as r*****8@example.com.

> **Note** A narrower scope also narrows the data. A company administrator for ABC Traders Pvt Ltd sees only that company's deliveries, and an application administrator only that application's.

PART 2

Daily work

Watch what is being sent, find a message, put right what went wrong, answer customers and decide approvals.

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8	Approvals	23

CHAPTER 5

Dashboard

The dashboard answers one question at a glance: is messaging healthy right now?

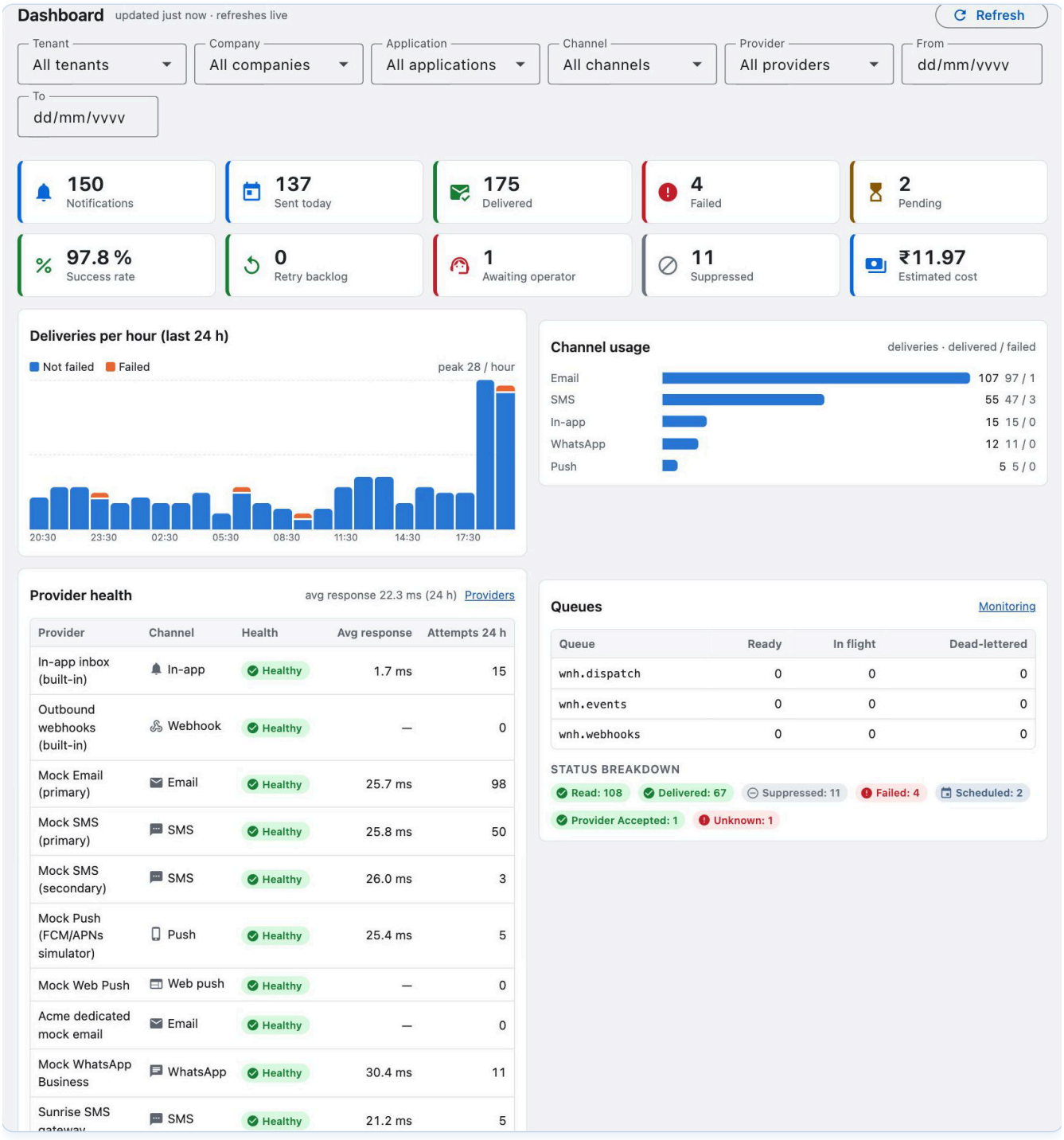


Figure 11: The dashboard: filters, summary cards, trends, provider health and queues.

5.1 Narrowing what you see

Use the filters across the top: **Tenant** (only if you can see more than one), **Company**, **Application**, **Channel**, **Provider**, **From** and **To**. The dashboard updates by itself when deliveries change, and at least once a minute.

5.2 Reading the cards

Selecting a card opens the **Notification explorer** filtered to that card, if your role can open the explorer. If it cannot, selecting a card does nothing.

Card	What it counts
Notifications	Notification requests received. Hover to see the number of deliveries.
Sent today	Sent so far today.
Delivered	Confirmed as delivered. Hover to see how many more were accepted by providers.
Failed	Deliveries that ended in failure.
Pending	Scheduled, queued, processing, submitted or retrying.
Success rate	(delivered + accepted) ÷ (delivered + accepted + failed). The card is green from 95 %, amber from 80 %, red below that.
Retry backlog	Deliveries waiting for an automatic retry.
Awaiting operator	Unknown-outcome deliveries waiting for a person to decide. See section 6.5 on page 19.
Suppressed	Blocked by consent, preference or suppression policy.
Estimated cost	Estimated provider cost. Opens **Usage & billing** if your role allows.

5.3 Reading the panels

- **Deliveries per hour (last 24 h)** shows volume, with failures in a contrasting colour. **peak** is the busiest hour.
- **Channel usage** shows deliveries per channel, then delivered / failed.
- **Provider health** lists each provider's health, average response and attempts in 24 hours. Select **Providers** to open that screen.
- **Queues** shows work waiting. A number under **Dead-lettered** needs attention: see section 21.1 on page 74.
- **Status breakdown** counts deliveries per status. Select a status to open the explorer filtered to it.

CHAPTER 6

Notification explorer

The explorer lists every delivery you are allowed to see. Use it to find out what happened to a message, and to put it right.

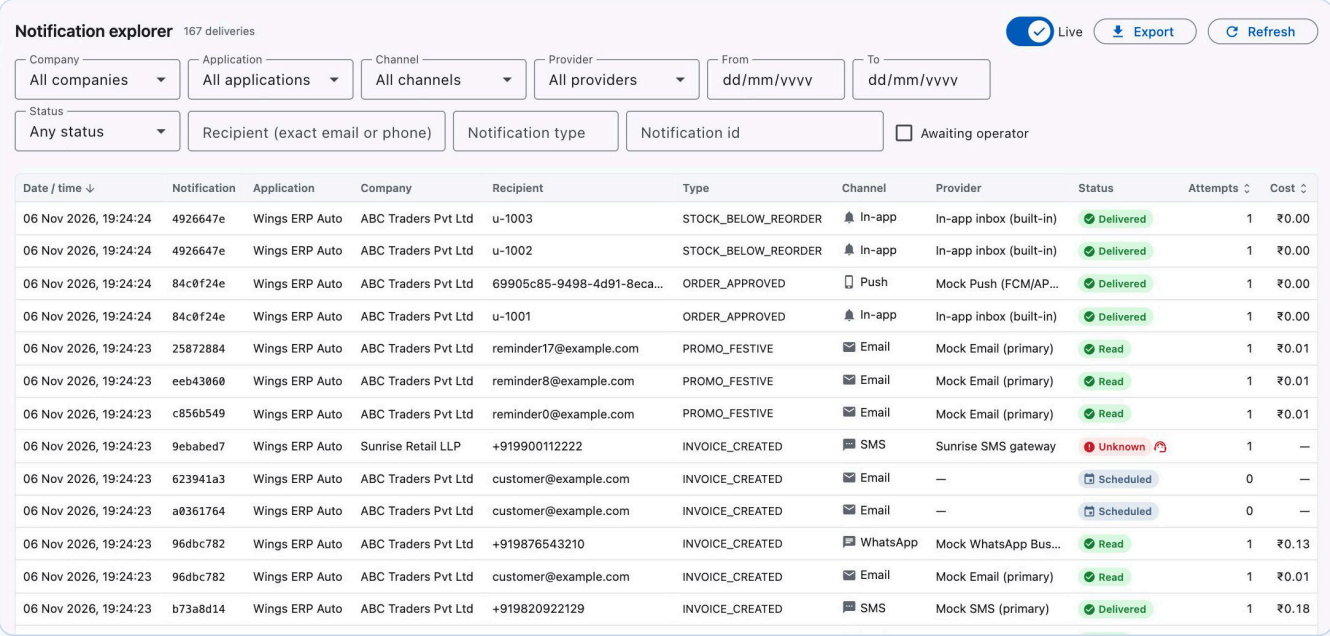


Figure 12: The Notification explorer, showing one row per delivery.

6.1 Finding a delivery

Who can do this: everyone who can open the explorer, which is every role except template designer and template approver.

1. Open **Notification explorer**.
2. Use any of the filters:
   - **Company**, **Application**, **Channel**, **Provider**, **From**, **To**, and **Tenant** if you see more than one.
   - **Status**, for example Failed.
   - **Recipient (exact email or phone)**: the full address, such as `customer@example.com` or `+919876543210`. Part of an address finds nothing.
   - **Notification type**, such as `PAYMENT_OVERDUE`. Lower case also works.
   - **Notification id**, if someone gave you one.
   - **Awaiting operator**, to list only deliveries that need a decision.
3. Press Enter after typing in a text box. The total at the top shows how many match.
4. Select **Date / time**, **Attempts** or **Cost** in the table heading to sort.

> **Tip** Leave **Live** switched on and the first page reloads when statuses change. Switch it off to hold the list still while you read.

6.2 Reading a delivery

Select a row. A panel opens on the right with the delivery's details and three tabs.

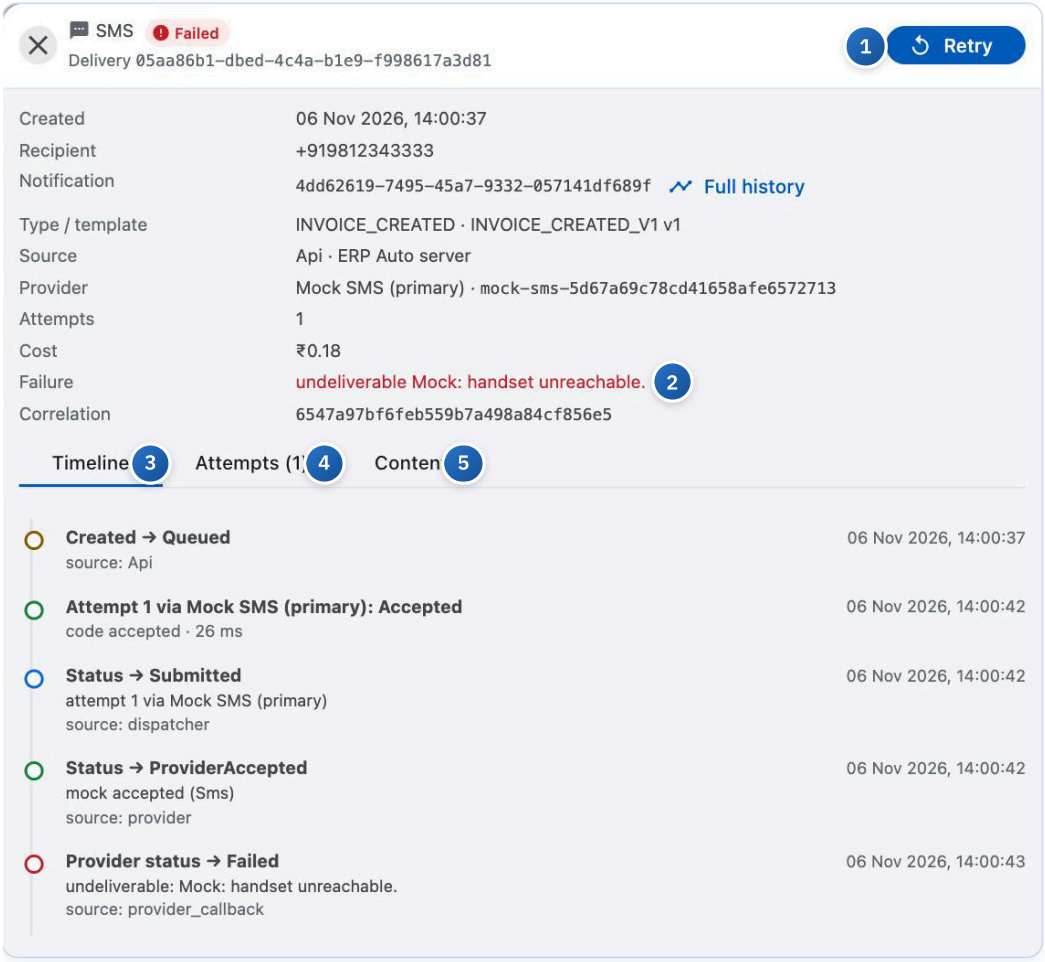


Figure 13: A failed SMS delivery, with its Timeline tab.

1. **Retry** appears when a retry is possible.
2. **Failure** shows the error code and reason.
3. **Timeline**: every event in order.
4. **Attempts**: each try at a provider.
5. **Content**: the message as rendered.

- The details list shows the recipient, type and template with its version, the source of the request, provider, attempts, cost and any failure. Links such as **Fallback of** open the related delivery.
- **Timeline** shows who or what changed the status. **source:** tells you whether it came from the hub, the provider or an operator.
- **Attempts** shows the outcome of each try, the response code, the time taken and the provider's sanitised response.
- **Content** shows the rendered message. **Content hidden** means your role cannot read message bodies. **Content purged** means the retention period has passed.
- **Full history** lists all deliveries of the same notification, with attempts and events counted. Addresses are masked in this list.

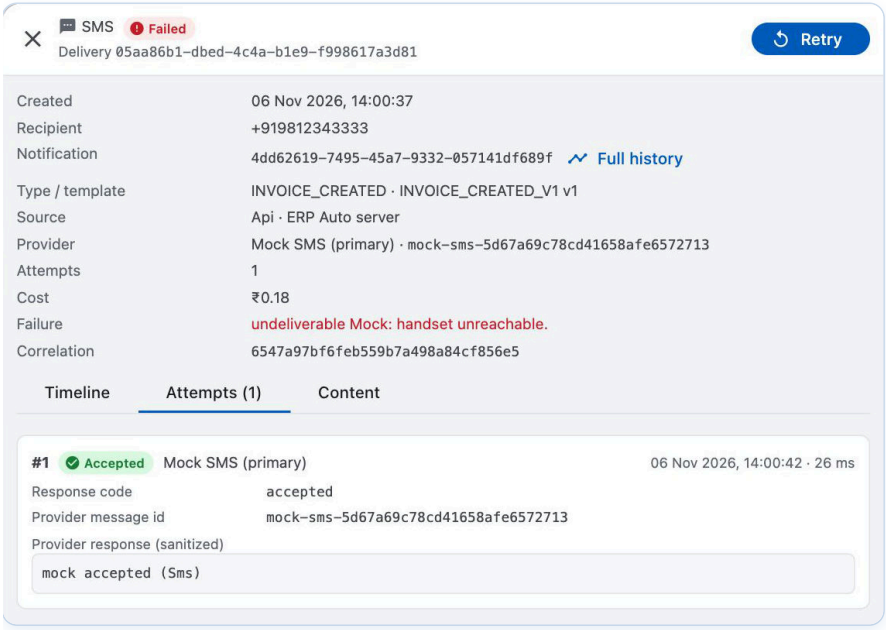


Figure 14: The Attempts tab, with the provider's sanitised response.

Attempt outcome	What it means
Accepted	The provider took the message.
Transient failure	The provider did not take it, and trying again may work. The hub retries or falls back by itself.
Permanent failure	The provider refused it for good, for example a bad address. Retrying will not help until the cause is fixed.
Ambiguous	Nobody knows whether the provider took it, for example a timeout after sending. See section 6.5 on page 19.
Throttled	The provider asked the hub to slow down. This does not count as a failed attempt.

6.3 What each status means

A delivery normally moves through Queued Processing Submitted Provider accepted Sent Delivered Read. Not every channel reaches every status: some email and SMS providers never confirm delivery, and only some channels report that a message was read.

Needs attention	What to do
Failed	Read the failure, fix the cause, then retry (section 6.4 on page 19).
Unknown	Decide with **Resolve** (section 6.5 on page 19).
Expired	It ran out of time before it could be sent. Retry if it is still useful.
Suppressed	The hub chose not to send it. Read the reason on the Timeline, then see Chapter 12 on page 44. Do not retry.
Retry pending	The hub will try again by itself. You can also select **Retry**.

All statuses are in Appendix A on page 78.

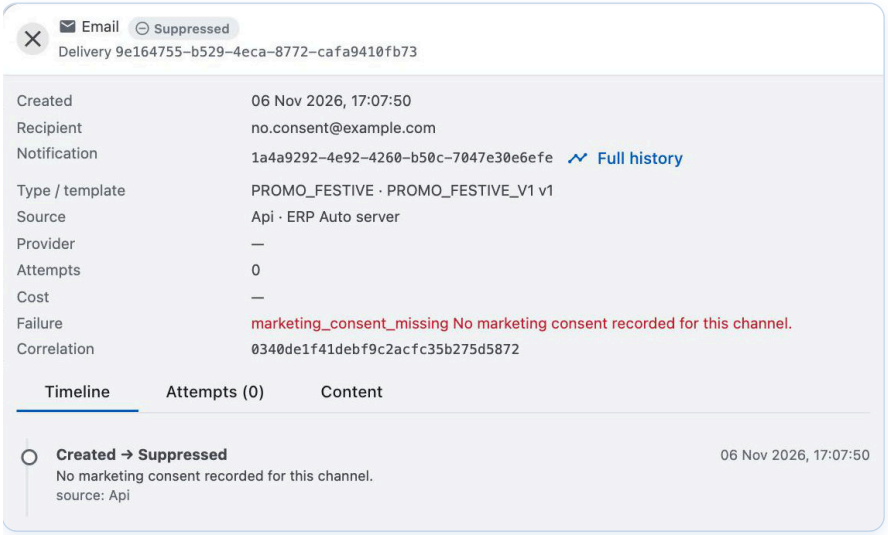


Figure 15: A suppressed delivery: the Failure line gives the reason, and there are no provider attempts.

6.4 Retrying a failed delivery

Who can do this: platform, tenant, company and application administrators, and notification operators.

1. Find the delivery and open it. **Retry** appears for Failed, Expired and Retry pending.
2. Read the **Failure** line. If the address is wrong or blocked, fix that first, or the retry will fail the same way.
3. Select **Retry**. You see `Retry scheduled (RetryPending). Earlier attempts stay on the timeline.`
4. Watch the panel: the status and Timeline update by themselves.

> **Note** A retry keeps the same delivery. Earlier attempts stay on the timeline as a permanent record, and a new attempt is added below them.

6.5 Resolving an unknown outcome

Sometimes a connection fails just after a message was handed over, so nobody knows whether the provider took it. The hub will not resend by itself when that could send a duplicate. The delivery stops at Unknown, is marked Awaiting operator, and raises a warning alert.

Who can do this: platform, tenant, company and application administrators, and notification operators.

1. On the **Dashboard**, select the **Awaiting operator** card. Or tick **Awaiting operator** in the explorer.
2. Open the delivery.
3. Check with the provider or the recipient whether the message arrived. The **Provider** line and the **Attempts** tab give you the reference to quote.
4. Select **Resolve** and choose:
   - **Resend (accept duplicate risk)**: it did not arrive, or a duplicate does no harm. The delivery is sent again.
   - **Mark as sent**: the provider did accept it. Nothing is resent.
   - **Mark as failed**: it did not go, and you do not want to resend. Nothing is resent.
5. Enter a **Comment for the audit trail**, then select **Resend** or **Confirm**.

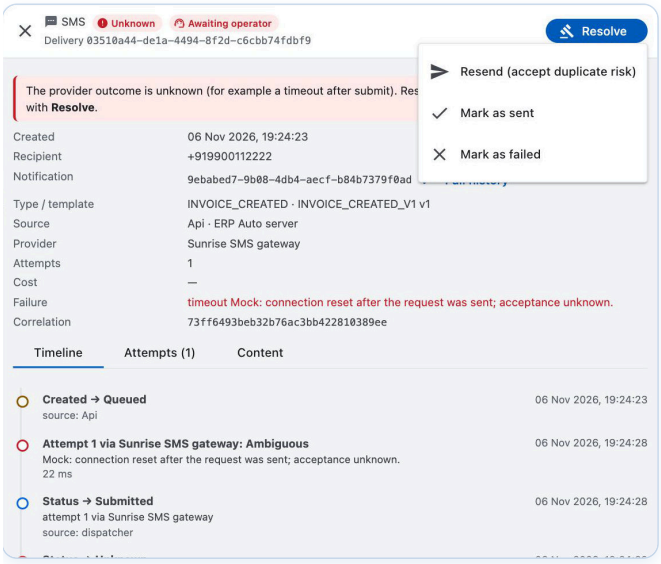


Figure 16: An unknown outcome and the Resolve menu.

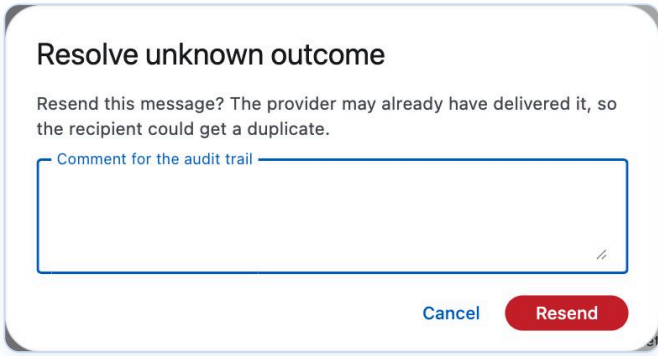


Figure 17: Resending asks you to accept the risk of a duplicate.

> **Warning** Resending an invoice twice is usually harmless. Resending a payment reminder or a one-time code may not be. When in doubt, check with the provider first.

6.6 Exporting to a file

1. Set the filters. The export contains what the filters select.
2. Select **Export**, then **CSV** or **Excel (.xlsx)**. The file downloads with a name such as `deliveries-20261106-1406.csv`.

> **Note** Email addresses and phone numbers are masked in exports, for every role. Times in the file are UTC, so they differ from the local times on the screen.

CHAPTER 7

Inbox

When customers reply on WhatsApp, SMS or email, their messages arrive here as conversations, and you can answer from the console.

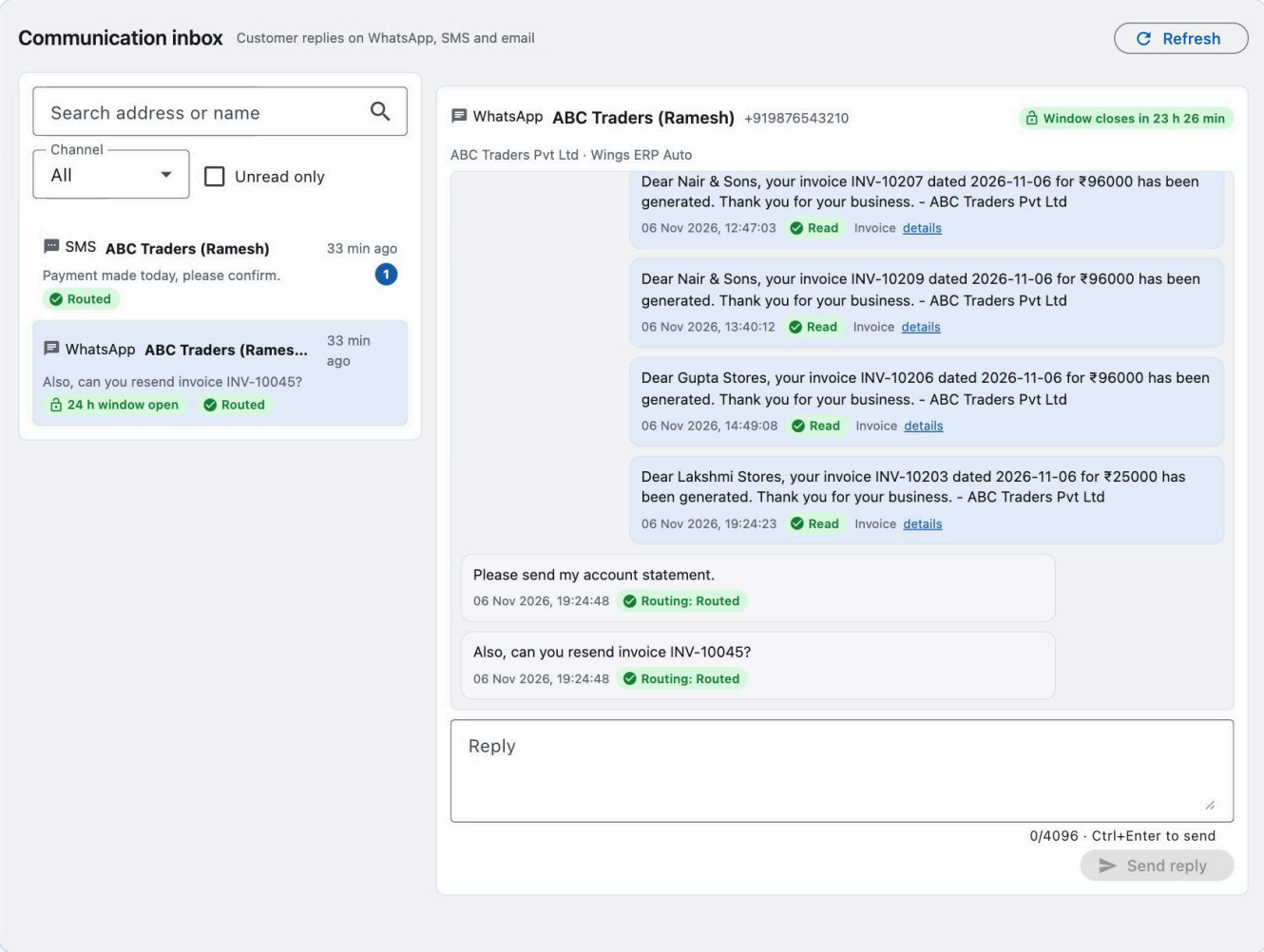


Figure 18: A WhatsApp conversation. Your own messages are on the right, with their delivery status.

7.1 Reading conversations

Who can do this: everyone who can open the Inbox. Replying needs a role with inbox rights: platform, tenant and company administrators, and notification operators. Auditors and application administrators read only.

1. Open **Inbox**. Conversations are on the left, most recent first. A number shows unread messages.
2. Narrow the list with **Search address or name** (press Enter), **Channel** or **Unread only**.
3. Select a conversation. It is marked as read if your role can reply.

Each conversation shows its routing to your application: Routed means the application received the reply, Pending means it has not yet, and No subscriber means no application is listening for replies (see Chapter 18 on page 66).

7.2 Replying to a customer

1. Open the conversation.
2. Type in **Reply**, up to 4096 characters.

3. Select **Send reply**, or press Ctrl+Enter (Cmd+Enter on a Mac). You see `Reply queued`, and the message appears with its status.
4. Select **details** under a message to open its delivery in the explorer.

7.3 The WhatsApp 24-hour window

WhatsApp allows free-form replies only within 24 hours of the customer's last message. The conversation heading counts down, for example Window closes in 23 h 26 min.

After that it shows Window closed, and the reply box is turned off. Send an approved WhatsApp template instead, or wait for the customer to write again.

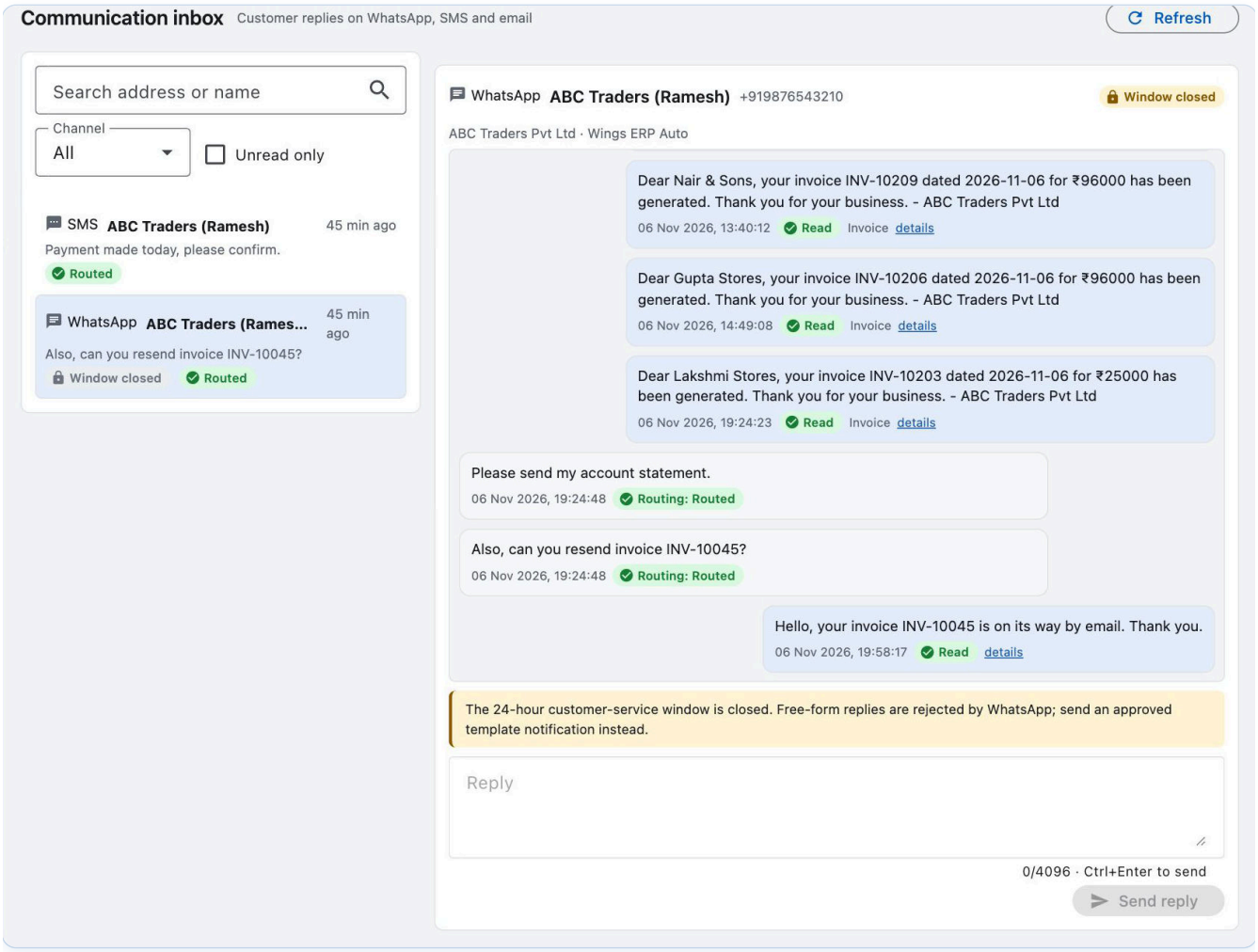


Figure 19: After 24 hours the reply box is switched off.

Note A reply is added to a conversation only when the hub can match it to a notification or customer. A message on a shared provider that cannot be matched does not appear in the Inbox. It raises a warning alert on **System monitoring** instead.

Note Platform administrators also see **Simulate reply**. It is a test helper and works only on a test or development hub.

CHAPTER 8

Approvals

Important changes need a second person. This is called maker-checker: whoever asked for the change cannot approve it.

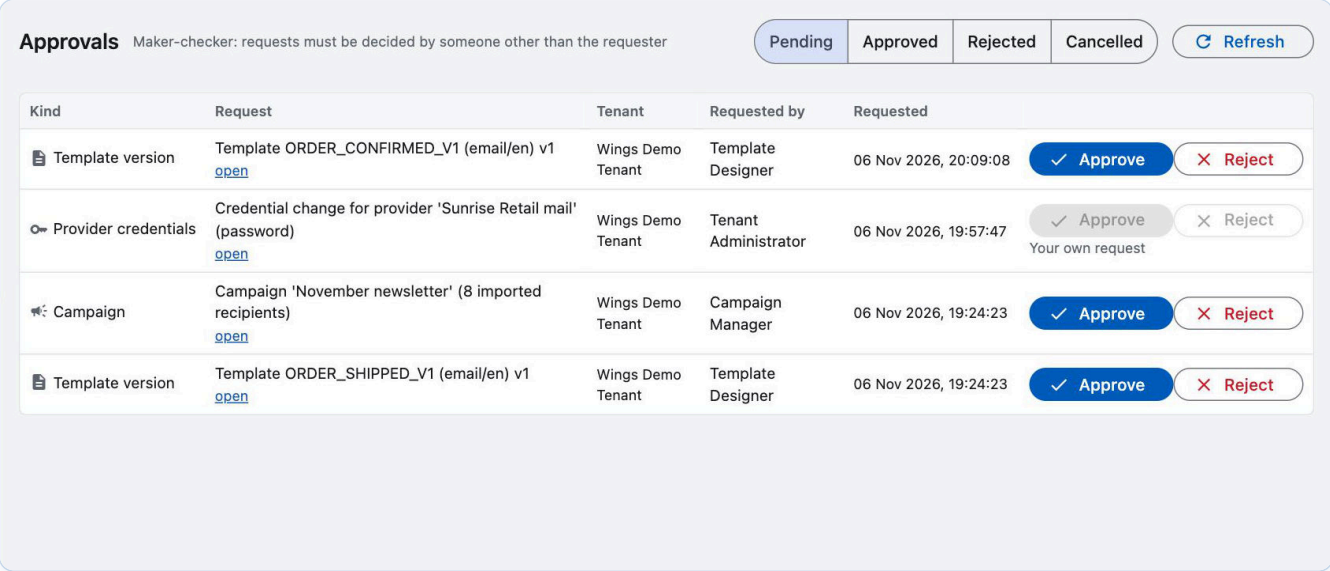


Figure 20: Pending approvals. You cannot decide your own request: it is marked Your own request.

Kind	What approving does	Who can decide
Template version	Activates the version. New notifications use it at once.	Template approver, company, tenant and platform administrators
Campaign	Schedules the campaign, or starts it if it has no schedule.	Company, tenant and platform administrators
Provider credentials	Replaces the stored credentials of a provider account.	Company, tenant and platform administrators
AI notification	Sends a message that an AI agent asked to send.	Company, tenant and platform administrators

A template approver sees only template versions on this screen. Other roles see the kinds they can decide.

8.1 Approving or rejecting a request

1. Open **Approvals**. **Pending** is selected.
2. Select **open** under a request to review it first. Each kind opens the screen where you can read the detail.
3. Select **Approve** or **Reject**.
4. Add a comment. It is optional when you approve, and a **Reason** is required when you reject.
5. Confirm. The request leaves the list.

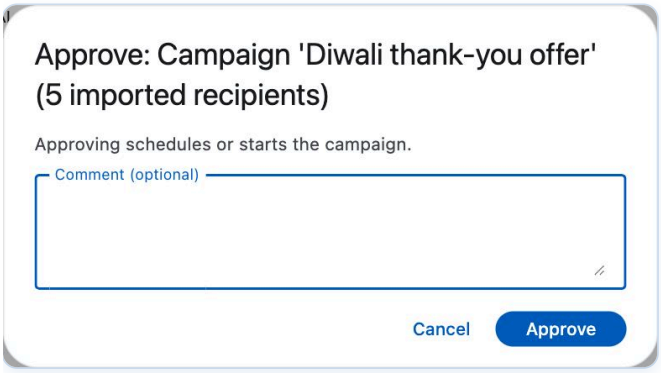


Figure 21: Approving tells you what will happen.

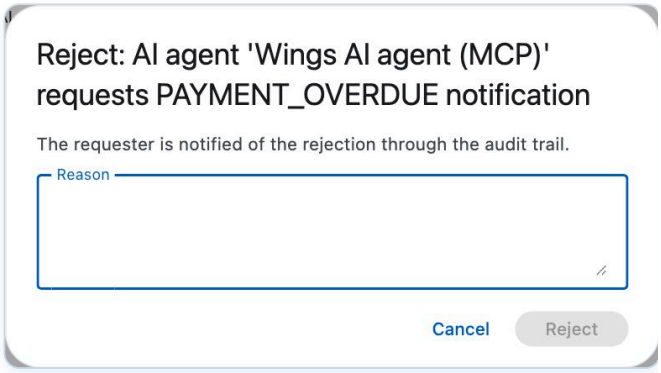


Figure 22: Rejecting needs a reason.

To review past decisions, select **Approved**, **Rejected** or **Cancelled**. A **Decision** column shows the outcome, when it happened and the comment.

Note AI agents can send only notification types that allow it. If a type also needs human approval, the agent's request waits here until someone decides. Rejecting tells the requester through the audit trail. No message goes to the recipient.

PART 3

Content and automation

Write messages, decide when they go out and to whom, and keep every recipient's choices respected.

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12	Recipients	44
13	Schedules and holiday calendars	49

CHAPTER 9

Templates

A template holds the wording of one message, for one channel and one language. Every change creates a version, and a second person approves it before it goes live.

Who can do this: template designers and platform, tenant, company and application administrators edit. Template approvers and platform, tenant and company administrators approve, activate and retire. Campaign managers and auditors can read. Operators do not see this screen.

9.1 Finding a template

The tree on the left runs from tenant to company, application, category and template. A company or application template overrides the tenant-wide one for that scope.

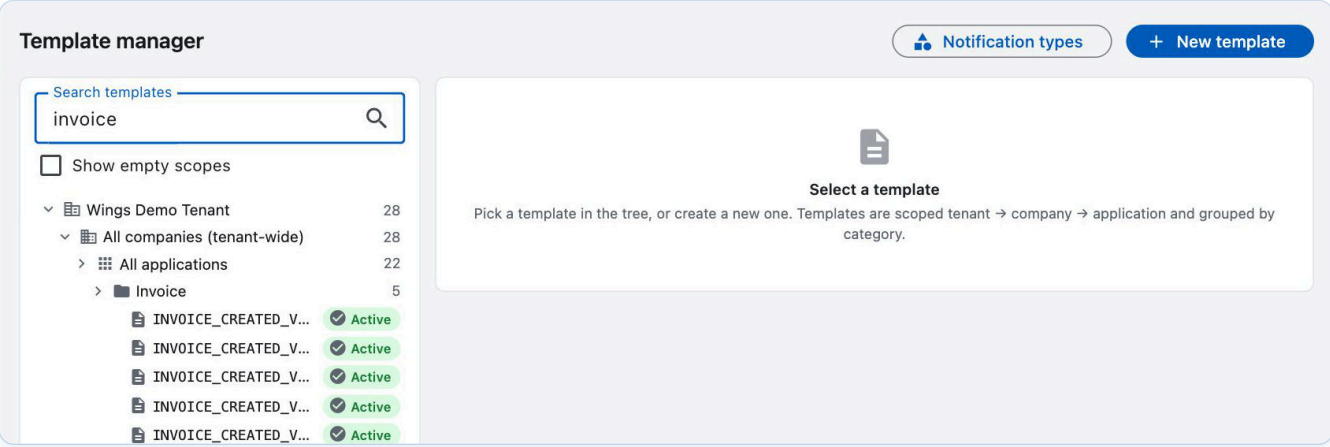


Figure 23: The template tree, filtered with Search templates.

- Use **Search templates** to match a code, name or channel.
- Tick **Show empty scopes** to see companies and applications that have no templates yet.
- Each template shows its status: Draft, In review, Approved, Active, Retired, Rejected.

9.2 Creating a template

1. Select **New template**. If you selected a place in the tree first, the scope is filled in for you.
2. Choose the **Tenant**, **Company** and **Application**. Leave **All companies (tenant-wide)** and **All applications** to share it.
3. Enter a **Code**, such as `PAYMENT_RECEIPT_V1`, which is changed to capitals, and a **Name**.
4. Enter a **Category** to group it in the tree, choose the **Channel**, and enter the **Locale**, such as `en`, `hi` or `ta-IN`.
5. For email, enter a **Subject**.
6. Enter a first version in **Body**. You can polish it in the editor.
7. Select **Create draft**. The template opens in the editor as Draft.

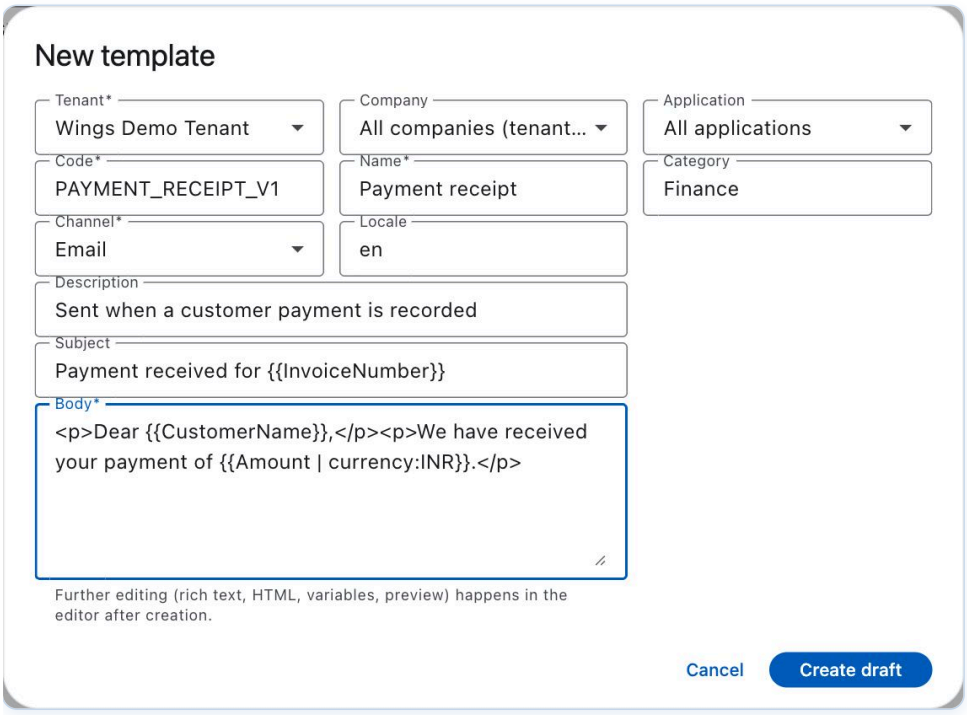


Figure 24: The New template dialog.

9.3 Editing a template

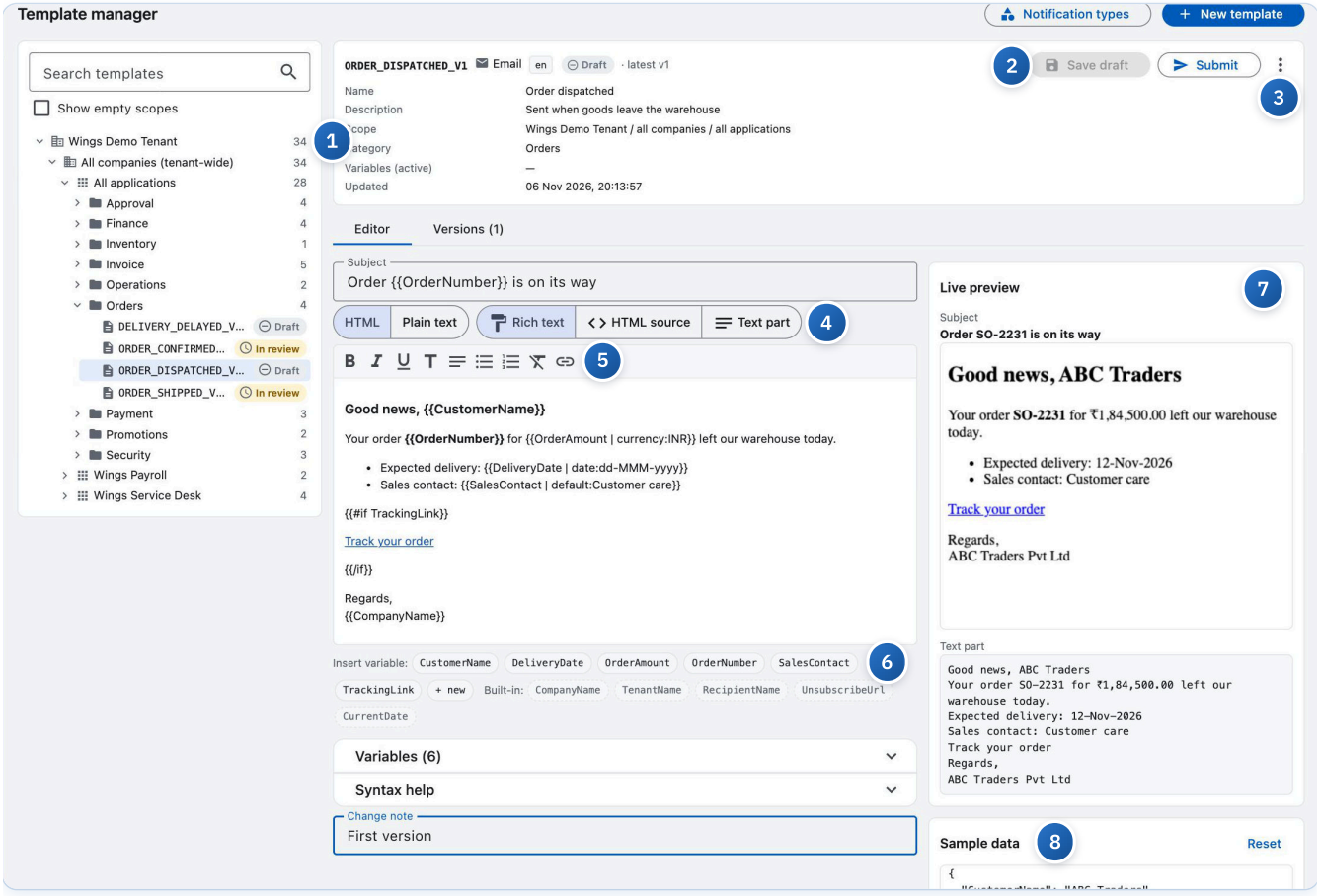


Figure 25: The template editor.

1. **Tree.** Select a template to open it.
2. **Save draft** and **Submit**.
3. **More actions:** test send, clone, retire.
4. **Format.** HTML or plain text; rich text, HTML source or text part.
5. **Toolbar** for rich text.
6. **Variable chips** insert at the cursor.
7. **Live preview** updates as you type.
8. **Sample data** feeds the preview.

To edit a template

1. Select the template. An orange **Unsaved changes** chip appears as soon as you edit.
2. For email, choose **HTML** or **Plain text**. For HTML, choose how to work:
   - **Rich text**: use the toolbar: **Bold**, **Italic**, **Underline**, **Heading**, **Paragraph**, **Bulleted list**, **Numbered list**, **Clear formatting** and **Link**.
   - **HTML source**: edit the HTML directly.
   - **Text part**: the plain-text alternative. Leave it empty and the hub generates one from the HTML.
3. Insert variables with the chips under the editor. Dashed chips are built-in values the hub supplies when sending.
4. Check **Live preview**. Change **Sample data** to try other values. **Reset** rebuilds it from the variables' sample values.
5. Enter a **Change note**, then select **Save draft**.

> **Note** Editing an active template does not affect what is being sent. Saving creates a new draft version, and the active version keeps sending until the new one is approved.

> **Note** Links can point to a web address or a variable: in the **Insert link** box enter `https://…` or `{{PaymentLink | url}}`.

> **Warning** Template HTML cannot contain scripts, event handlers, `javascript:` links or frames. Saving `<script>` fails with `Template HTML contains a disallowed construct: '<script'`.

9.4 Variables and filters

Variables are placeholders in double braces, such as `{{CustomerName}}`. The sending application supplies the values. Filters format them: `{{Amount | currency:INR}}` shows ₹18,400.00.

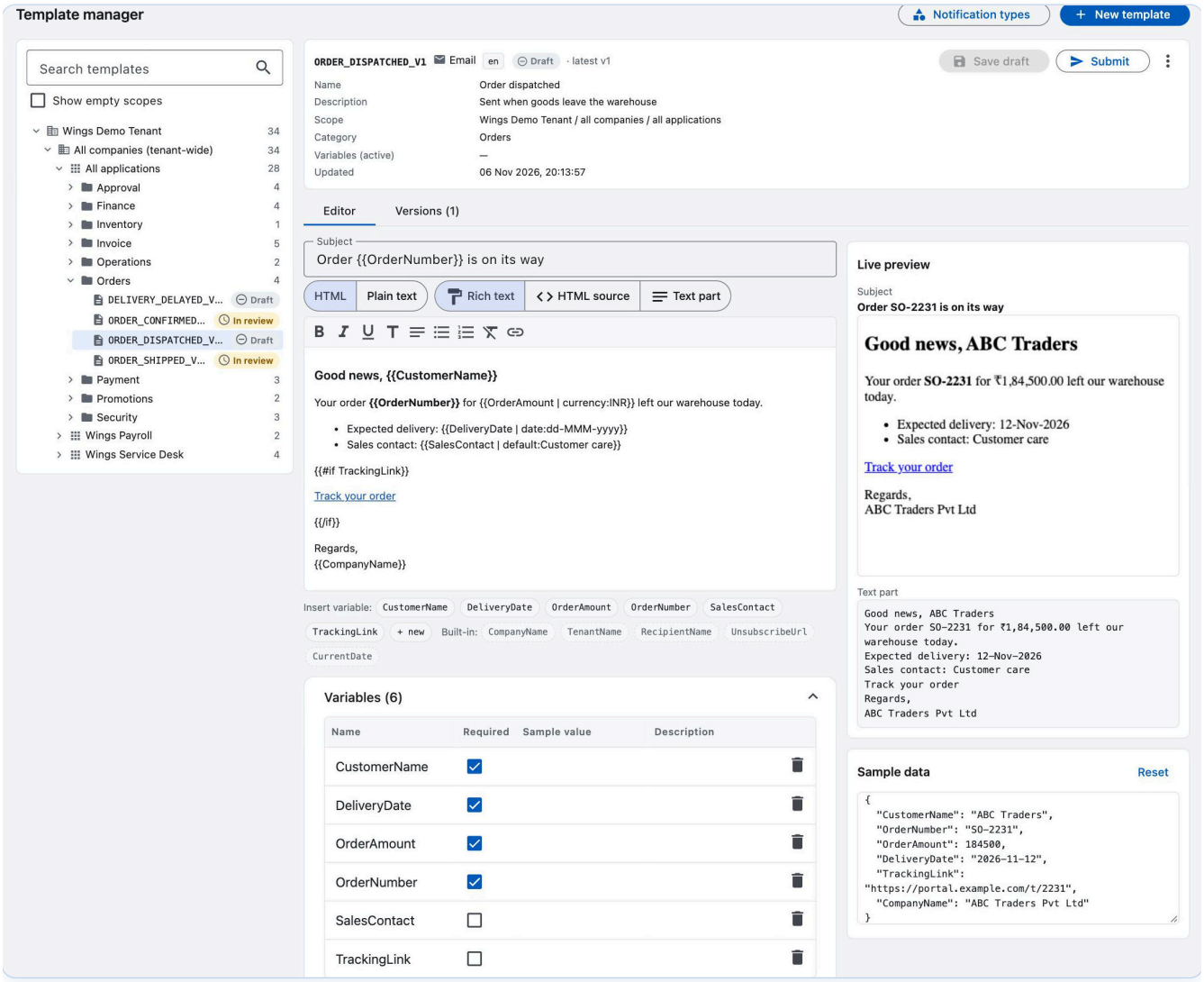


Figure 26: The Variables panel: required flags, sample values and descriptions.

- Open **Variables** to set each one's **Required** flag, **Sample value** and **Description**.
- A variable you use but have not listed shows `used in the content; added as required on save`.
- The preview warns **Missing:** for a required value with no sample, and **Not used by the template:** for a sample nothing uses.
- Built-in variables need no declaring: `CompanyName`, `TenantName`, `RecipientName`, `UnsubscribeUrl`, `CurrentDate`.
- Open **Syntax help** for the filters and blocks. Appendix B on page 79 lists them all.

9.5 Sending a test

Who can do this: anyone who can edit templates, including template designers.

1. Select **More actions** (⋮), then **Test send**.
2. Choose the **Company**, **Application** and **Version**.
3. Enter the test address, which depends on the channel: **Email**, **Mobile (E.164)**, **WhatsApp (E.164)** or **User id**.
4. Check **Variables (JSON)**, then select **Send test**.

Only the active version is delivered. For any other version the dialog says `Not active: a rendered preview is returned instead of a delivery.` and shows the result. For an active version it shows `Accepted as notification …` and the delivery's status.

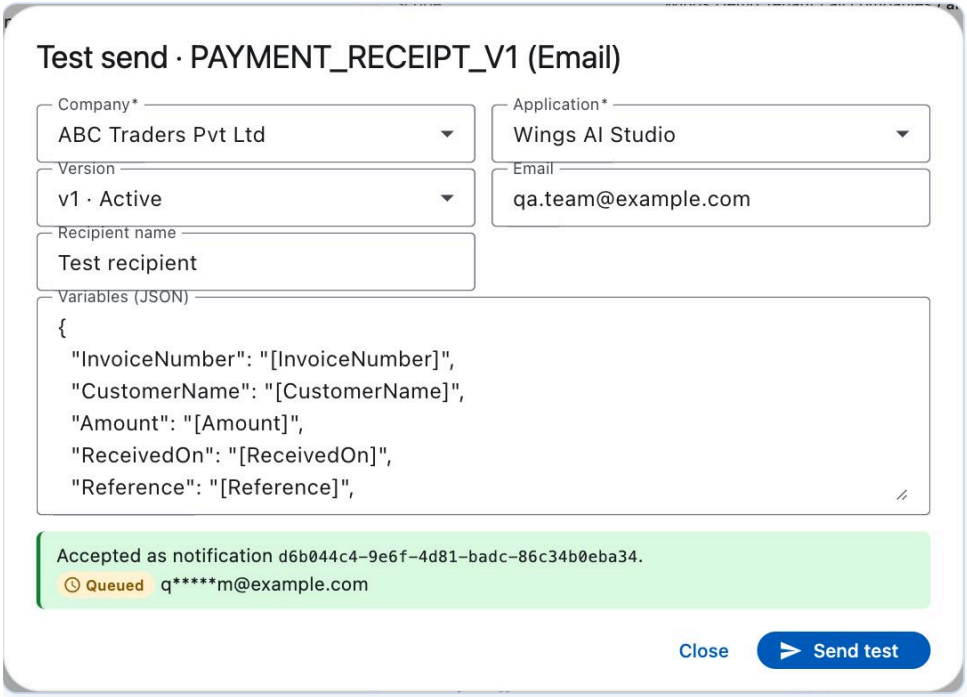


Figure 27: A test of the active version, accepted for delivery.

9.6 Submitting, approving and activating

Who	Step	Status afterwards
Designer	Saves, then selects **Submit** and confirms. The version becomes read-only.	In review
Approver	Selects **Approve** on the **Versions** tab or the **Approvals** screen, and confirms.	Active
Approver	Or selects **Reject** and enters a reason.	Rejected
Designer	Sees the reason at the top of the editor. Saving turns it back into a draft: fix it and submit again.	Draft

Note You cannot approve a version you created or submitted. The buttons are switched off, with the reason `Maker–checker: you created or submitted this version`.

Note Approving always activates the version as well, so you will rarely see Approved in normal work.

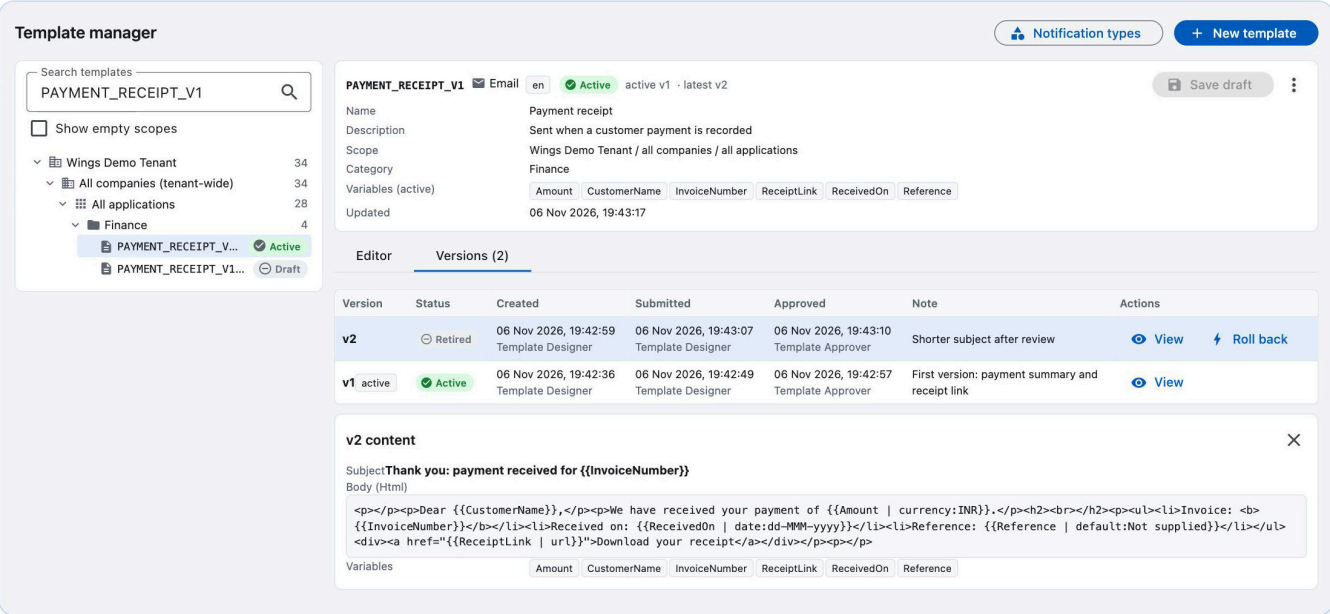


Figure 28: The Versions tab, with one version's content open. Every version stays on record.

The **Versions** tab lists who created, submitted and approved each version. Select **View** to read its content. People without user-management rights, such as template approvers, see an internal id in place of the person's name.

9.7 Rolling back or retiring

Who can do this: template approvers and platform, tenant and company administrators.

To roll back to an earlier version

1. Open the template's **Versions** tab.
2. On the earlier Retired version, select **Roll back**. An Approved version shows **Activate** instead.
3. Select **Activate** to confirm. The current version is retired, and new notifications use the one you chose.

Past deliveries are not changed: each records the version it used.

To retire a template

1. Select **More actions** (⋮), then **Retire template**.
2. Select **Retire** to confirm.

> **Warning** Notifications that use a retired template fail until a version is activated again. Check that no rule, schedule or application still uses it. To bring it back, use **Roll back** on the **Versions** tab.

9.8 Cloning for a translation

Translations are separate templates with the same code and a different locale. Cloning is also how you make a company or application override.

1. Open the template, then select **More actions** (⋮), **Clone / translate**.
2. Change the **Locale**, such as `hi`, and the **Name**. For an override, choose a narrower **Company** or **Application**.
3. Select **Clone**. A new draft opens. Translate it, then submit it for approval as usual.

9.9 Managing notification types

A notification type names a kind of message and sets its defaults. Select **Notification types** at the top of the screen.

1. To change one, select its row. To add one, select **New**.
2. Fill in the **Tenant**, **Application (optional)**, **Code**, **Name** and **Category**.
3. Choose the **Message class**, **Default channels**, **Default template code** and **Default priority**.
4. Tick the options you need, then select **Create** or **Save**.

Option	Effect
Mandatory (ignores opt-outs)	Transactional messages of this type go out even if the recipient turned the channel off.
Sensitive (codes masked in logs)	For one-time codes and the like. AI agents cannot send these types.
AI agents may send	Lets a connected AI agent request this type.
AI sends need human approval	Each AI request waits in **Approvals**.

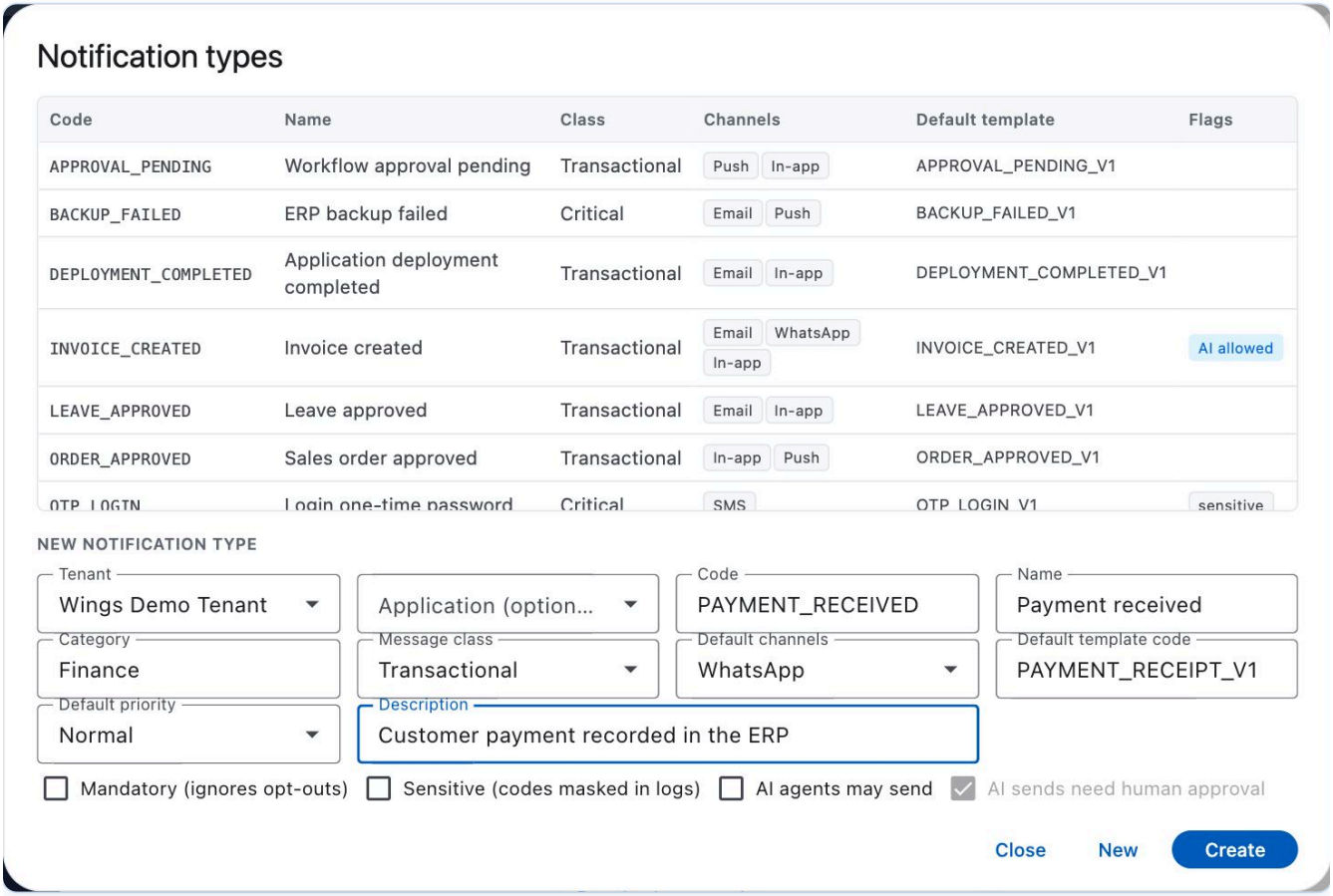


Figure 29: The Notification types dialog.

9.10 SMS templates and DLT

The editor shows the encoding and length as you type, for example `GSM-7 · 123 chars · 1 segment (160/segment)`. The figure turns red above three segments. Characters outside the GSM-7 set switch the message to UCS-2, which fits only 70 characters per segment, or 67 when split. The preview shows the exact count once variables are filled in.

For India, enter the DLT content template id in **DLT template id (India)**. The rendered text must match the registered template exactly. The DLT entity id goes on the sender identity (section 14.4 on page 56).

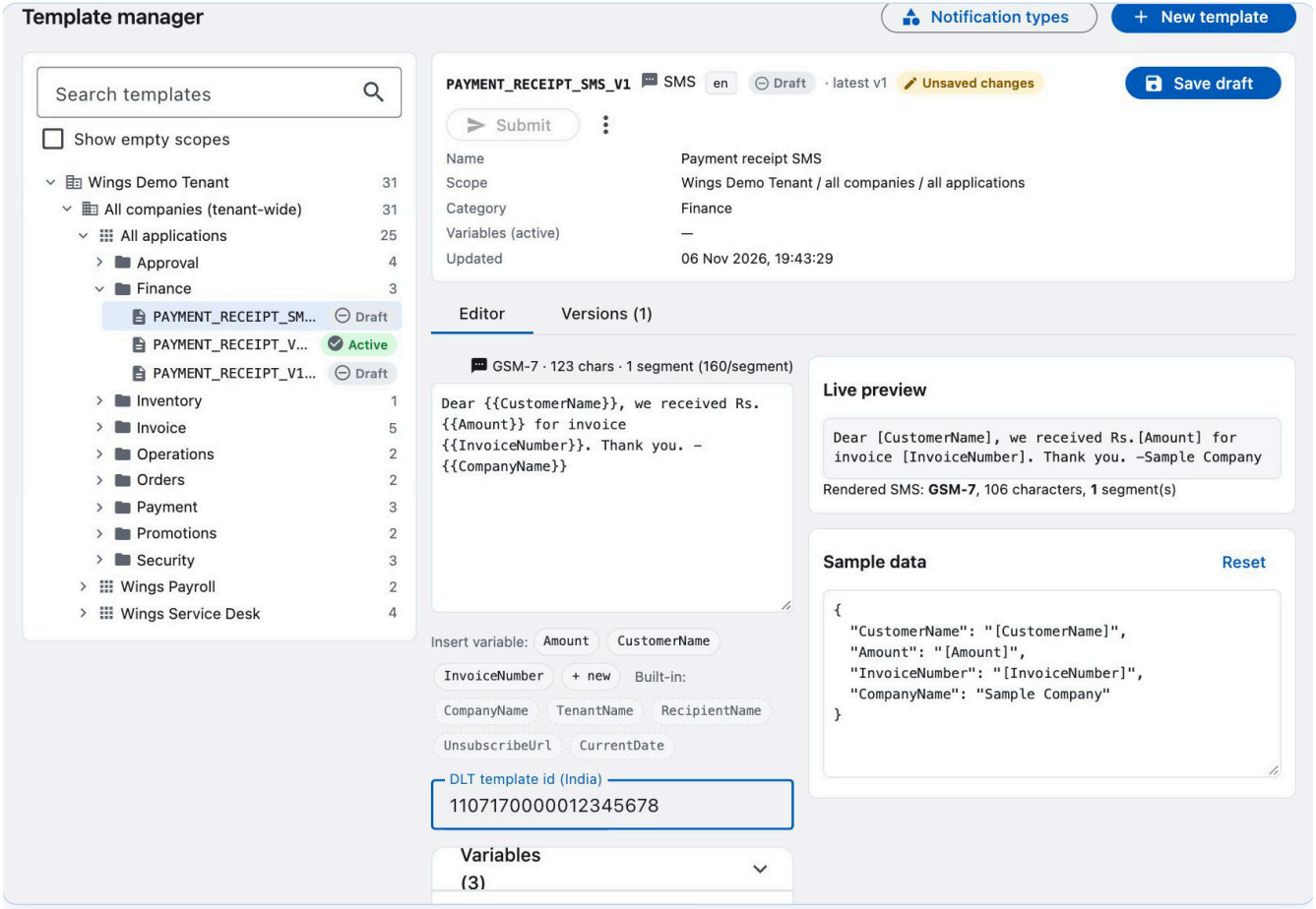


Figure 30: An SMS template, with the segment counter and DLT template id.

9.11 WhatsApp templates

Messages your business starts on WhatsApp must use a template approved by Meta. Write the body with normal variables: the hub converts them to WhatsApp's numbered placeholders.

1. Under **WhatsApp provider template**, enter the **Provider template name** in lower case with underscores, such as `invoice_created_v1`, and the **Language**, such as `en_US`.
2. Choose the **Category**: **Utility**, **Marketing** or **Authentication**.
3. Optionally set a **Header** (**Text**, **Image**, **Document** or **Video**; media headers need a **Media URL variable**), a **Footer**, and up to ten buttons: **Quick reply**, **URL** or **Phone**.
4. Select **Save draft**.
5. Select **More actions** (⋮), then **Submit to WhatsApp**. Choose the **WhatsApp provider account** and **Version**, then **Submit**. The result appears on the **Versions** tab as **WA:** and the status.

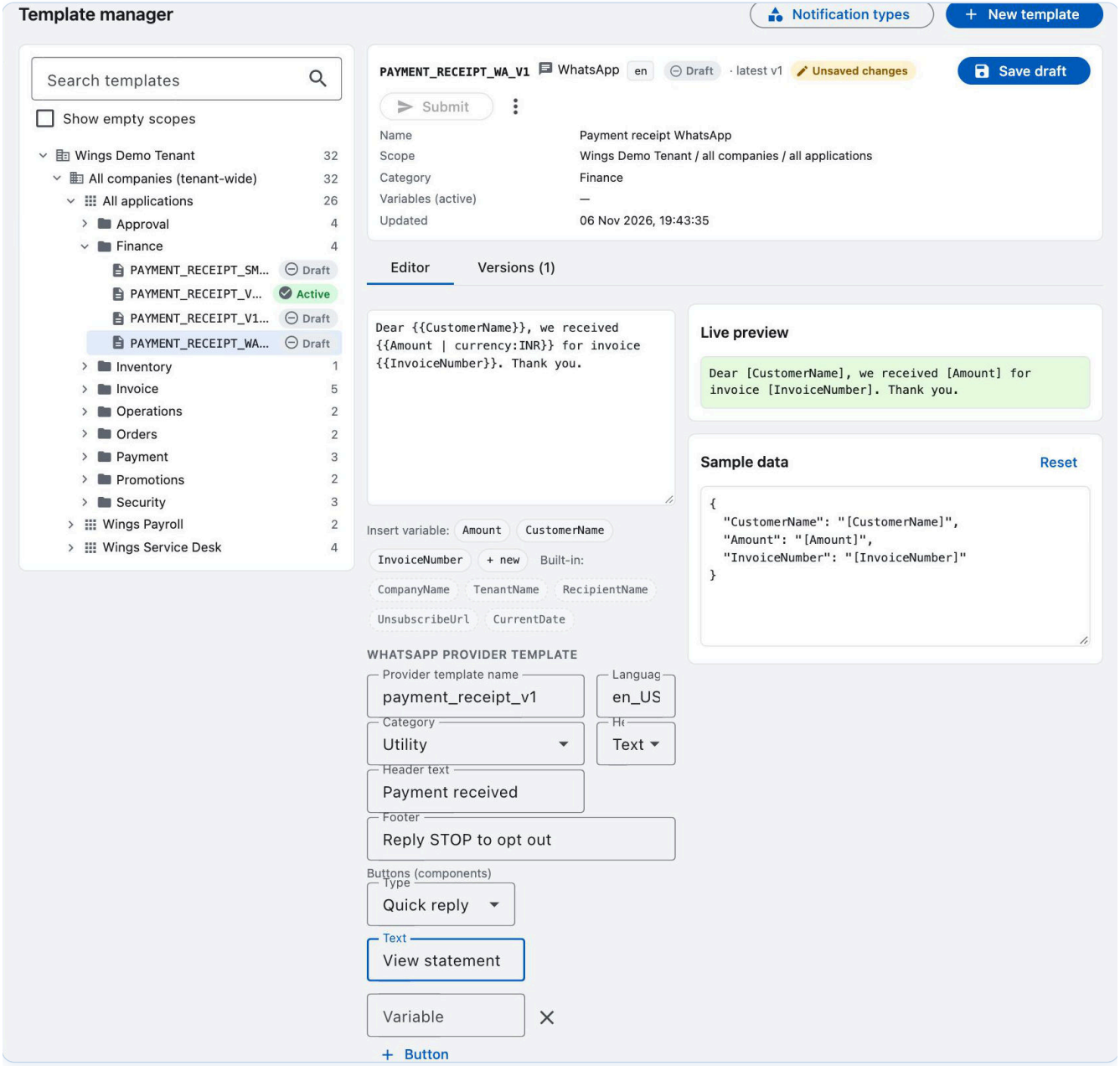


Figure 31: A WhatsApp template with header, footer and a quick-reply button.

> **Note** If no WhatsApp account is listed, ask an administrator to add one under **Providers**. A Mock provider answers `The provider account does not support template submission.`

CHAPTER 10

Rule designer

Rules let your applications report what happened, such as “payment overdue”, and leave the hub to decide who hears about it and how. Each rule reads: when this event happens, and these conditions are true, tell these recipients on these channels.

Who can do this: platform, tenant, company and application administrators change rules. Template designers and auditors can read rules, and can run a simulation.

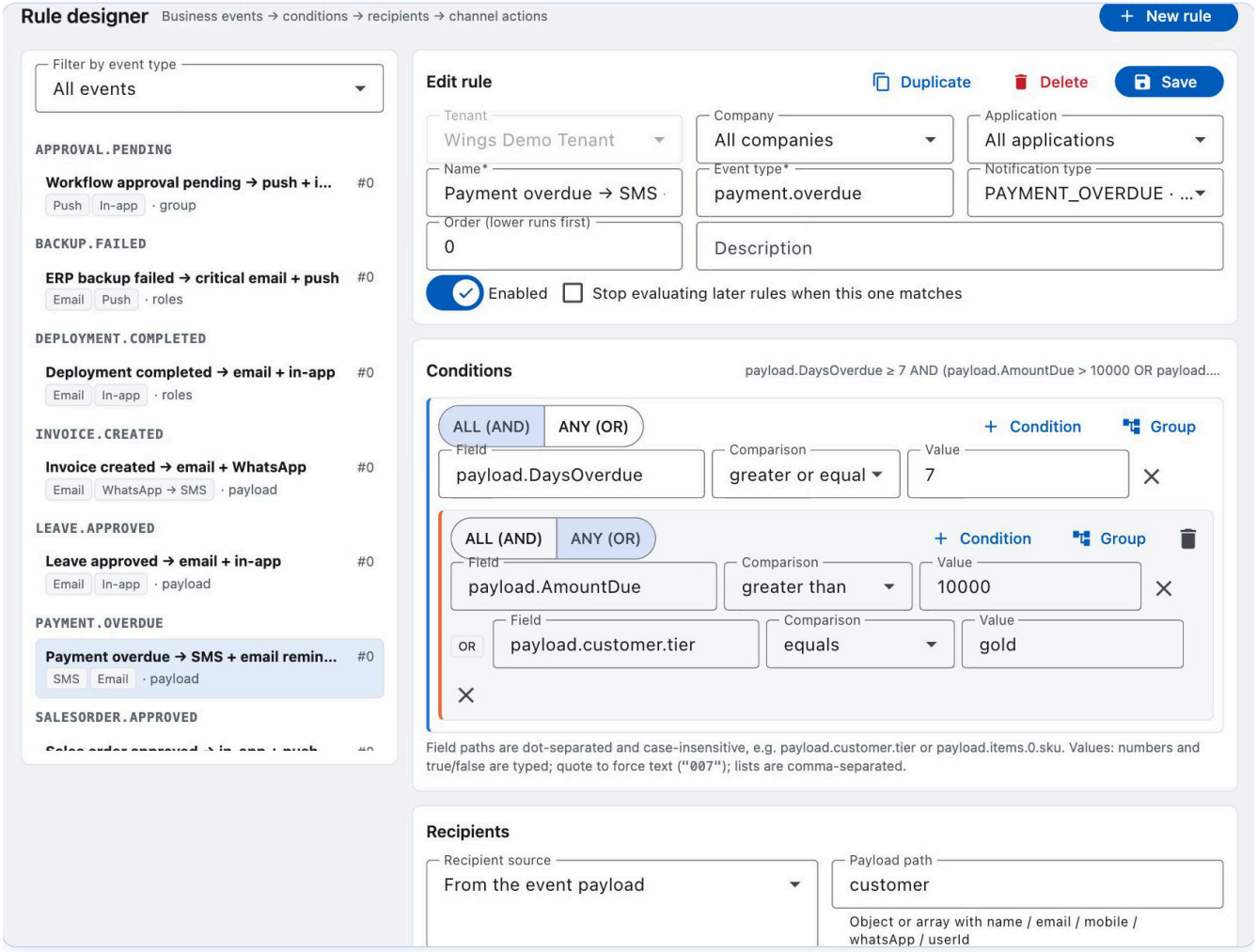


Figure 32: The Rule designer: rules grouped by event on the left, the selected rule on the right.

10.1 Creating a rule

1. Select **New rule**, or open a similar rule and select **Duplicate**. A copy starts disabled, with `(copy)` in its name.
2. Set the **Tenant**, **Company** and **Application** the rule covers.
3. Enter a **Name** and the **Event type**, such as `payment.overdue`. Suggestions appear as you type.
4. Choose a **Notification type**, and set **Order (lower runs first)**.
5. Leave **Enabled** on. Tick **Stop evaluating later rules when this one matches** if this rule should be the last one to run.
6. Add conditions, recipients and channel actions as described below, then select **Save**.

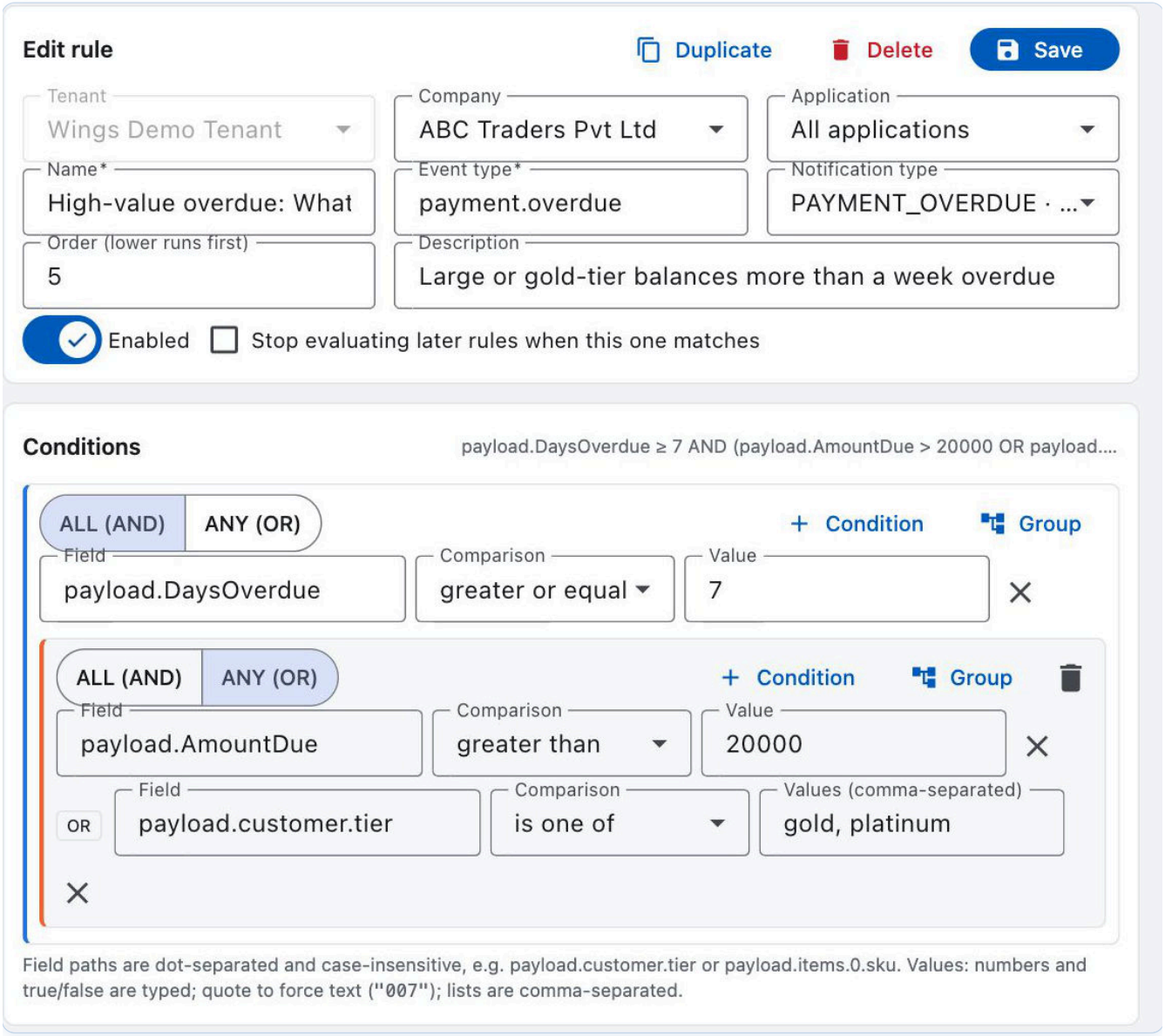


Figure 33: A rule with nested conditions: at least 7 days overdue, and either a large amount or a gold or platinum customer.

> **Note** A rule cannot be saved without a name, event type and template. The message is `The rule is invalid. Name is required. Event type is required. Every action needs a template code.`

10.2 Building conditions

Conditions decide whether the rule fires. With none, it shows `No conditions: always matches`. A summary at the top of the panel reads the logic back in one line, such as `payload.DaysOverdue ≥ 7 AND (payload.AmountDue > 20000 OR payload.customer.tier in [gold, platinum])`.

1. Choose **ALL (AND)** or **ANY (OR)** for the group.
2. Select **Condition**. Enter the **Field**, such as `payload.DaysOverdue`, choose the **Comparison**, and enter the **Value**.
3. Select **Group** to nest another AND/OR group. Groups can go up to ten levels deep.

Comparisons	Notes
equals, not equal, greater than, greater or equal, less than, less or equal	Numbers and `true` / `false` compare by type. Put text in quotes to force it, such as `"007"`.

Comparisons	Notes
contains, does not contain, starts with, ends with	For text.
is one of, is none of	Values are comma-separated, such as `gold, platinum`.
exists, does not exist, is empty, is not empty	No value needed.

Field paths are dot-separated and ignore case, for example `payload.customer.tier` or `payload.items.0.sku`.

10.3 Choosing recipients and channels

Recipient source	Then enter
From the event payload	**Payload path**: an object or list holding `name`, `email`, `mobile`, `whatsApp` or `userId`, such as `customer`.
Specific users	**User ids (comma-separated)**.
Users with roles	**Roles (comma-separated)**, such as `Manager, StoreKeeper`. These are the roles held by contacts, not console roles.
Recipient group	**Group**, from section 12.6 on page 47.
All company users	Nothing more.

Under **Channel actions**, each line sets a **Channel** and **Template**. Select **Action** to add another. Set **Fallback** to try a second channel when the first fails, expires or is suppressed, with a **Fallback template** or **Same code**.

10.4 Priority, retry, escalation and quiet hours

Open these panels under the channel actions when you need them.

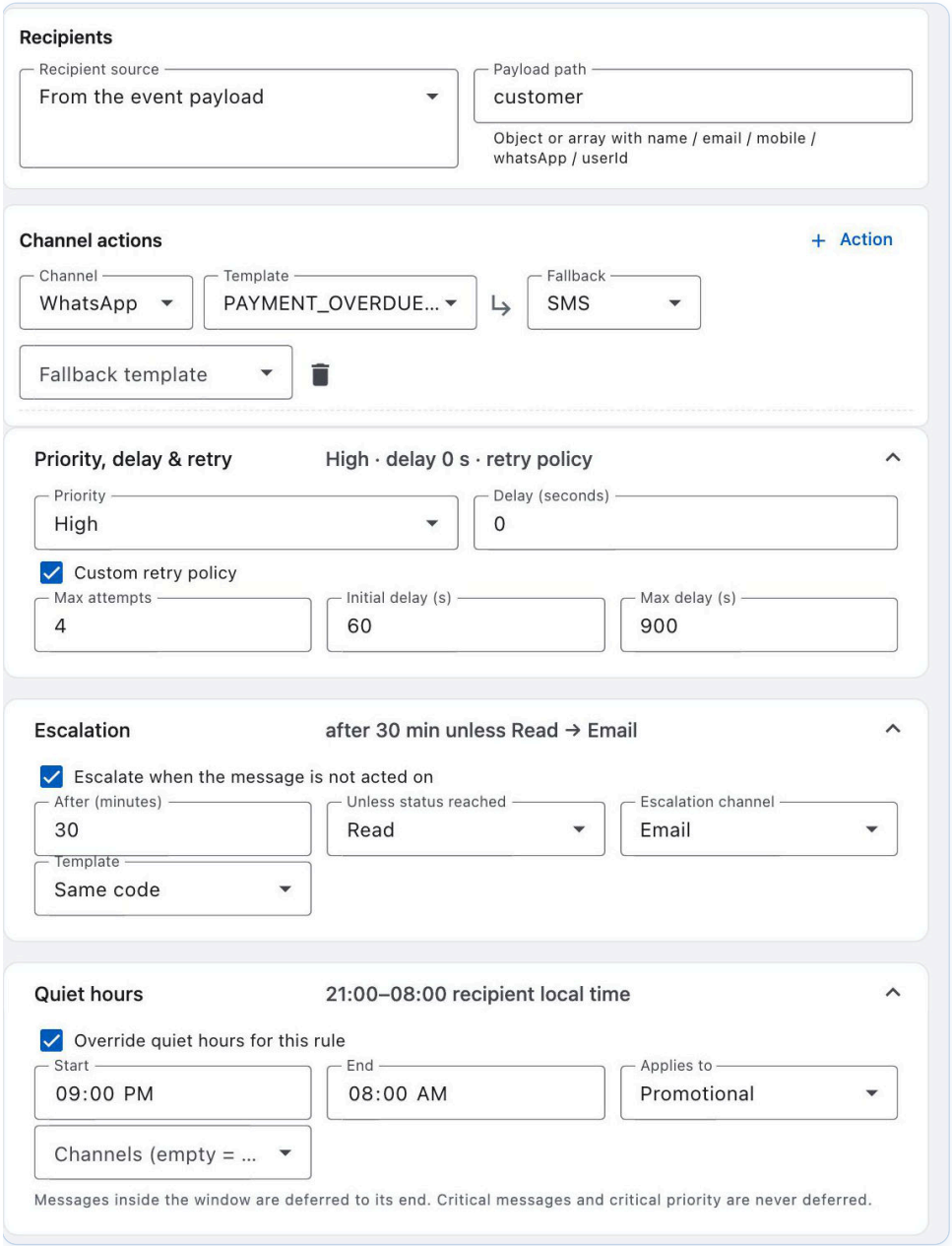


Figure 34: Recipients, a channel action with an SMS fallback, and the three optional panels opened.

Panel	Settings
Priority, delay & retry	**Priority** overrides the type's default. **Delay (seconds)** holds the message back. **Custom retry policy** sets **Max attempts**, **Initial delay (s)** and **Max delay (s)**.
Escalation	Tick **Escalate when the message is not acted on**, then set **After (minutes)**, **Unless status reached**, **Escalation channel** and **Template**. For example: send an email after 30 minutes unless the WhatsApp was read.
Quiet hours	Tick **Override quiet hours for this rule**, then set **Start**, **End**, **Applies to** and **Channels (empty = all except in-app)**, in the recipient's local time. Messages in the window wait until it ends. Critical messages and critical priority are never delayed.

10.5 Simulating a rule

A simulation shows exactly what would happen. Nothing is sent.

1. Save the rule. Simulation uses saved rules: `save first to test your edits`.
2. Under **Simulation**, choose the **Company** and **Application**.

3. Edit **Sample payload (JSON)** to resemble a real event.
4. Tick **Only this rule (saved version, even if disabled)** to test it alone, or leave it clear to see every rule for the event.
5. Select **Run simulation**.

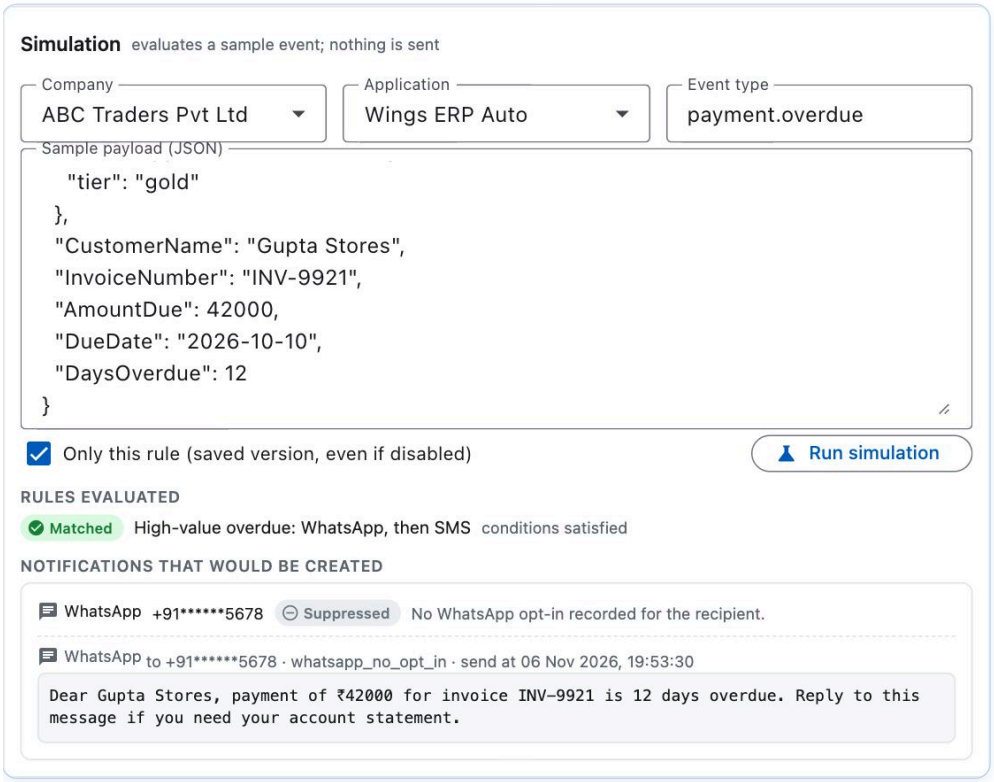


Figure 35: A simulation: the rule matched, but this customer has no WhatsApp opt-in.

Rules evaluated shows Matched or No match for each rule. **Notifications that would be created** lists each delivery, the consent decision, the send time and the rendered text. In the example above, a real event would then go by SMS through the fallback.

> **Tip** Before switching on a rule, simulate three payloads: one that should match, one that should not, and one customer with no consent.

To remove a rule, open it and select **Delete**. Events no longer trigger it.

CHAPTER 11

Campaigns

A campaign sends one message to many people at a controlled pace. A manager prepares it, and someone else approves it.

Who can do this: campaign managers and platform, tenant and company administrators prepare campaigns. Administrators at those levels approve them. Operators, auditors and application administrators can look.

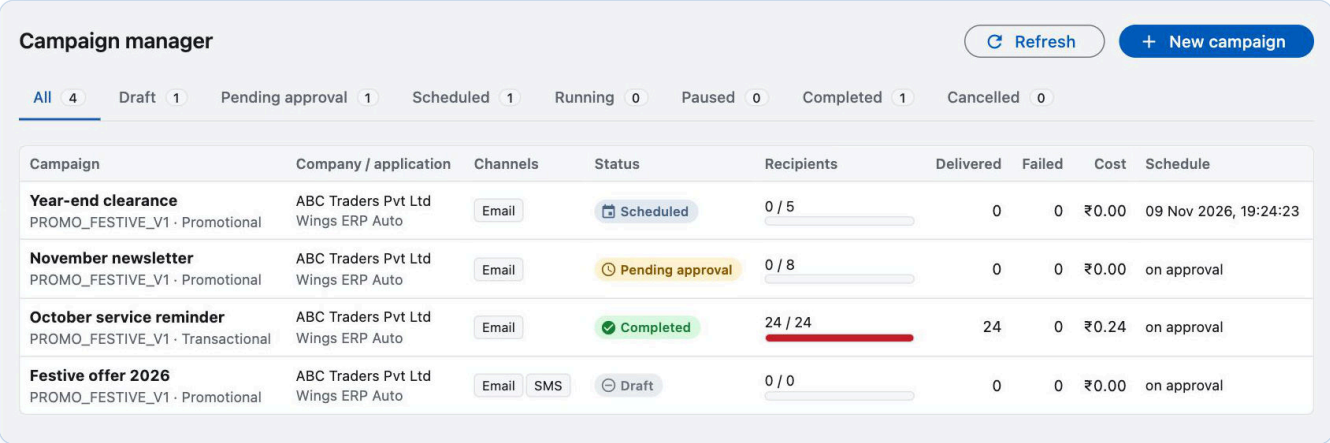


Figure 36: Campaign manager. Tabs filter by status: All, Draft, Pending approval, Scheduled, Running, Paused, Completed, Cancelled.

11.1 Creating a campaign

1. Select **New campaign**.
2. Choose the **Company** and **Application**, and enter a **Name**.
3. Choose the **Channels** and **Template code**. The list shows active templates for the channels you chose, and each channel needs one.
4. Choose the **Notification type** and **Message class**. **Promotional** needs marketing consent and honours unsubscribes.
5. Set **Schedule (local time, optional)**. Leave it empty to start on approval.
6. Set **Throttle (messages/minute)**. The default is 600.
7. Choose who receives it: **Recipient groups**, a **Segment**, an imported list, or a mix.
8. Enter **Template variables (one per line: Name=value)**. Imported columns override these for each recipient.
9. Select **Save draft**.

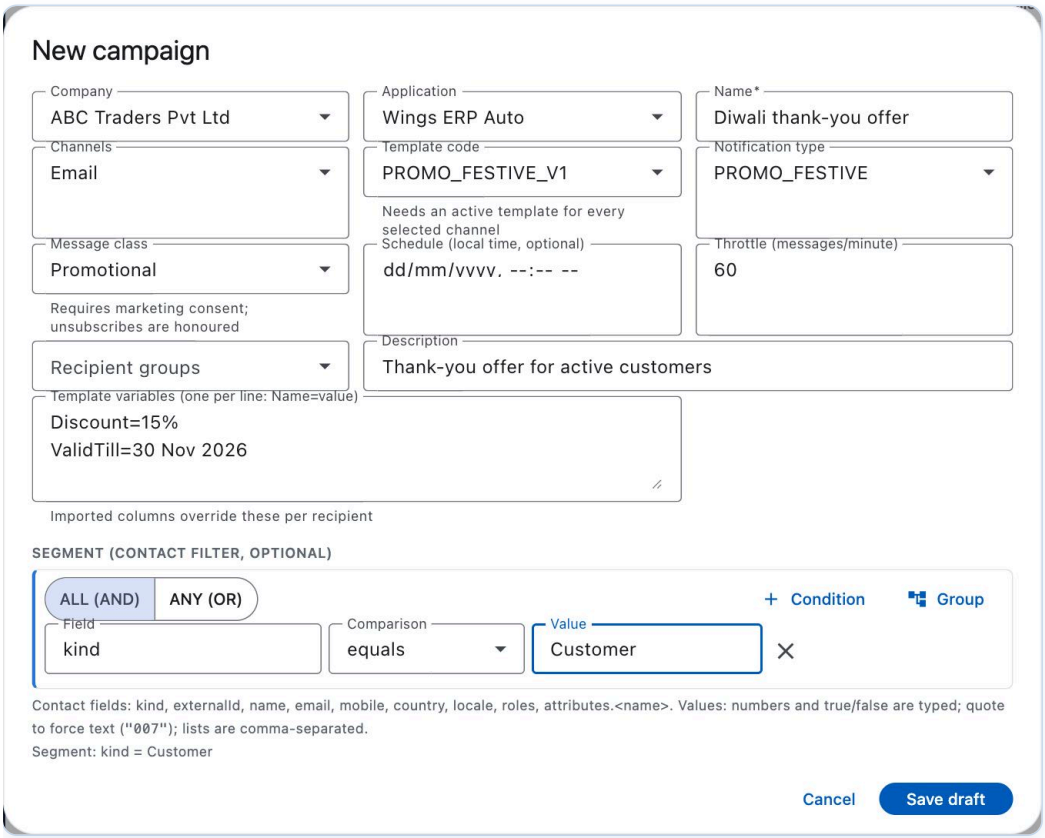


Figure 37: The New campaign dialog, with a segment that selects customers.

11.2 Importing recipients

1. Prepare a `.csv` or `.xlsx` file with the columns `email`, `mobile`, `name` and `externalId`, plus a column for each template variable.
2. Open the draft campaign and select **Import recipients**.
3. Choose the file. The result appears at the top of the campaign, for example `Import: 5 imported, 1 rejected.`, with each bad row listed.
4. Fix the rejected rows and import them in a new file. Imported rows are added to the list.

```
email,mobile,name,externalId,CustomerName
anita@example.com,+919800000101,Anita Rao,C-2001,Anita Rao
bharat@example.com,+919800000102,Bharat Shah,C-2002,Bharat Shah
```

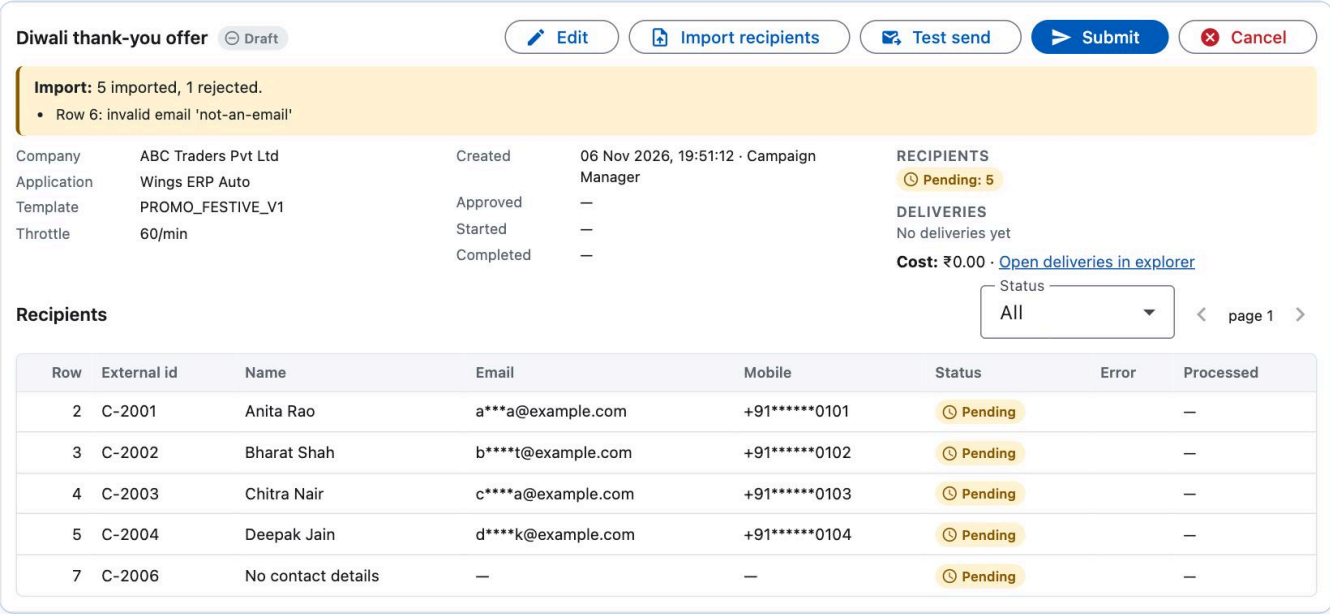


Figure 38: An import result: one bad email was rejected and the other rows are listed as recipients.

> **Note** Each row needs an `email`, `mobile` or `externalId`. A row with only an `externalId` is accepted, and is Skipped when the campaign runs if it has no usable address. Row numbers count the heading as row 1. A second import adds to the list; repeats in the same file are rejected as `duplicate recipient`.

11.3 Testing a campaign

Select **Test send** on a draft and enter one **Recipient email or mobile (E.164)**. The result appears on the campaign as `Test accepted as …`.

The test uses the campaign's **Template variables** and has no import columns to draw on. If the template needs a value such as `CustomerName`, add it there, or the test fails with `The notification could not be rendered.` Consent rules apply to tests too: a promotional test to an address with no marketing consent is accepted but shows Suppressed.

11.4 Submitting and approving

1. The manager selects **Submit**, then **Confirm**. The campaign becomes Pending approval. A campaign with no audience is refused: `Add recipients (import, groups or a segment) before submitting.`
2. An approver selects **Approve** on the campaign, or uses **Approvals** (section 8.1 on page 23).
3. With a schedule, it waits as Scheduled. Without one, it starts shortly after approval and shows Running.

You cannot approve a campaign you created: `Maker–checker: you created this campaign`.

11.5 Following and controlling a campaign

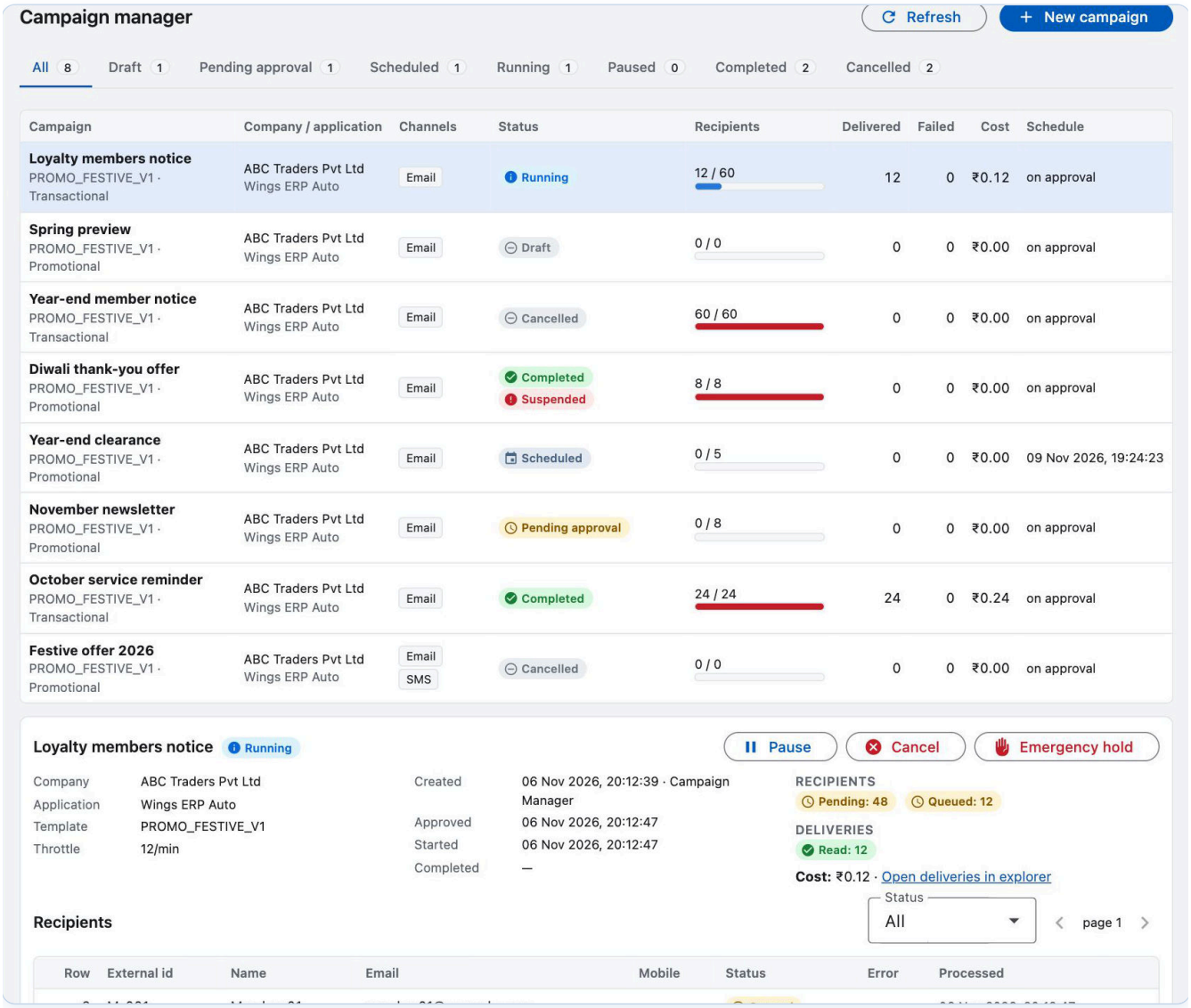


Figure 39: A running campaign, with progress, recipients and controls.

Select a campaign to see progress, recipient and delivery counts, cost, and a table of recipients you can filter by **Status:** Pending Queued Skipped Failed. The **Error** column gives the reason, such as `No marketing consent recorded for this channel.` Select **Open deliveries in explorer** to see each delivery.

Button	What it does
Pause, **Resume**	Stops and restarts sending at once. Appears while the campaign is Running or Paused.
Emergency hold, **Lift hold**	Holds pending messages, not lost, and needs a **Reason**. The campaign shows Suspended beside its status.
Cancel	Pending deliveries are cancelled. This cannot be undone.

Note A small campaign can finish in seconds, before you can pause it. Set a low throttle if you want time to watch one. If a campaign finishes while on hold, it keeps showing Suspended until you select **Lift hold**, because some of its messages may still be held. Cancelling a campaign clears its hold.

CHAPTER 12

Recipients

This screen holds who you can reach and what they agreed to: contacts, preferences, consent, suppressions and groups.

Who can do this: campaign managers and platform, tenant, company and application administrators change recipients. Auditors read with addresses masked. Operators, designers and approvers do not see this screen.

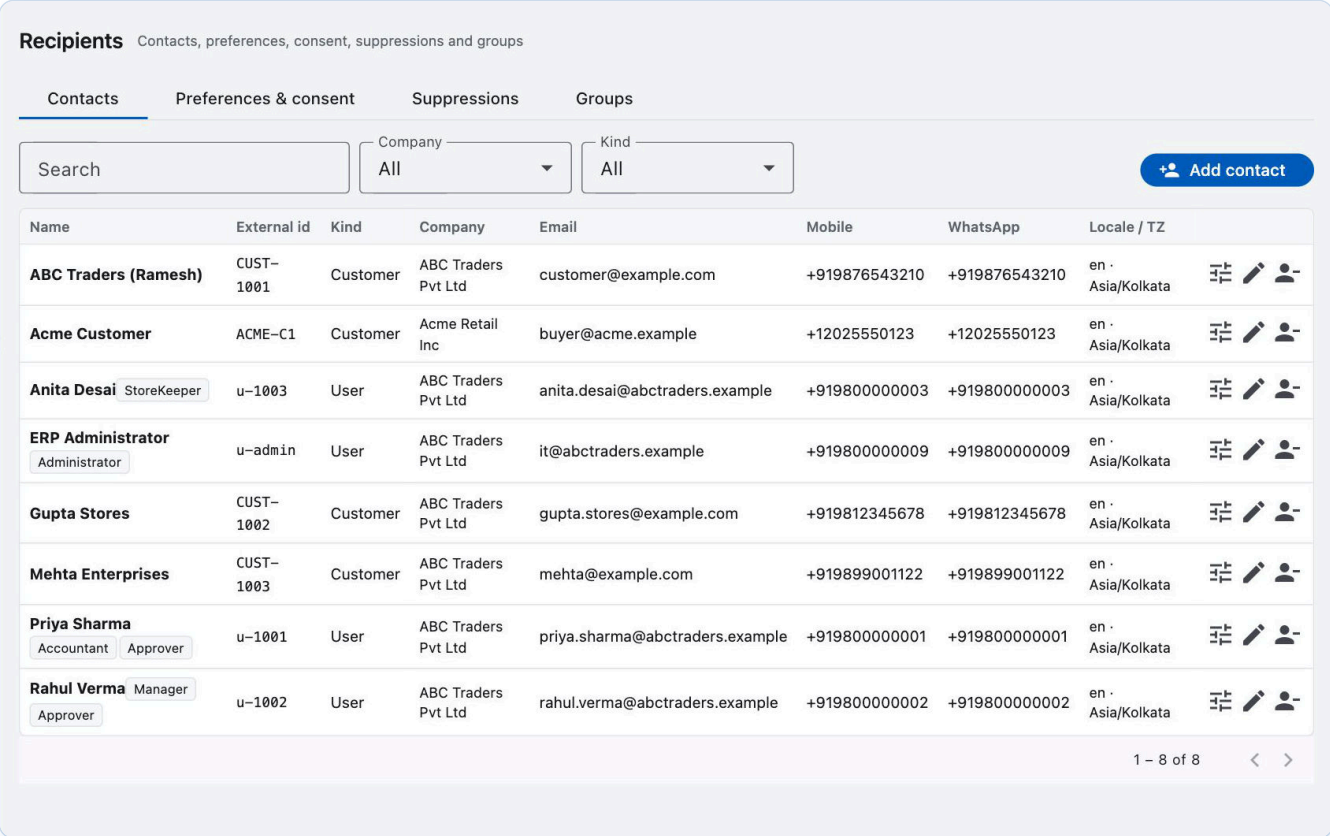


Figure 40: The Contacts tab.

12.1 Adding and editing contacts

Your applications normally keep contacts up to date. Use this screen for corrections and one-off additions.

1. On **Contacts**, select **Add contact**.
2. Choose the **Company** and **Kind**: **User**, **Customer** or **Other**.
3. Enter the **External id** your application uses, and the **Name**.
4. Enter the **Email**, **Mobile (E.164)** and **WhatsApp (E.164)**. E.164 means the full international form, such as `+919845012345`.
5. Enter the **Time zone**, **Locale**, **Country** and **Roles (comma-separated)** if you know them. Then select **Save**.

To change a contact, use the pencil icon. **Company**, **Kind** and **External id** are fixed. Stored addresses appear as hints: leave a field empty to keep its value. Find contacts with **Search** (id, name, email or mobile, then Enter), **Company** and **Kind**.

12.2 Erasing a person's data

1. On the contact's row, select the **Erase personal data** icon.

2. Read the warning, then tick **I understand this is permanent (right to erasure).**
3. Select **Erase**. You see `Personal data erased`.

> **Warning** This cannot be undone. Name, email, phone numbers and attributes are removed, addresses in delivery history are replaced by hashes, message bodies are purged and devices are unregistered. The contact also disappears from the list.

12.3 Preferences and consent

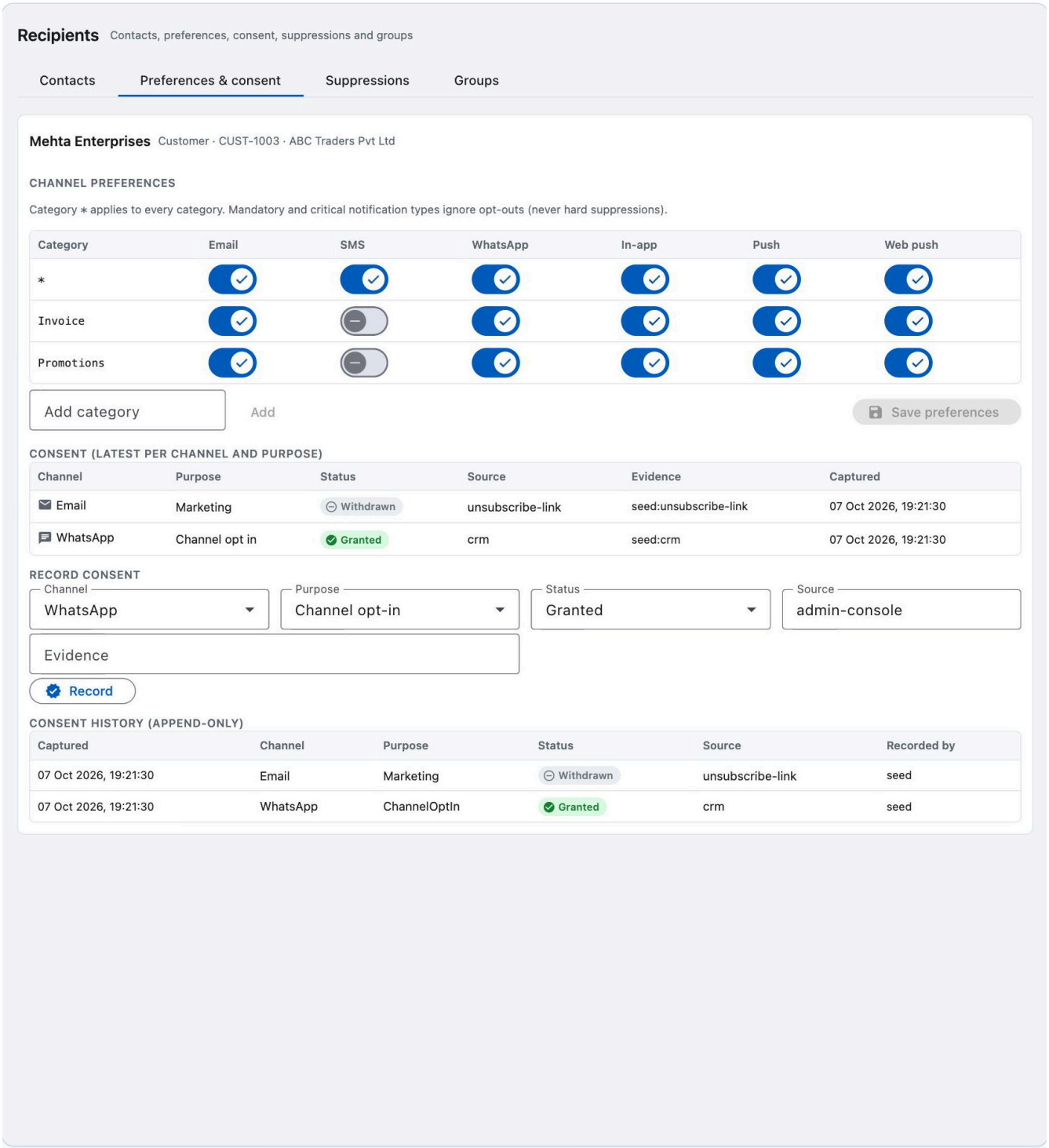


Figure 41: Preferences & consent for one contact.

To change a contact's preferences

1. On **Contacts**, select the row or its **Preferences and consent** icon. The **Preferences & consent** tab opens.
2. Use the switches under **Channel preferences**. Category `*` applies to every category. A specific category beats `*`.
3. To add a category, enter it in **Add category** and select **Add**.
4. Select **Save preferences**.

To record consent or a withdrawal

1. Under **Record consent**, choose the **Channel**.
2. Choose the **Purpose**: **Channel opt-in** (needed for WhatsApp) or **Marketing**.
3. Choose the **Status**: **Granted** or **Withdrawn**.
4. Enter the **Source**, such as `web-form`, and the **Evidence**, such as a form or call reference.
5. Select **Record**.

Consent (latest per channel and purpose) shows what applies now. **Consent history (append-only)** keeps every change, and cannot be edited.

12.4 The consent rules

The hub applies these rules to every message, and checks them again at send time, so an opt-out is honoured even for a message accepted earlier.

Class	Preferences	Marketing consent	WhatsApp opt-in	Unsubscribe	Hard bounce or invalid address	Complaint or opt-out
Transactional	Honoured, unless the type is mandatory	Not needed	Required	Ignored	Blocks	Blocks
Promotional	Honoured	Required, if the tenant requires it	Required	Blocks	Blocks	Blocks
Critical	Ignored	Not needed	Required	Ignored	Blocks	Ignored

A blocked message shows ⊖ Suppressed in the explorer, with one of these reasons:

Reason shown	What to do
`whatsapp_no_opt_in`	Record a WhatsApp **Channel opt-in** if the person agreed.
`marketing_consent_missing`	Record **Marketing** consent for that channel if the person agreed.
`marketing_consent_withdrawn`	The person withdrew. Do not send, unless they grant it again.
`preference_opt_out`	The person switched the channel off for that category.
`suppressed_…`	The address is on the suppression list. See below.

12.5 Suppressions

A suppression blocks an address, and the hub adds most of them itself when a provider reports a hard bounce, complaint or invalid address. Each one blocks according to the table above.

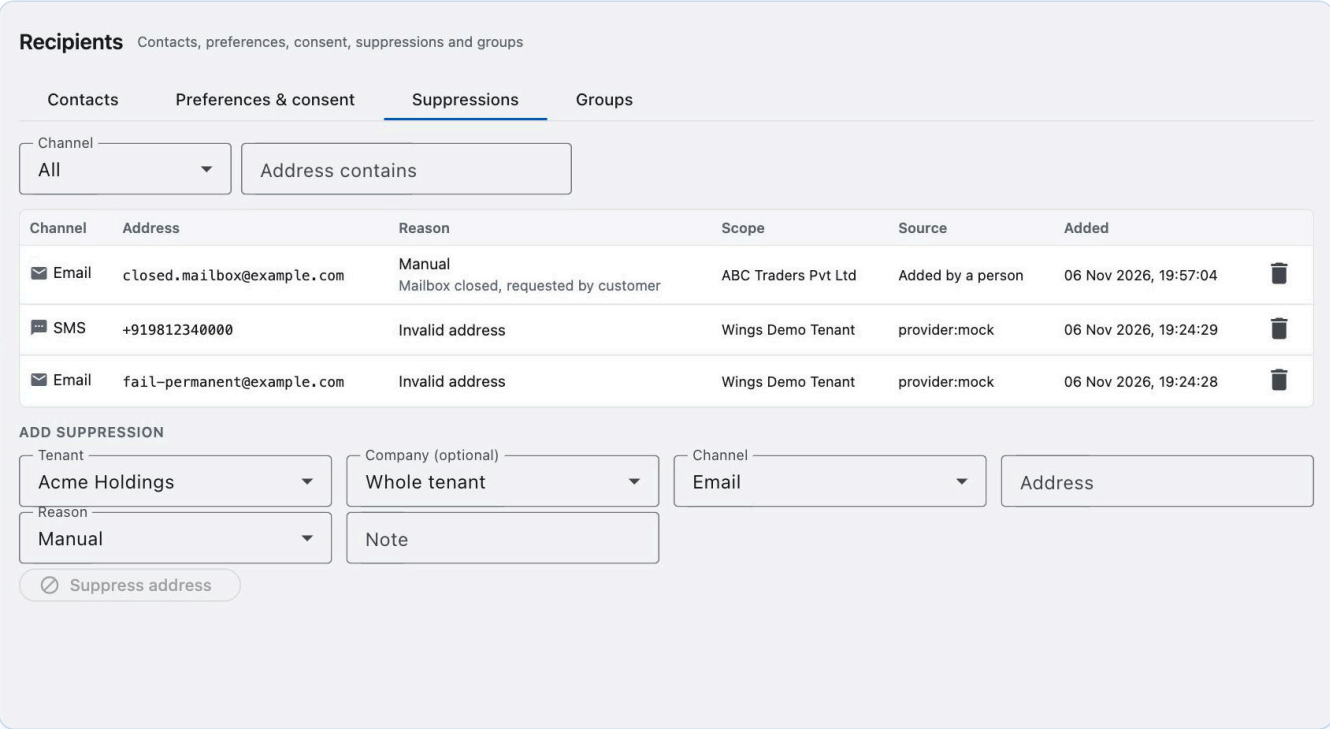


Figure 42: The Suppressions tab.

To suppress an address

1. Open **Suppressions**. Under **Add suppression**, choose the **Tenant** and, if it applies, the **Company (optional)**.
2. Choose the **Channel**, enter the **Address**, choose the **Reason**, and add a **Note**.
3. Select **Suppress address**.

> **Note** The suppression applies to the company you chose, or to the whole tenant if you chose **Whole tenant**. The **Scope** column shows which, and your **Note** appears under the **Reason**. Suppressions the hub adds itself after a bounce or complaint always apply to the whole tenant.

To remove a suppression

1. Find it with **Channel** and **Address contains** (press Enter).
2. Select the delete icon, then **Remove**. Messages to that address are allowed again.

12.6 Groups

Groups name a set of contacts for rules and campaigns. A **Static** group lists its members. A **Dynamic** group works them out from a filter each time.

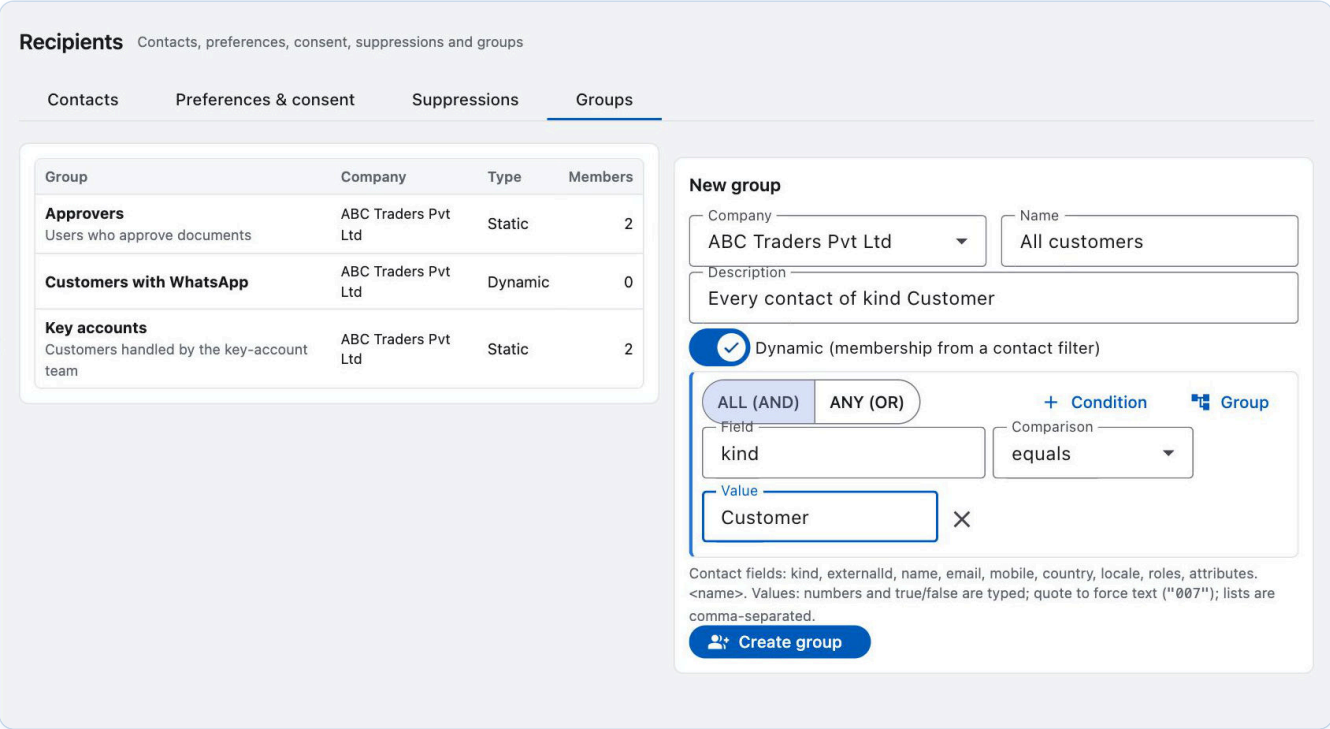


Figure 43: Creating a dynamic group.

1. Open **Groups**. Under **New group**, choose the **Company**, then enter a **Name** and **Description**.
2. For a static group, choose the **Members**. For a dynamic group, switch on **Dynamic (membership from a contact filter)** and build the filter as for rule conditions. Fields include `kind`, `externalId`, `name`, `email`, `mobile`, `country`, `locale`, `roles` and `attributes.<name>`.
3. Select **Create group**. Select a group in the list to see its members.

CHAPTER 13

Schedules and holiday calendars

Schedules send a notification on a repeating pattern or as a series of reminders, in a time zone you choose, and can skip weekends and holidays.

Who can do this: platform, tenant, company and application administrators create schedules. Campaign managers, operators and auditors can look. Calendars need tenant or platform administrators.

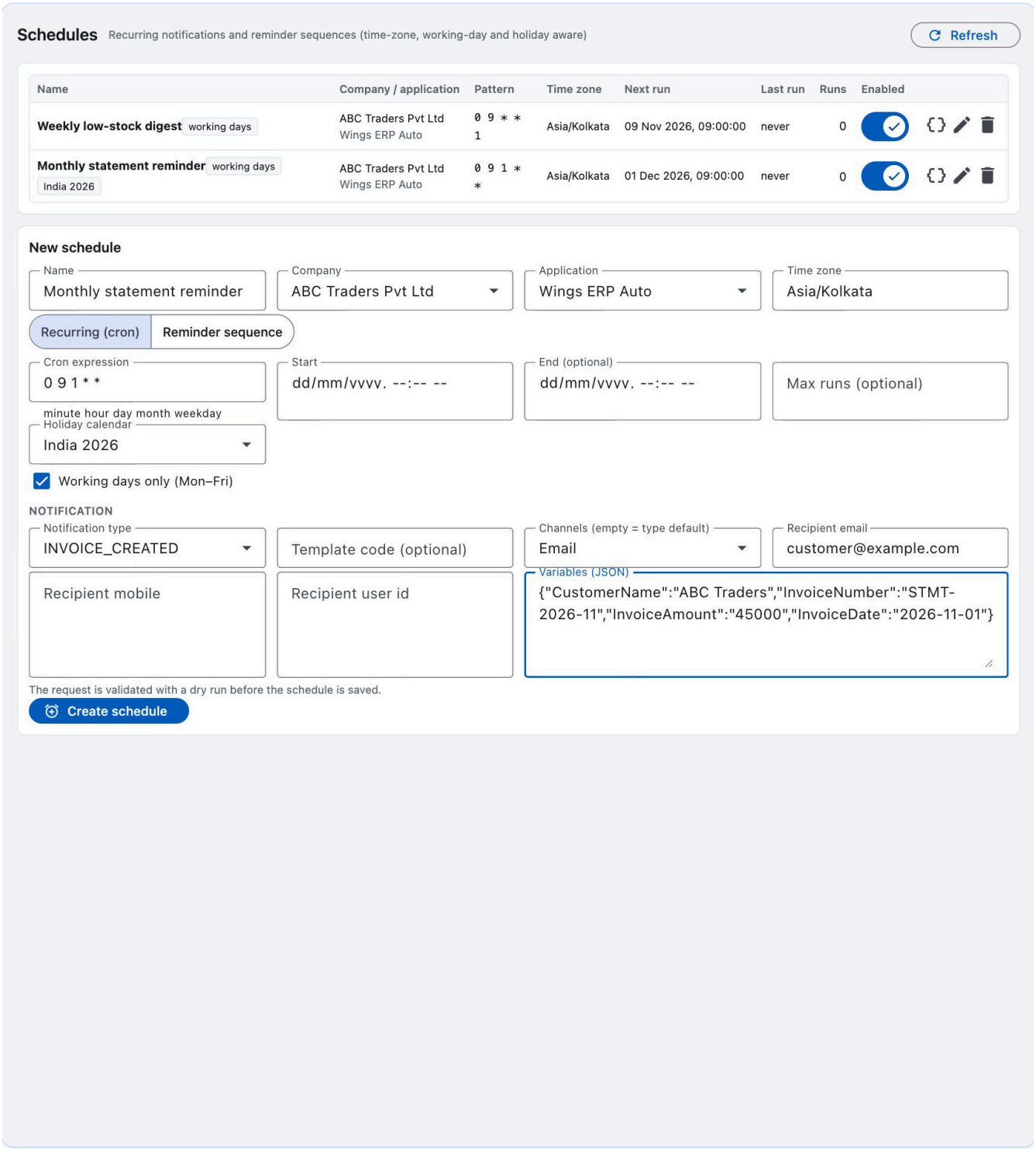


Figure 44: The Schedules screen, with a new recurring schedule filled in.

13.1 Creating a schedule

1. Under **New schedule**, enter a **Name**, and choose the **Company**, **Application** and **Time zone**.
2. Choose the kind:
   - **Recurring (cron)**: enter a **Cron expression** in the order minute, hour, day, month, weekday. `0 9 1 * *` means 09:00 on the first of each month, and `0 9 * * MON-FRI` means 09:00 on weekdays.
   - **Reminder sequence**: enter **Offsets in minutes**. `0, 1440, 4320` sends at the start, a day later and three days later.
3. Optionally set **Start**, **End (optional)** and **Max runs (optional)**.
4. To skip weekends, tick **Working days only (Mon–Fri)**. To skip holidays as well, choose a **Holiday calendar**.
5. Under **Notification**, choose the **Notification type** or enter a **Template code (optional)**, the **Channels (empty = type default)**, and at least one of **Recipient email**, **Recipient mobile** and **Recipient user id**.
6. Enter the **Variables (JSON)**, then select **Create schedule**. The hub checks it with a dry run, then shows the next run.

> **Note** A holiday calendar has an effect only when **Working days only (Mon–Fri)** is ticked.

What you see	What to do
`Invalid cron expression: …`	Check the five fields and their order.
`None of the requested channels could be addressed for the recipient(s).`	Add a recipient address that suits the channels.

13.2 Managing schedules

The table shows the pattern, time zone, next and last run, and the number of runs. Use **Enabled** to pause and restart a schedule. The braces icon (**View request**) shows what will be sent. The delete icon removes it, and future runs are cancelled.

To change a schedule

1. Select the edit icon on its row. The form below fills in, under the heading **Edit** and the schedule's name.
2. Make your changes. The **Company** and **Application** cannot change: to move a schedule, delete it and create it again.
3. Select **Save changes**. The hub checks it again with a dry run and shows the next run. Select **Cancel** to leave it as it was.

13.3 Creating a holiday calendar

1. Open **Holiday calendars**.
2. Under **New calendar**, enter the **Name** and **Country**, and choose the **Tenant**. Only platform administrators can choose **Platform-wide**.
3. Enter **Dates (one per line: yyyy-mm-dd Name)**, such as `2027-01-26 Republic Day`.
4. Select **Create calendar**.

Lines the hub cannot read are listed as `Unrecognised lines: …`, and the calendar is not saved.

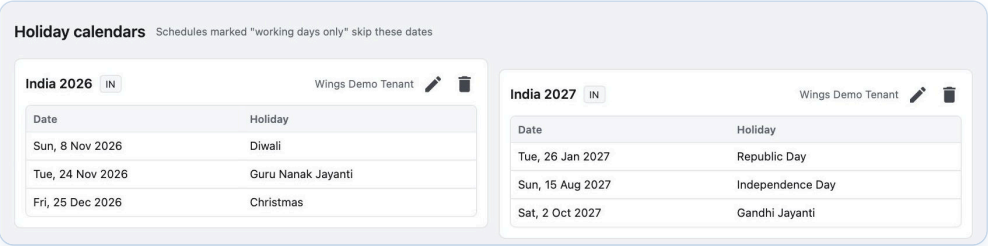


Figure 45: Holiday calendars, each with edit and delete icons.

13.4 Changing or deleting a calendar

1. To change one, select its edit icon, change the **Name**, **Country** or **Dates (one per line: yyyy-mm-dd Name)**, then select **Save changes**. Schedules that use the calendar are rescheduled straight away, so a run on a new holiday moves.
2. To delete one, select its delete icon and confirm with **Delete**. A calendar that a schedule uses cannot be deleted: choose another calendar for that schedule first.

PART 4

Setup and administration

Connect providers, set up customers, applications and people, watch costs and system health, and keep a record of everything.

14	Providers	53
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16	Applications and credentials	62
17	Users and roles	64
18	Webhooks	66
19	Usage and billing	68
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21	System monitoring	73
22	Emergency suspension	75
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CHAPTER 14

Providers

A provider account connects the hub to a service that carries messages. The hub chooses between accounts for each delivery and moves to another if one fails.

Who can do this: platform, tenant and company administrators add, edit, test and suspend. Only a platform administrator can create a **Shared (all tenants)** account. Application administrators, operators and auditors read only.

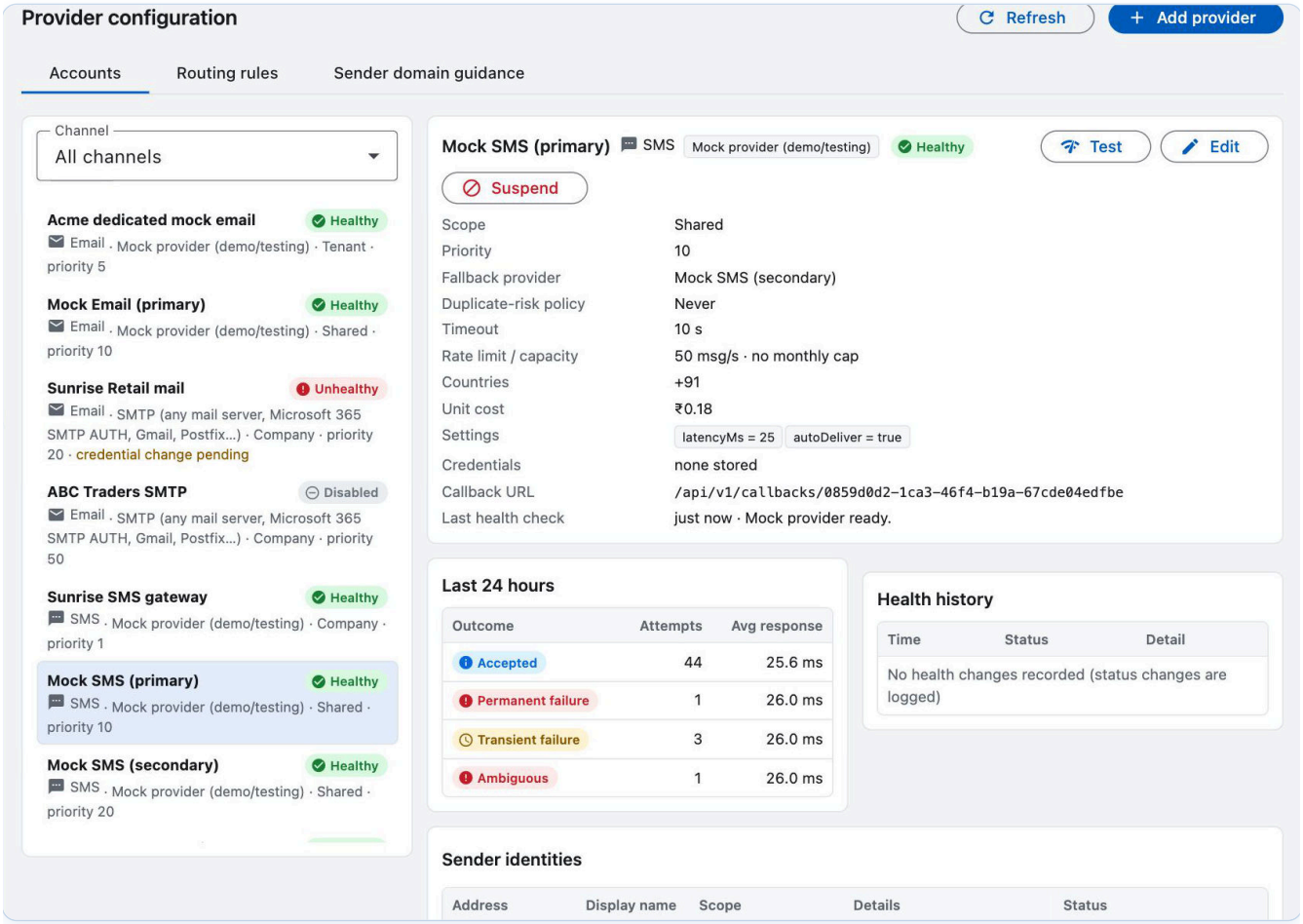


Figure 46: Provider configuration: accounts on the left, the selected account on the right.

14.1 How the hub chooses a provider

1. It looks at company accounts, then tenant accounts, then shared accounts, if the tenant allows them.
2. It applies any routing rules, then drops accounts that are suspended, unhealthy, wrong for the destination country or over their monthly capacity.
3. It picks by **Priority** (lower is preferred), then by cost.

Sandbox applications use only Mock accounts.

14.2 Adding a provider account

1. Select **Add provider**.
2. Enter a **Name**, and choose the **Provider type** and **Channel**. The type cannot be changed later.
3. Choose the **Scope**: **Tenant** or **Company**, and which one. Platform administrators can also choose **Shared (all tenants)**.

4. Fill in **Settings** and **Credentials** from the provider's own console. Secrets are encrypted and never shown again.
5. Under **Routing & limits**, set the **Priority (lower is preferred)**, **Fallback provider**, **Duplicate-risk policy** and any limits.
6. Tick **Enabled**, then select **Save**.

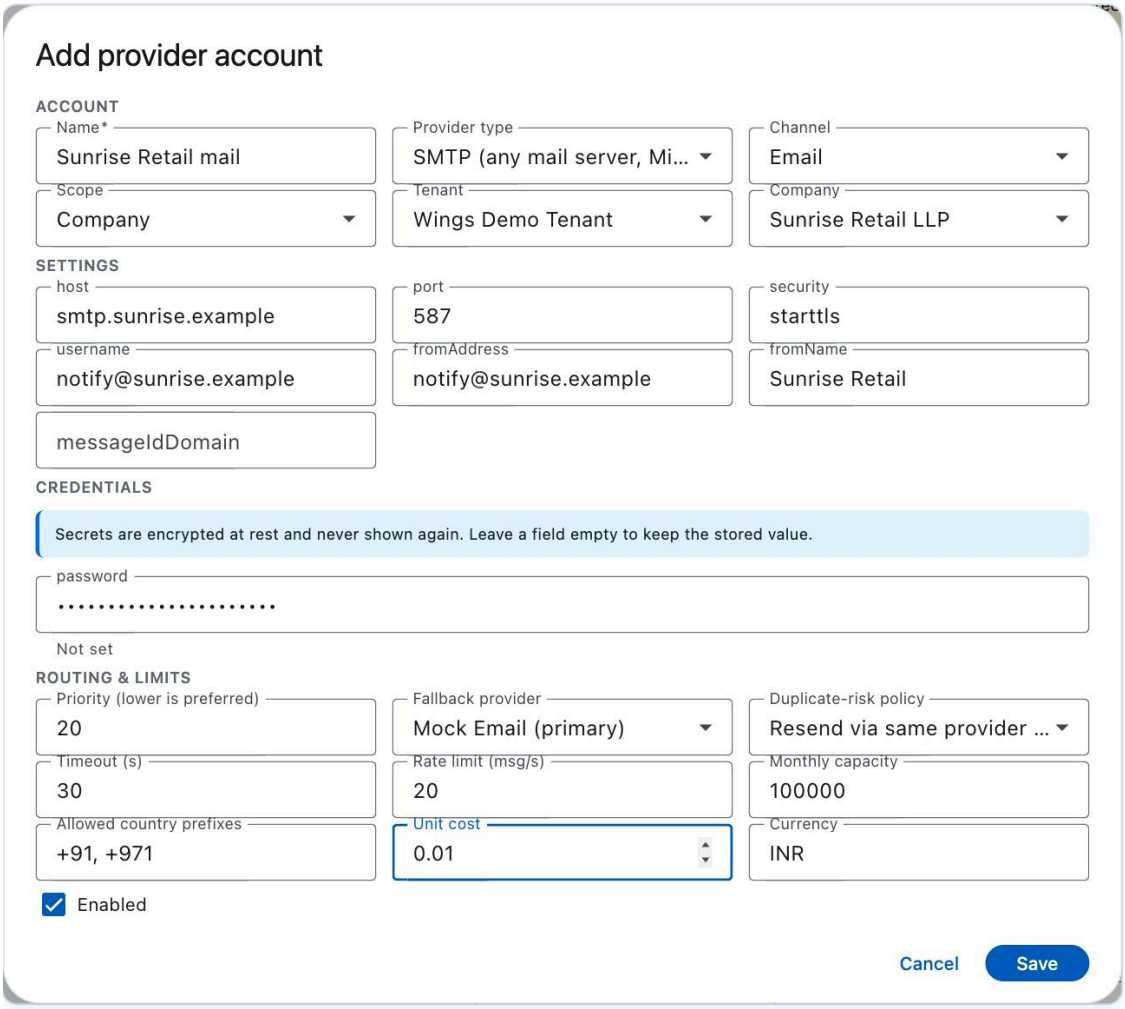


Figure 47: Adding an SMTP account. The password is hidden as you type.

Provider type	Channel	What you need
SMTP (any mail server, Microsoft 365 SMTP AUTH, Gmail, Postfix...)	Email	Host, port, security, username, from address; password. No delivery receipts.
Amazon SES (v2 API)	Email	Region, configuration set, from address; access keys or an instance role.
Microsoft 365 (Graph sendMail, app-only)	Email	Tenant id, client id, sender mailbox; client secret.
Twilio Programmable Messaging	SMS	Account SID, messaging service or from number; auth token.
MSG91 (India, TRAI DLT)	SMS	Sender id, route, DLT entity id; auth key.
Generic HTTP SMS gateway (configurable request)	SMS	URL, method, body template and response paths; API key.
WhatsApp Business Platform — Meta Cloud API	WhatsApp	Phone number id, business account id, verify token; access token, app secret.
WhatsApp via 360dialog (Cloud API compatible BSP)	WhatsApp	Base URL; API key.

Provider type	Channel	What you need
Firebase Cloud Messaging (HTTP v1)	Push, web push	Project id; service account JSON.
Apple Push Notification service (token-based, HTTP/2)	Push	Team id, key id, bundle id, environment; private key.
Standards Web Push (VAPID)	Web push	VAPID public key, subject; VAPID private key.

The built-in **In-app inbox** and **Outbound webhooks** accounts need no setup. Each provider's details are in `docs/providers.md`.

Duplicate-risk policy	After an unknown outcome
Never resend after unknown outcome	Waits for an operator (section 6.5 on page 19). Safest for most messages.
Resend via same provider only	Resends through the same account, which can spot duplicates.
Duplicates acceptable	Retries and falls back freely. Use this where a repeat does no harm.

14.3 Testing an account and setting its callback

1. Select the account, then **Test**.
2. Optionally enter an address in **Send a test message to (optional)**. Leave it empty to check connectivity only.
3. Select **Run test**. The result appears as **Connectivity test:**, green for healthy and red otherwise.
4. Copy the **Callback URL** from the account details into the provider's console, so that delivery receipts, bounces and replies reach the hub.

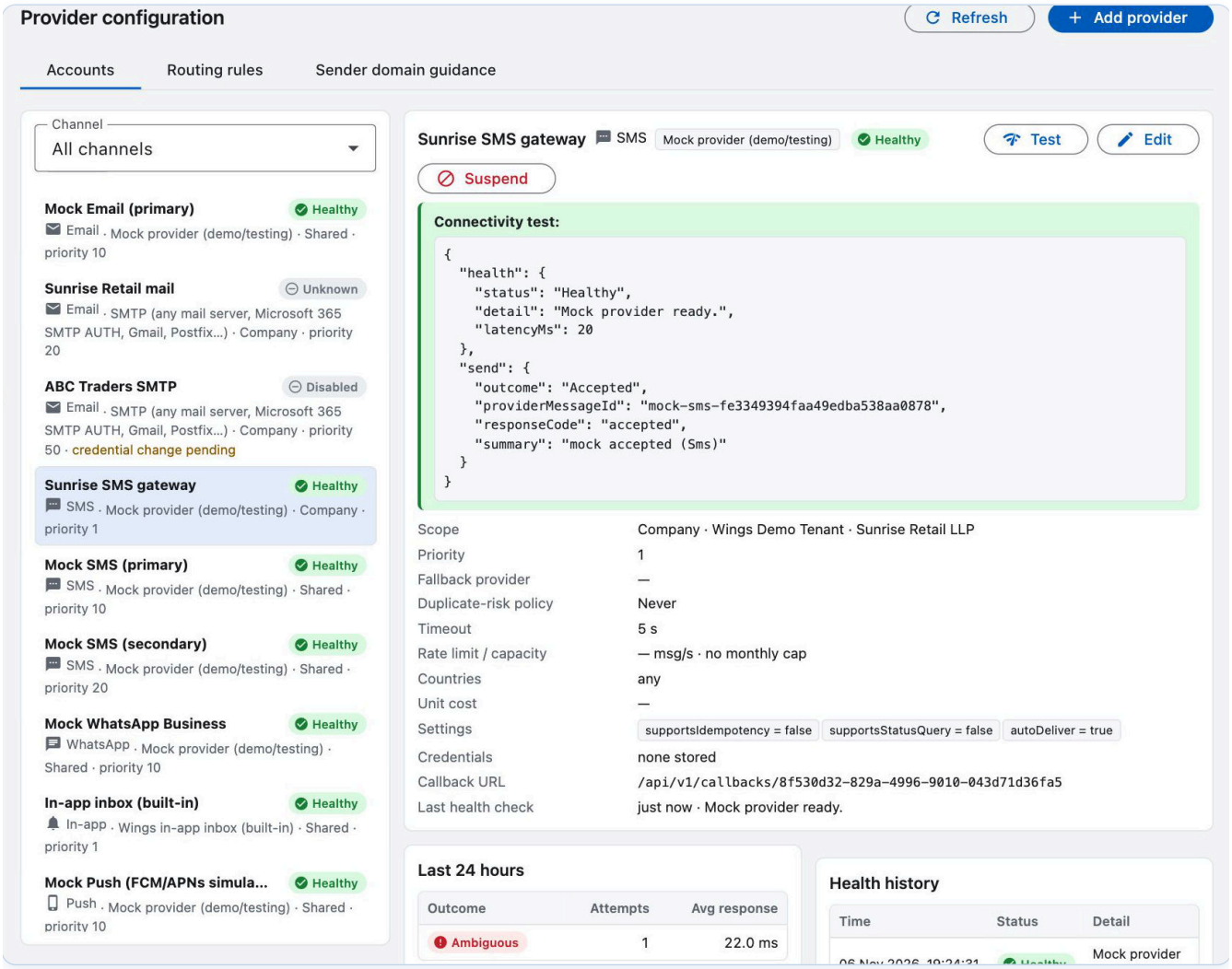


Figure 48: A connectivity test with a test message: the provider reports Healthy and Accepted.

14.4 Adding a sender identity

1. Select the account and go to **Sender identities**.
2. Under **Add sender**, enter the address. The label fits the channel: **From address**, **Sender id / number** or **Address**.
3. Enter a **Display name**, and limit it with **Company (optional)** if needed.
4. Add the channel's details: **Reply-to** for email, **DLT entity id** for SMS in India, or **Phone number id** and **Business account id** for WhatsApp.
5. Tick **Default sender** if it applies, then select **Add sender**.

Use **Suspend** and **Resume** on a sender's row to stop and restart it at once.

14.5 Pinning traffic with routing rules

Routing rules send a channel to a chosen account, optionally for one country or message class. Without a matching rule, enabled healthy providers are tried by priority.

1. Open the **Routing rules** tab.
2. Choose the **Channel** and **Provider**.
3. Choose the **Tenant** and **Company**. Platform administrators can choose **Platform-wide**.
4. Optionally enter a **Country prefix**, such as `+91`, and choose a **Message class**. Set the **Order**.

5. Select **Add rule**. Use the delete icon to remove a rule.

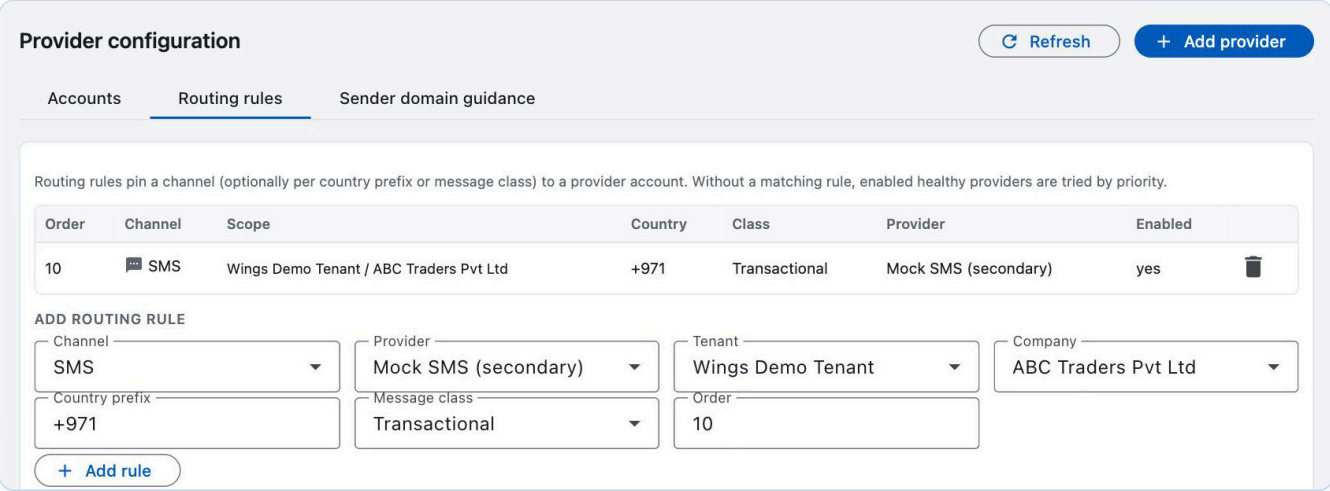


Figure 49: A routing rule for +971 numbers.

14.6 Setting a fallback and suspending an account

Select **Edit** on an account and set **Fallback provider**: if this account has an outage, the hub moves to that account for the same channel.

To stop an account at once, select **Suspend**, enter a **Reason** and confirm. Traffic moves to other providers. Select **Resume** to bring it back. Both are audited.

14.7 Changing credentials

Credentials of an existing account are protected by maker-checker.

1. Select the account, then **Edit**. Each stored secret shows `•••••••• (stored)`.
2. Enter a new value only for what is changing. Leave the rest empty.
3. Select **Save**. You see `Settings saved. Credential changes take effect after approval by another administrator.`
4. The account shows **credential change pending**, and the stored credentials stay in use.
5. Another administrator opens **Approvals** and approves it. The new credentials then replace the old ones.

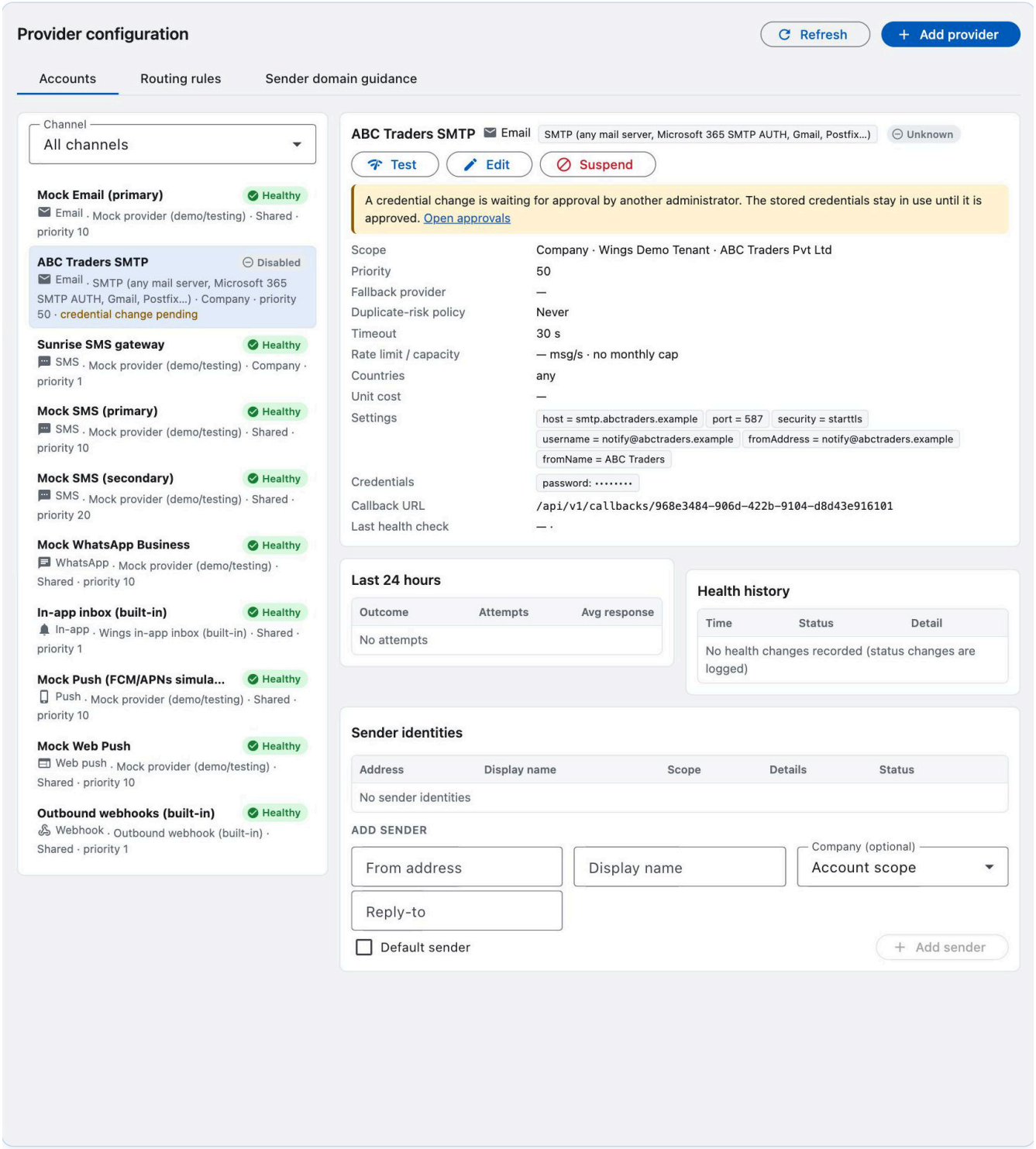


Figure 50: An account waiting for a credential change to be approved, with its settings and callback URL.

14.8 Sender domain guidance

For email to reach the inbox, your sending domain needs SPF, DKIM and DMARC records.

1. Open the **Sender domain guidance** tab.
2. Enter the **Sender domain**, such as `mail.example.com`, and select **Show DNS records**.
3. Give the records to whoever manages your DNS. Replace the placeholders with your provider's values.

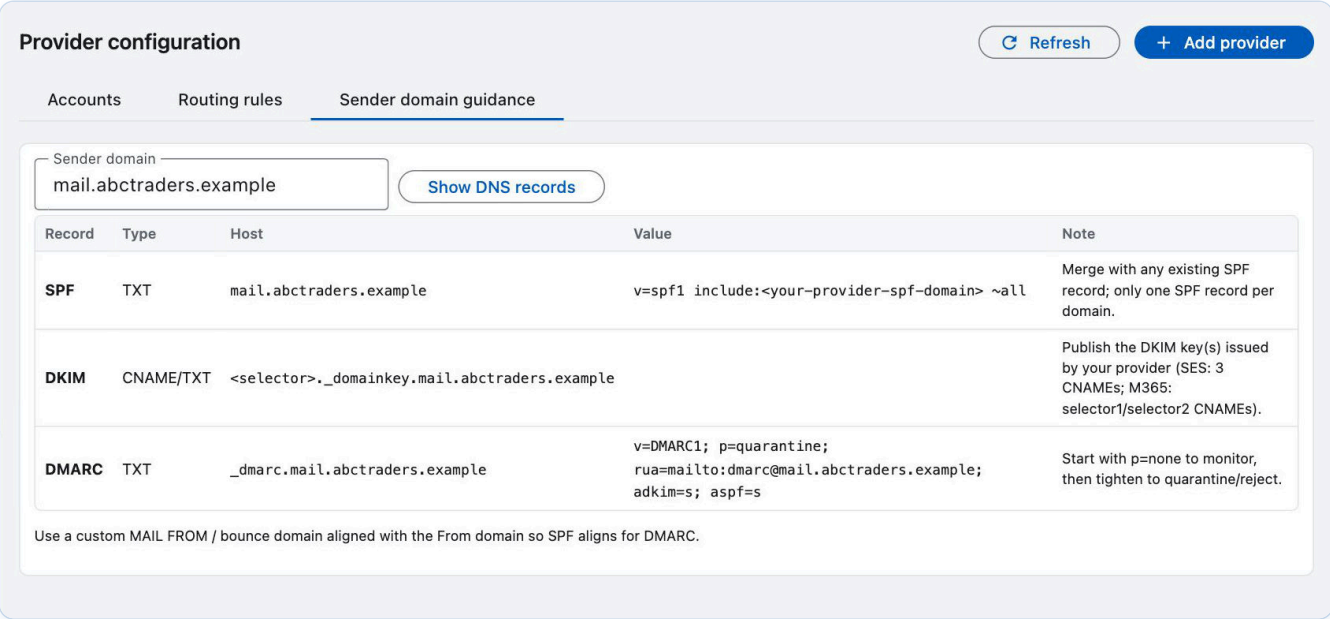


Figure 51: The DNS records for a sender domain.

Tip Use a separate subdomain for bulk and promotional mail, and start DMARC at `p=none` while you watch the reports.

CHAPTER 15

Tenants and companies

A tenant is a customer on the hub, and companies are the businesses inside it.

Who can do this: only platform administrators create tenants. Tenant administrators add and edit companies and edit their tenant. Other roles see this screen read-only.

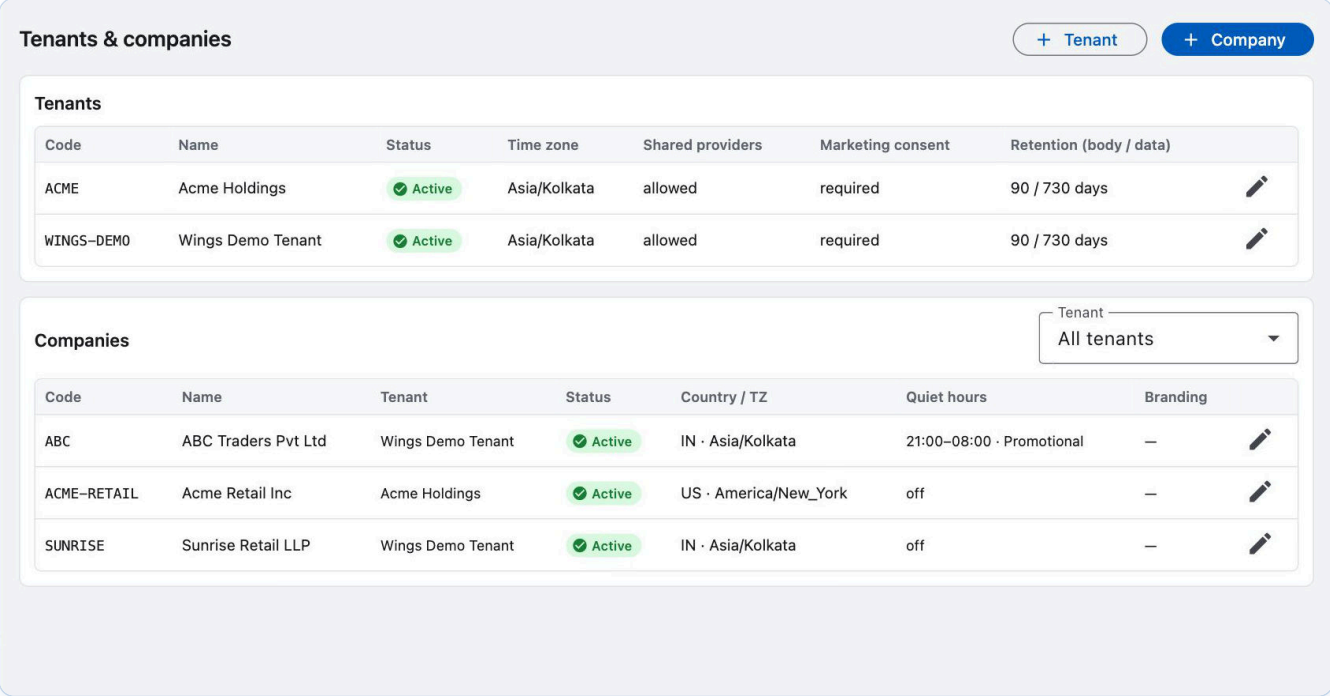


Figure 52: Tenants & companies.

15.1 Creating or editing a tenant

1. Select **+ Tenant**, or the pencil icon on a tenant's row.
2. Enter the **Code**, which cannot be changed later, the **Name** and the **Default time zone**.
3. Set **Message body retention (days)**. After this many days message bodies are deleted and the record of delivery stays.
4. Set **Data retention (days)**, at least 30. After this many days delivery records are deleted.
5. Choose **Allow shared (platform) providers** and **Require marketing consent for promotional messages**.
6. Fill in **Branding** if you need it, then select **Save**.

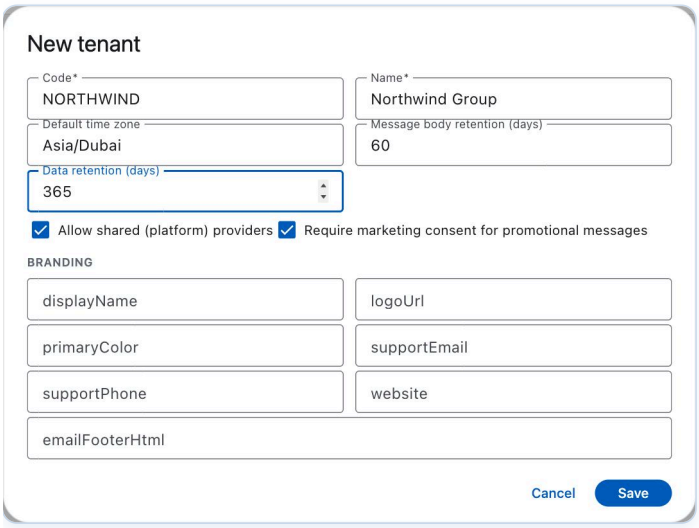


Figure 53: The New tenant dialog.

15.2 Creating or editing a company

1. Select **+ Company**, or the pencil icon on a company's row.
2. Choose the **Tenant**, and enter the **Code**, **Name**, **Country** and **Time zone (IANA)**, such as `Asia/Kolkata`.
3. Under **Quiet hours (recipient local time)**, tick **Enabled**, then set **Start**, **End**, **Applies to** and **Channels (empty = all except in-app)**. For example, 21:00 to 08:00 for promotional messages.
4. Fill in **Branding** to override the tenant's, then select **Save**.

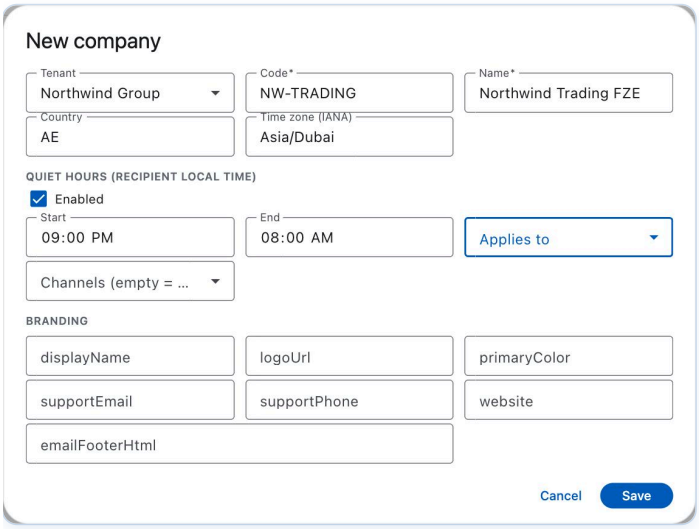


Figure 54: A company with quiet hours for promotional messages.

Messages inside quiet hours wait until the window ends. A rule can set its own quiet hours (section 10.4 on page 37).

Branding field	Use
`displayName`	The name shown to recipients.
`logoUrl`, `primaryColor`	Logo and accent colour.
`supportEmail`, `supportPhone`, `website`	Contact details.
`emailFooterHtml`	Footer added to emails.

CHAPTER 16

Applications and credentials

An application is a system that sends through the hub. It signs in with a client credential, and can send only for the companies you link it to.

Who can do this: platform and tenant administrators add and link applications. They and application administrators manage credentials.

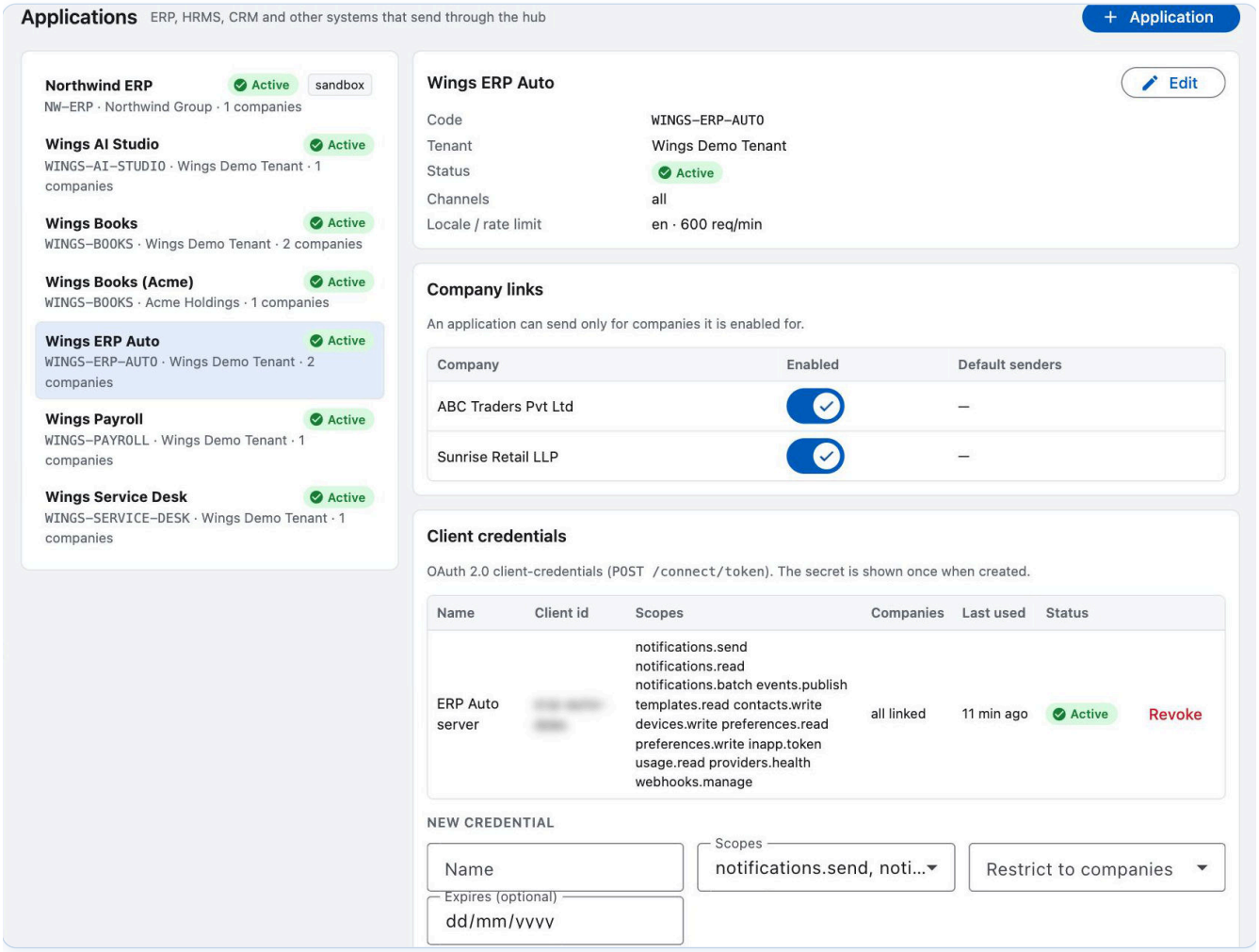


Figure 55: An application, its company links and its client credentials. The client id is blurred here.

16.1 Adding an application

1. Select **+ Application**.
2. Choose the **Tenant**, and enter the **Code** and **Name**.
3. Choose **Allowed channels (empty = all)**, the **Default locale** and the **Rate limit (requests/min per credential)**.
4. Tick **Sandbox (messages are not sent to real providers)** while it is being tested.
5. Select **Save**, then switch on each company under **Company links**. An application can send only for companies it is enabled for.

16.2 Creating and revoking a credential

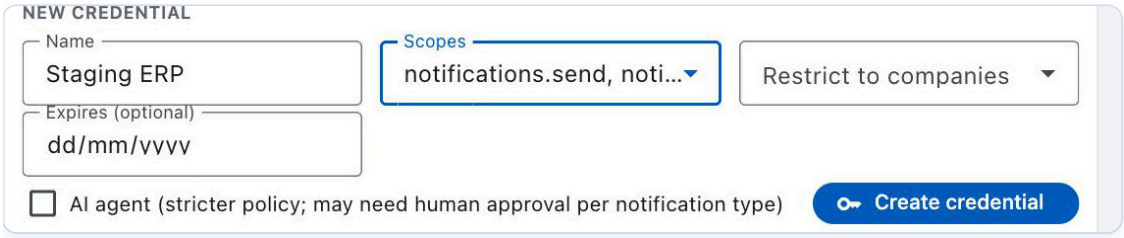


Figure 56: The New credential form.

1. Select the application. Under **New credential**, enter a **Name**, such as `Production ERP`.
2. Choose the **Scopes**. Remove any it does not need.
3. Optionally choose **Restrict to companies** and **Expires (optional)**.
4. Tick **AI agent (stricter policy; may need human approval per notification type)** only for an AI agent.
5. Select **Create credential**.
6. Copy the **Client id** and **Client secret**, store them safely, and select **I have stored it**.

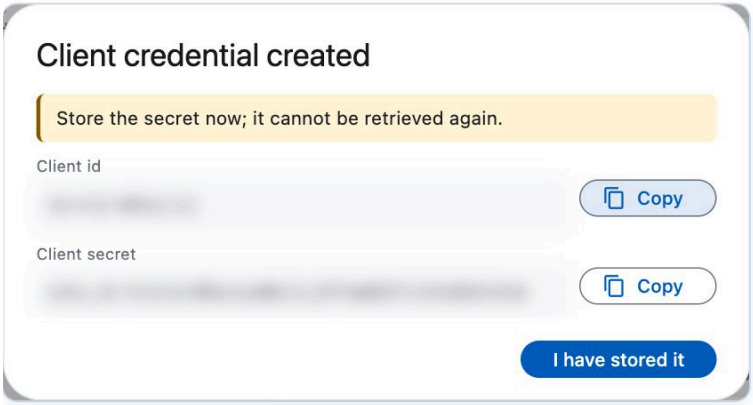


Figure 57: The secret is shown only once. Values are blurred here.

> ⚠ **Warning** The secret cannot be shown again. If it is lost or exposed, select **Revoke** and create a new credential. Revoking stops issued tokens within seconds and cannot be undone.

Scope	Allows
`notifications.send`, `notifications.batch`	Send, schedule, cancel and send in batches
`notifications.read`	Status, history and the delivery list
`events.publish`	Publish business events for rules
`templates.read`, `templates.write`	Read or manage the application's templates
`contacts.write`, `devices.write`, `preferences.read`, `preferences.write`	Recipients, devices, preferences and consent
`inapp.token`	Sign end users into the in-app inbox
`usage.read`, `providers.health`	Usage and reports; provider health
`webhooks.manage`	Delivery and reply callbacks
`mcp`	The AI agent connection

CHAPTER 17

Users and roles

People get into the console through a user account and one or more role grants. Each grant has a scope: grant the narrowest that works.

Who can do this: platform and tenant administrators. A tenant administrator cannot grant **Platform administrator**, or grant anything outside the tenant.

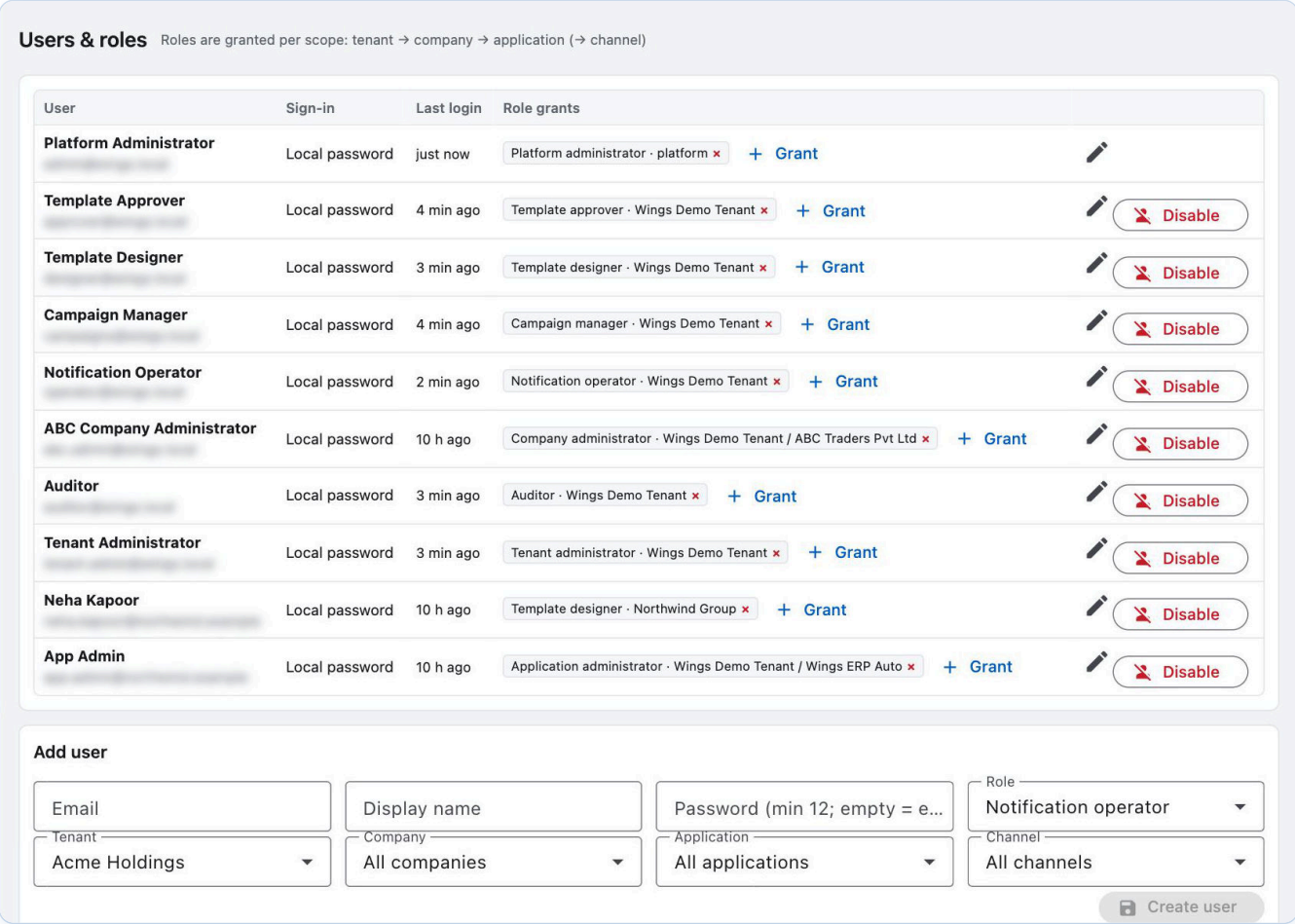


Figure 58: Users & roles: each person's sign-in type, last login and role grants, with edit and disable buttons. Email addresses are blurred here.

17.1 Adding a user

1. Under **Add user**, enter the **Email** and **Display name**.
2. Enter a **Password (min 12; empty = external OIDC)**. Leave it empty for someone who signs in through your organisation's identity provider.
3. Choose the **Role**.
4. Set the scope: **Tenant**, **Company**, **Application** and **Channel**. Leave **All companies**, **All applications** and **All channels** for the widest.
5. Select **Create user**. Give the person their password by a safe route.

Each email address can belong to one user only. If it is already in use you see `A user with this email already exists. Use Grant on their row to add a role.`, and the existing account is not changed.

17.2 Granting or removing a role

1. To grant, select **Grant** on the person's row. The form changes to **Grant a role to ...**.
2. Choose the role and scope, then select **Grant role**. Select **Add a different user** to return to the blank form.
3. To remove, select the **×** on the role and confirm with **Remove**.

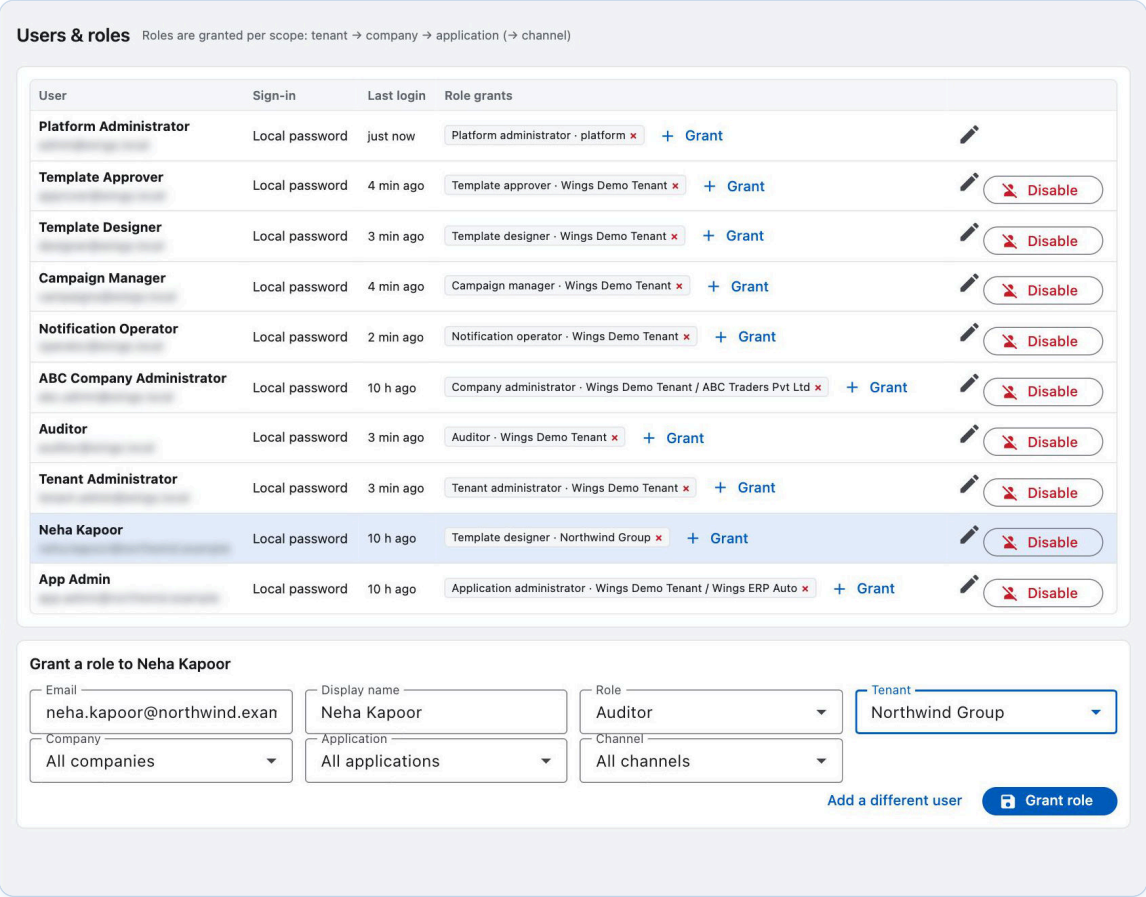


Figure 59: Granting a second role to a user.

17.3 Changing an account

1. To rename someone or set a new password, select the edit icon on their row. Change the **Display name** or enter a **New password** of at least 12 characters, then select **Save**. Leave the password empty to keep the current one.
2. To stop someone signing in, select **Disable** and confirm. They are signed out at once. Their roles are kept, so **Enable** brings the account back as it was.

> **Note** These buttons appear only where you manage every role the person holds, so a tenant administrator cannot change a platform administrator or someone who also works in another tenant. You cannot disable your own account. Someone who signs in through your identity provider has no password here.

CHAPTER 18

Webhooks

Webhooks tell your applications what happened, as it happens: a status changed, a notification finished, a customer replied. Each call is signed so the application can trust it.

Who can do this: platform, tenant, company and application administrators.

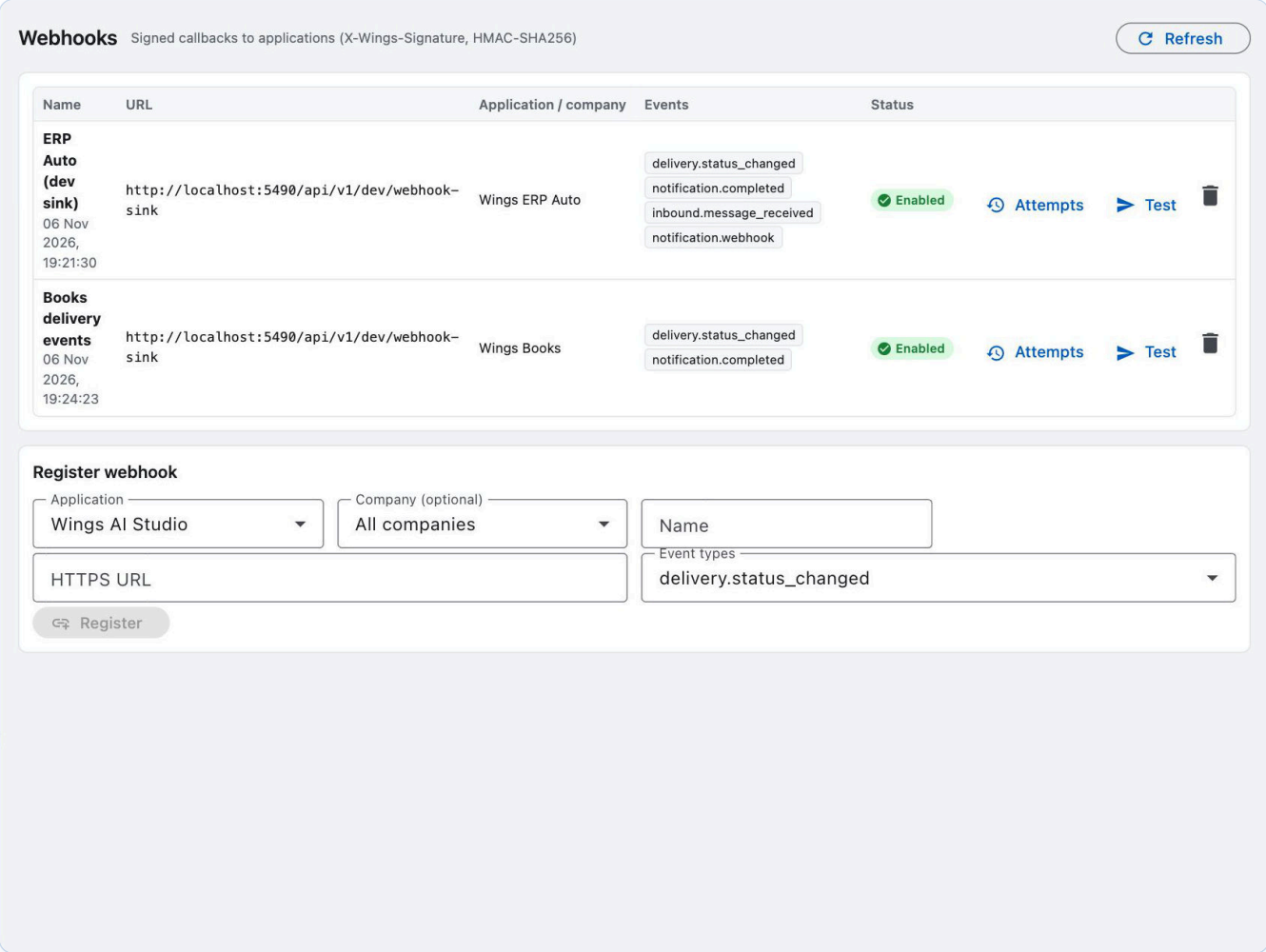


Figure 60: Webhooks: registered endpoints and the form to add one.

18.1 Registering a webhook

1. Under **Register webhook**, choose the **Application** and, optionally, **Company (optional)**.
2. Enter a **Name** and the **HTTPS URL** of the application's endpoint.
3. Choose the **Event types**, or **All events (*)**.
4. Select **Register**.
5. Copy the **Signing secret** and give it to the application's developers. It is shown only once.

Event type	Sent when
`delivery.status_changed`	A delivery's status changes
`notification.completed`	All of a notification's deliveries have finished
`inbound.message_received`	A customer replies
`push.interaction`	Someone interacts with a push message

Event type	Sent when
`inapp.action`	Someone acts on an in-app message
`notification.webhook`	A message is sent on the Webhook channel

Note URLs must be https and cannot point to a private network, except on a development system set up to allow it.

18.2 Testing and watching a webhook

- **Test** sends a test event. You see `Test event … queued`, and the attempts list opens.
- **Attempts** shows each call: time, event, Delivered or Failed, HTTP code, duration and error.
- The delete icon stops events to that address.

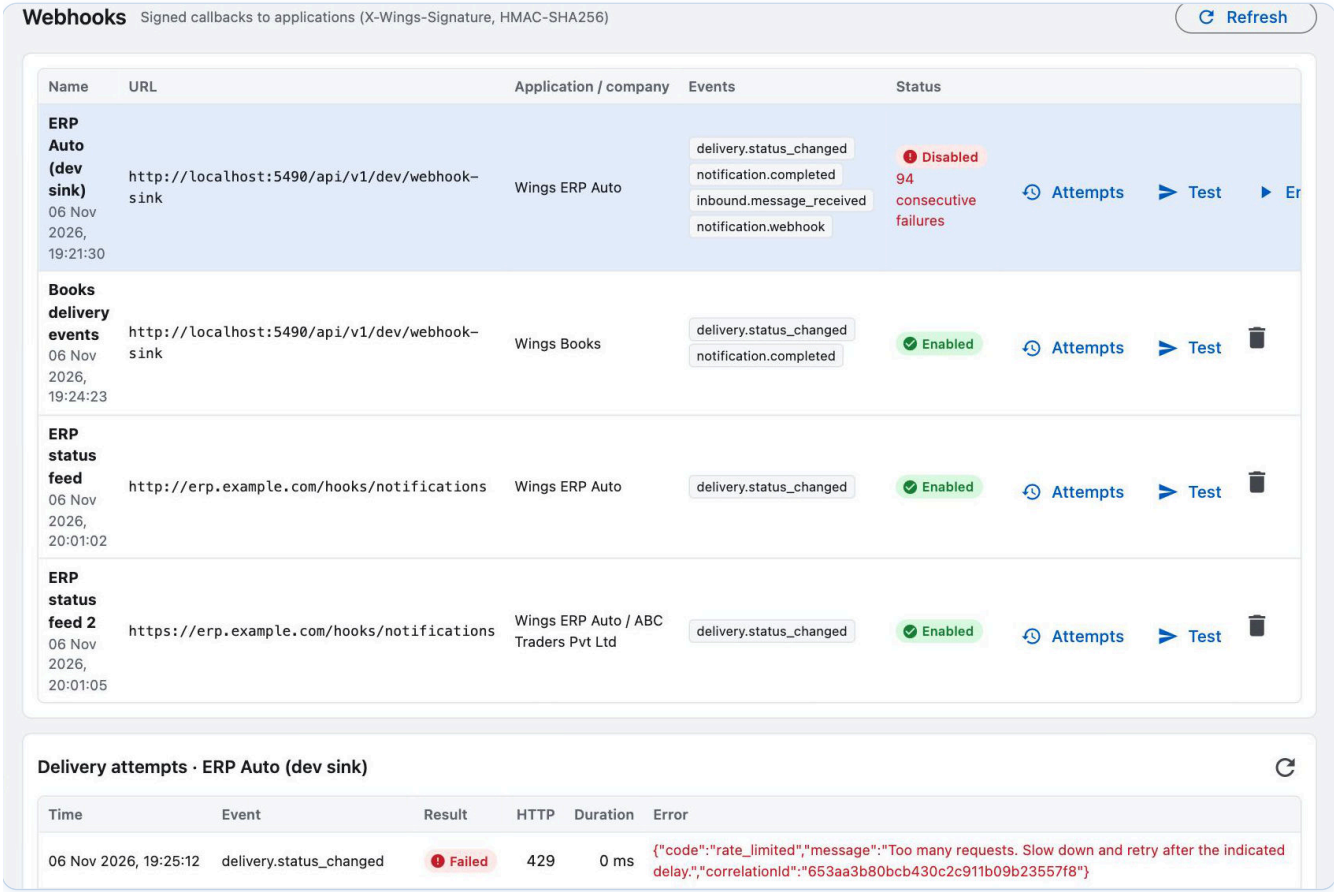


Figure 61: Attempts for a disabled webhook. HTTP 429 means the receiver refused the calls.

Failed calls are retried, up to 8 times with growing gaps. After 100 failures in a row the hub disables the webhook and raises an alert. Fix the receiver, then select **Enable**.

CHAPTER 19

Usage and billing

See what was sent and what it cost, and set limits.

Who can do this: everyone with this screen can read it: administrators, campaign managers and auditors. Tenant and platform administrators set quotas. Only platform administrators add credits or change rates.

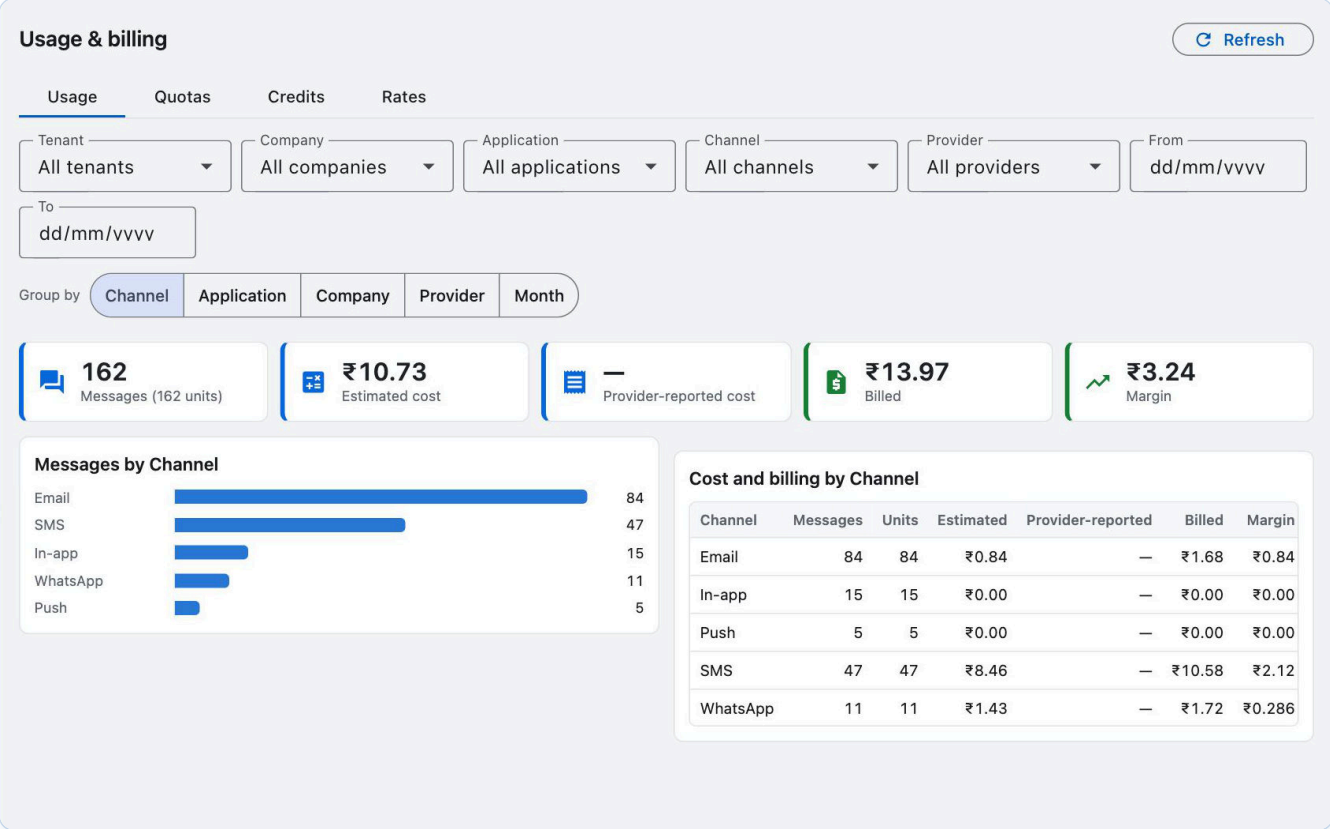


Figure 62: The Usage tab, grouped by channel.

19.1 Reading usage

Use the filters, then **Group by**: **Channel**, **Application**, **Company**, **Provider** or **Month**.

Figure	What it means
Messages (units)	Messages sent. A long SMS counts as several units.
Estimated cost	What the hub expects the providers to charge, from the cost rates.
Provider-reported cost	What providers said they charged, where they report it. Otherwise —.
Billed	What the customer is charged, from the billing rates.
Margin	Billed minus cost. Negative figures are shown in red.

19.2 Setting a quota

1. Open the **Quotas** tab.
2. Under **Set quota**, choose the **Tenant**, and optionally the **Company**, **Application** and **Channel**.
3. Enter the **Monthly limit** and **Alert threshold %**.
4. Select **Save quota**. Saving the same scope again updates it. Use the delete icon to remove one.

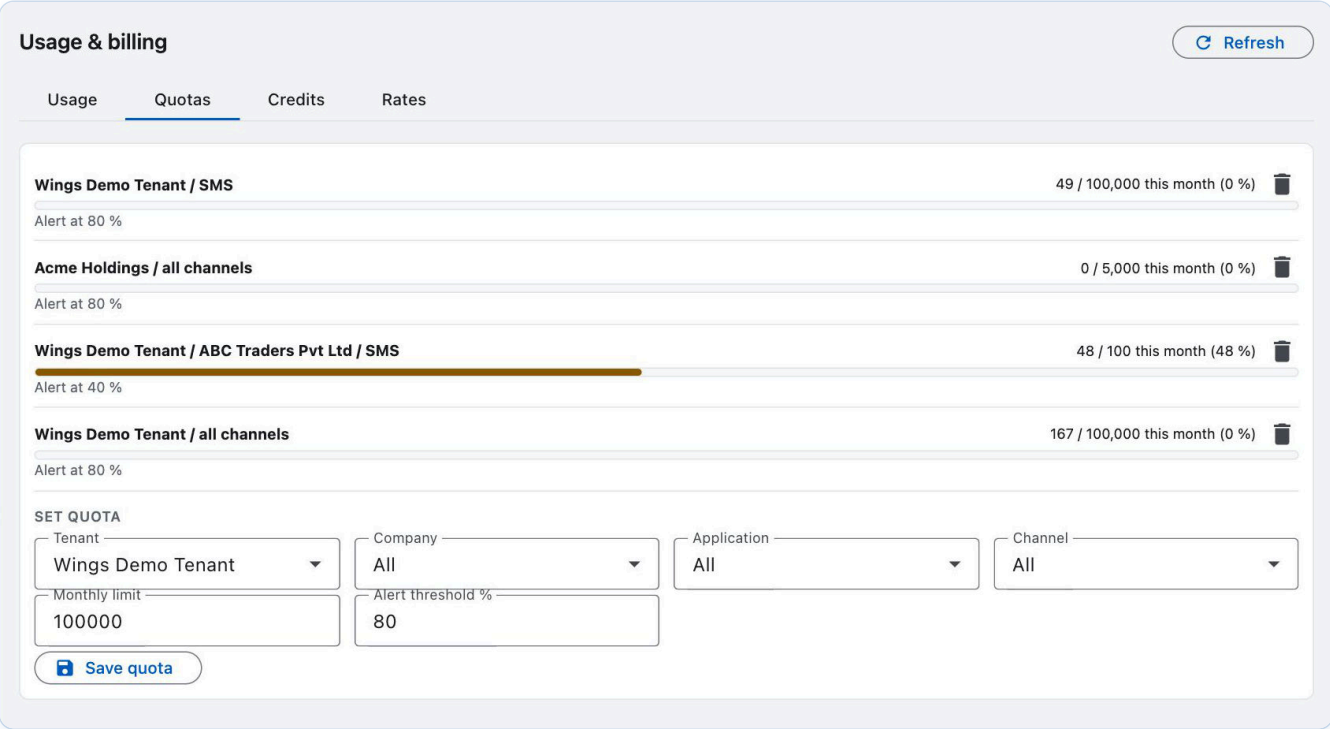


Figure 63: Quotas, with the use so far this month.

> **Warning** A quota is a hard stop. Once it is reached, the hub refuses new sends with `quota_exceeded`, for example `Monthly email quota for this company is exhausted (9/1).` The application sees this error, and nothing appears in the explorer.

19.3 Prepaid credits

Without a credit account, sending is not limited by balance. With one, each account shows its **Balance**, **reserved** and **available** amounts, and a Low balance chip below its threshold. Open **Recent transactions** to see each movement.

1. On the **Credits** tab, under **Top up**, choose the **Tenant** and **Company**, or **Tenant-wide**.
2. Enter the **Amount**, an optional **Low-balance threshold** and a **Note**.
3. Select **Add credits**. You see the new balance.

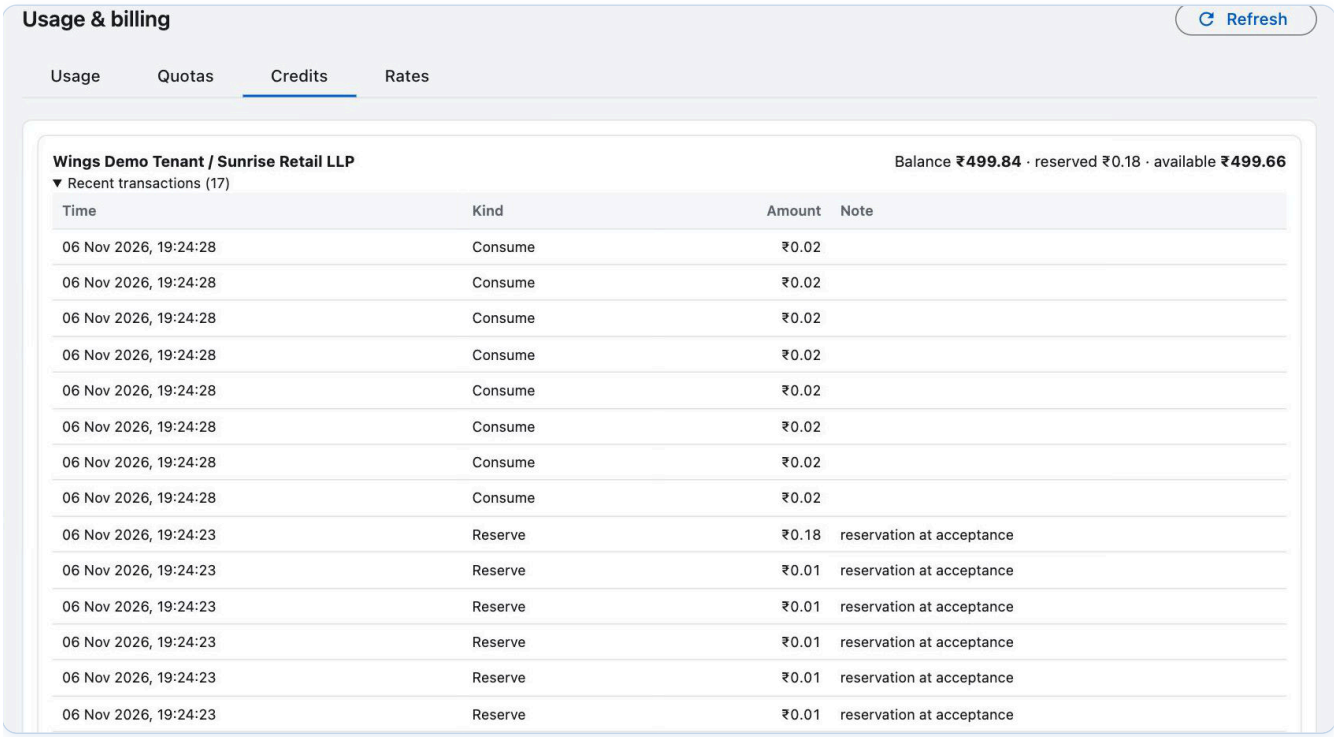


Figure 64: A prepaid credit account with its recent transactions.

19.4 Setting rates

The **Rates** tab has two tables.

- **Customer billing rates** set what customers pay. Choose the **Tenant** or **Default (all tenants)** and the **Channel**, then enter a **Unit price (optional)** or leave it empty to use cost plus **Margin %**. Select **Save billing rate**.
- **Provider cost rates** set what providers charge. Choose the **Channel** and **Provider**, optionally a **Country prefix** and **Message class**, and enter the **Unit cost**. Select **Add cost rate**. Each rate takes effect from when it is added.

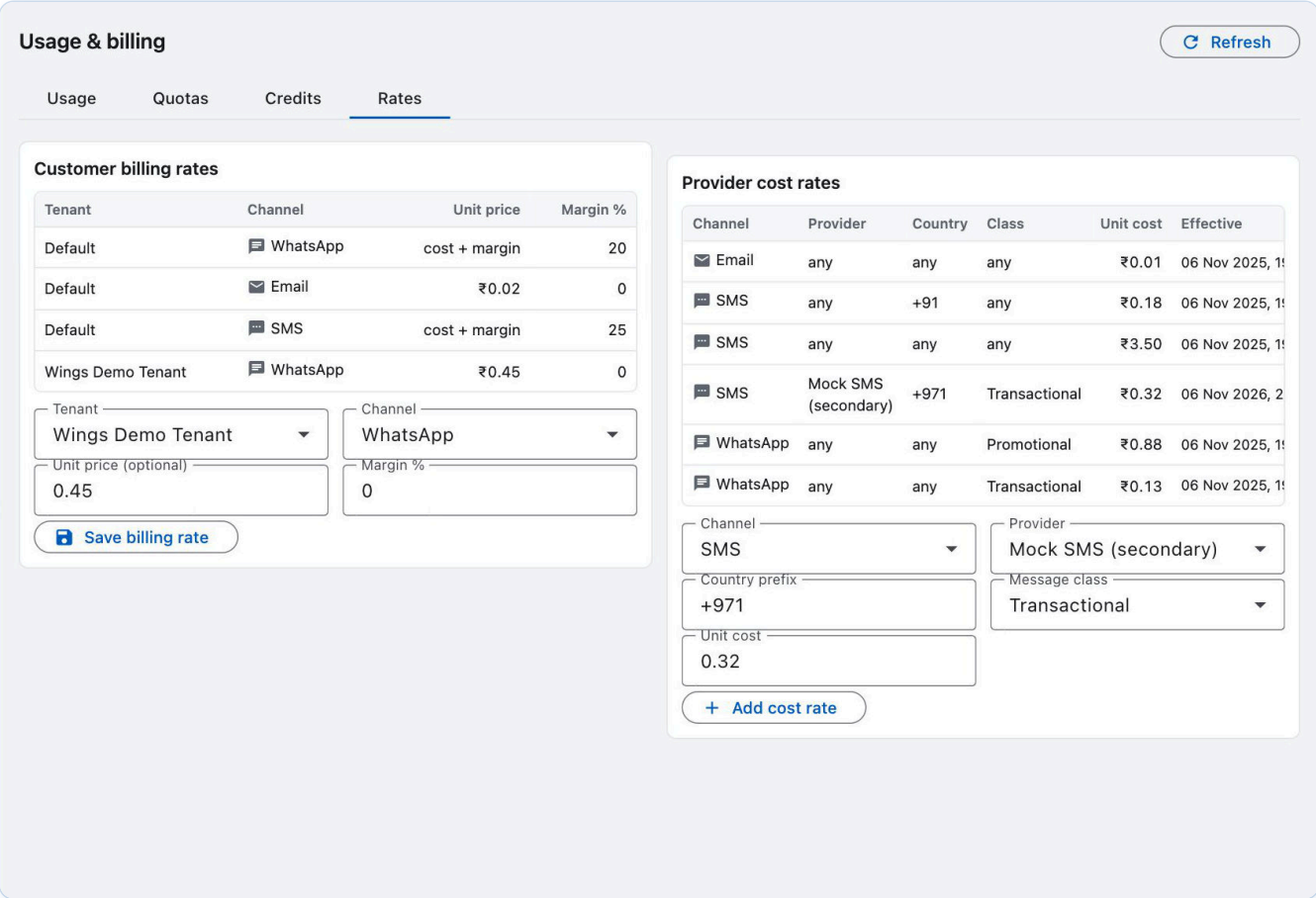


Figure 65: Billing rates and cost rates.

CHAPTER 20

Reports

Fifteen ready-made reports, each available as a table and a download.

Who can do this: the same people who can read **Usage & billing**.

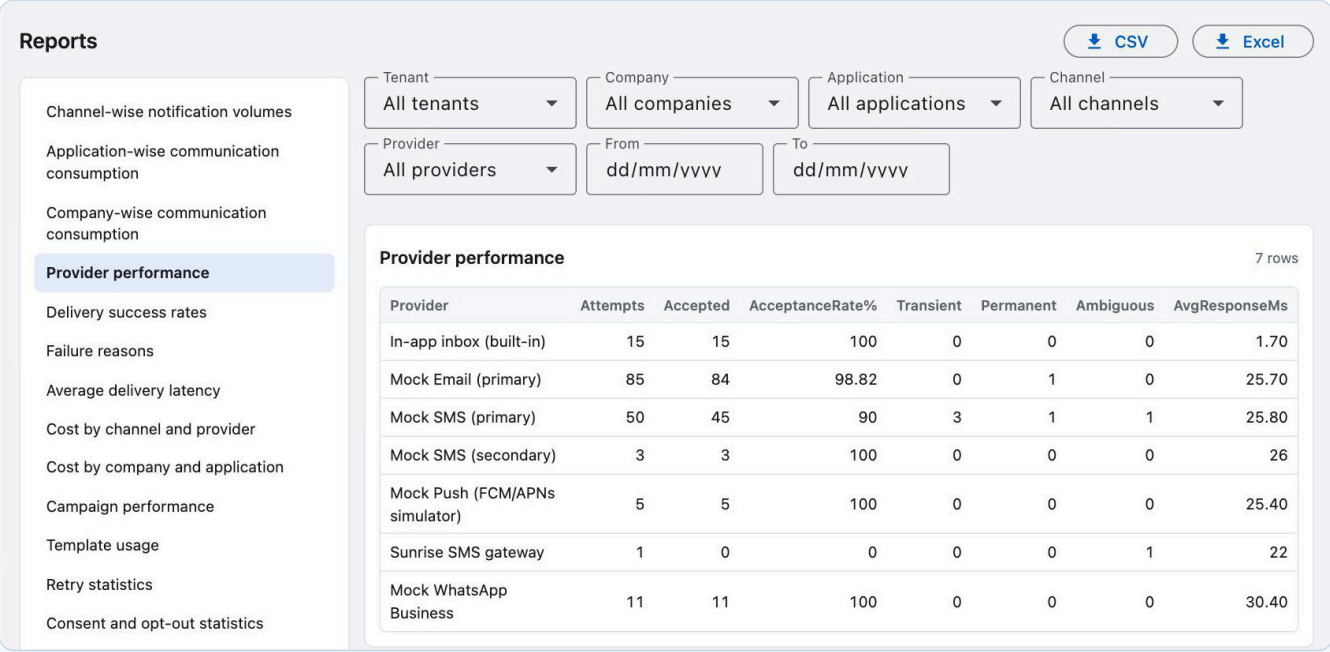


Figure 66: The Reports screen, showing Provider performance.

1. Open **Reports** and select a report on the left.
2. Narrow it with **Company**, **Application**, **Channel**, **Provider**, **From** and **To**, and **Tenant** if you see more than one.
3. Select a column heading to sort.
4. Select **CSV** or **Excel** to download what the filters select.

Report	Columns
Channel-wise notification volumes	Channel, Deliveries, Units
Application-wise communication consumption	Application, Channel, Deliveries, Cost
Company-wise communication consumption	Company, Channel, Deliveries, Cost
Provider performance	Provider, Attempts, Accepted, AcceptanceRate%, Transient, Permanent, Ambiguous, AvgResponseMs
Delivery success rates	Channel, Total, Delivered, AcceptedNotConfirmed, Failed, Suppressed, Read, DeliveryRate%, SuccessRate%
Failure reasons	Channel, Status, Reason, Count
Average delivery latency	Channel, Provider, Messages, AvgAcceptanceSec, AvgDeliverySec
Cost by channel and provider	Channel, Provider, Messages, Units, EstimatedCost, ProviderReportedCost, Billed
Cost by company and application	Company, Application, Messages, EstimatedCost, Billed
Campaign performance	Campaign, Status, Recipients, Deliveries, Delivered, Read, Failed, Suppressed, Cost
Template usage	Template, Version, Channel, Locale, Uses

Report	Columns
Retry statistics	Channel, Attempts, Deliveries, EventuallySucceeded
Consent and opt-out statistics	Type, Channel, Detail, Count
Queue backlog trends	Hour, Created, StillPending, AvgQueueWaitSec
Monthly usage comparison	Month, Channel, Messages, Billed

CHAPTER 21

System monitoring

This screen shows whether the hub itself is healthy. It refreshes every 15 seconds.

Who can do this: administrators, notification operators and auditors can read it. Everyone except auditors can acknowledge alerts. **Requeue** is for platform administrators, tenant administrators and notification operators who cover a whole tenant.

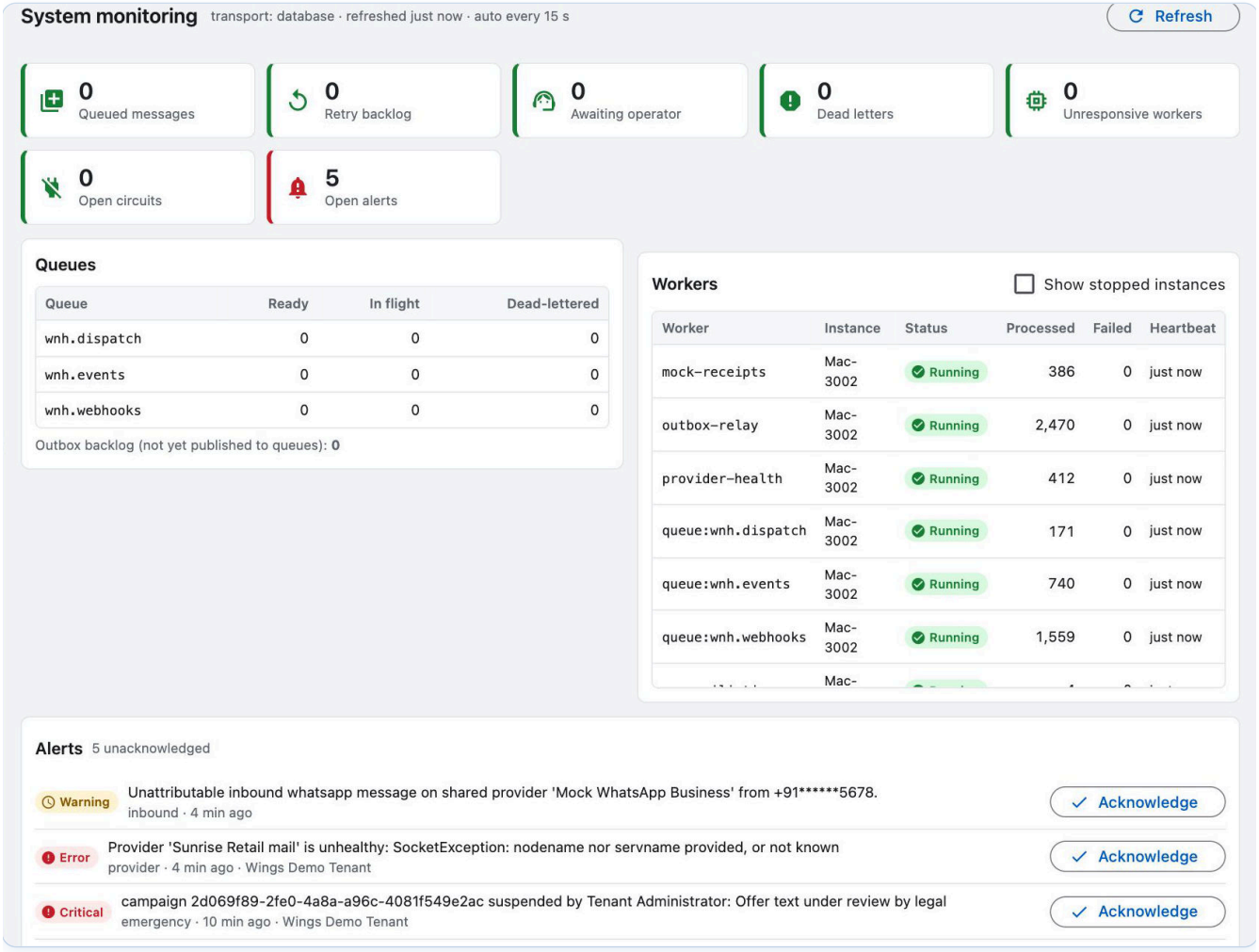


Figure 67: System monitoring: summary cards, queues, workers and open alerts.

Panel or card	What it tells you	What to do
Queued messages, Queues	Work waiting. **Outbox backlog** counts work not yet handed to a queue.	A number that keeps growing means workers are behind.
Retry backlog	Deliveries waiting to retry.	Open it to find a common cause, usually one provider.
Awaiting operator	Unknown outcomes.	See section 6.5 on page 19.

Panel or card	What it tells you	What to do
Dead letters	Items that used up their retries in a queue.	See below.
Unresponsive workers, Workers	The background processes, with work done, failures and the last heartbeat.	Any ⊖ Stopped or unresponsive worker that should be running is one for your support team.
Open circuits, Circuit breakers	A provider failing repeatedly is rested for a while. Closed is normal. Open until ... means traffic is going elsewhere.	Check the provider's status page. It closes by itself once the provider recovers.
Alerts	Things the hub wants a person to see, by severity.	Read it, deal with it, select **Acknowledge**.
Provider errors (24 h)	Error codes per provider.	Look for a sudden new code.

21.1 Requeuing a dead letter

1. Find the item under **Dead letters**. Read **Last error** and fix the cause if you can.
2. Select **Requeue**, then **Requeue** to confirm. You see `Requeued`.

If it fails again, it returns to this list. Dead letters are kept for 30 days.

CHAPTER 22

Emergency suspension

If something goes wrong, such as a compromised sender or a campaign sent in error, stop it here. Pending messages are held, not lost, and continue when you resume.

Who can do this: platform, tenant and company administrators. Only a platform administrator can suspend a tenant.

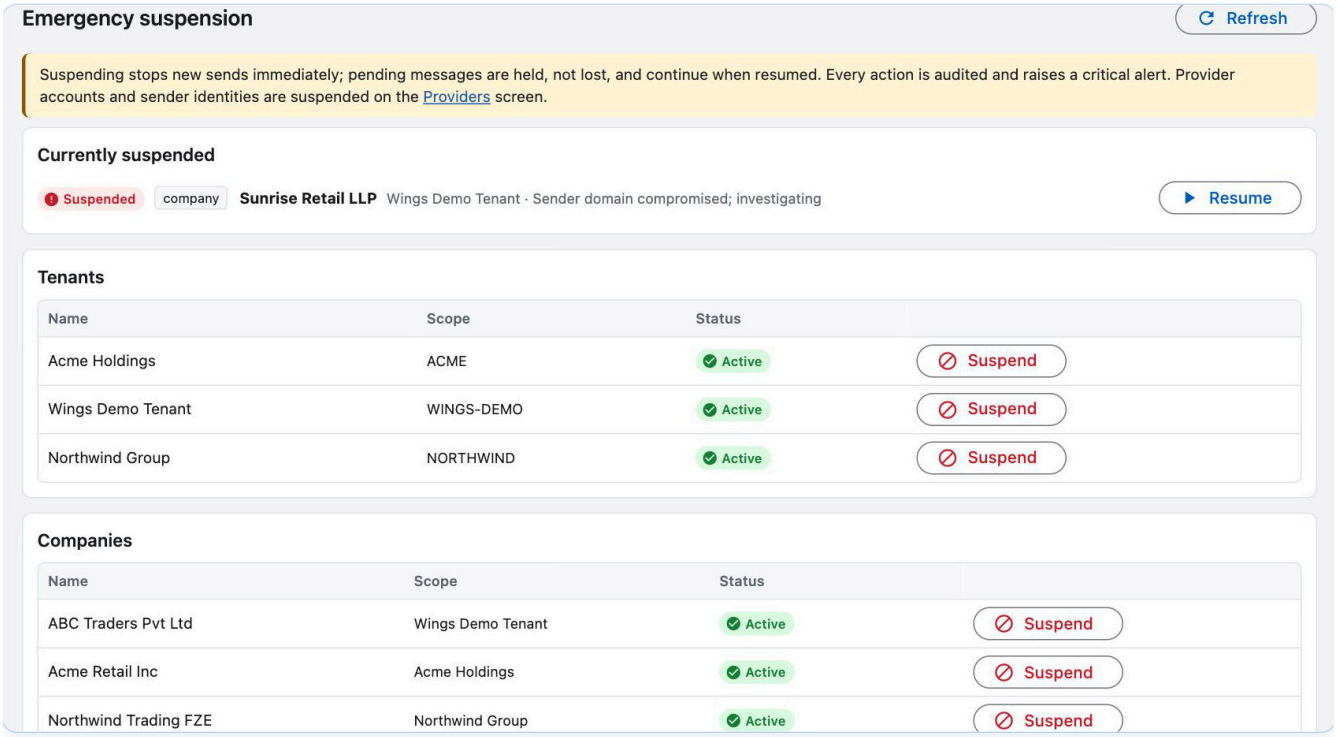


Figure 68: Emergency suspension, with one company suspended and listed under Currently suspended.

22.1 Suspending and resuming

1. Open **Emergency suspension**. It lists **Tenants**, **Companies**, **Applications** and **Active campaigns**.
2. Select **Suspend** on the row.
3. Enter a **Reason (audited)**. For a tenant, also tick **I understand this stops all messaging for every company in the tenant.**
4. Select **Suspend now**. It appears under **Currently suspended**.
5. When it is safe, select **Resume**, then **Resume** to confirm. Held messages continue.

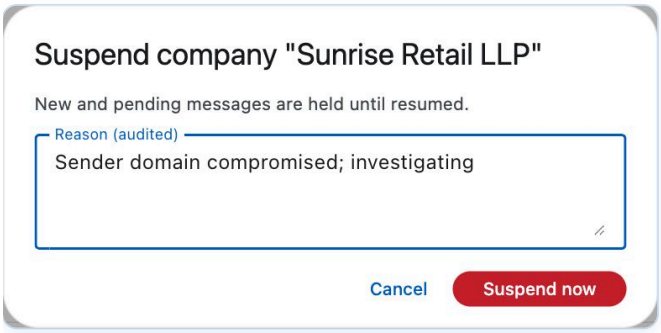


Figure 69: Suspending needs a reason.

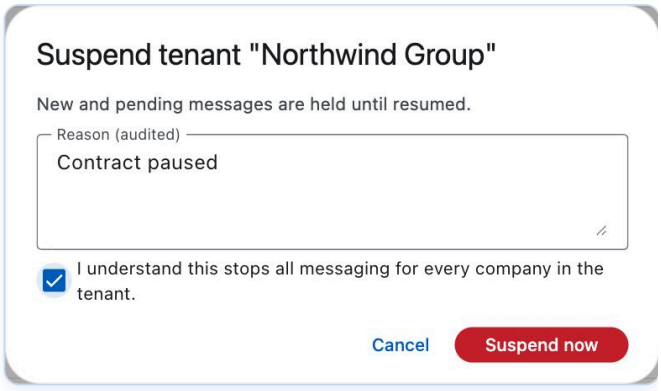


Figure 70: Suspending a tenant needs a second confirmation.

While a company is suspended, its applications are refused with `The company is suspended.` Every suspension raises a critical alert and is audited. Provider accounts and sender identities are suspended on the **Providers** screen (section 14.6 on page 57).

CHAPTER 23

Audit log

Every change made in the console or through the API is recorded: who, what, when and where.

Who can do this: auditors and platform, tenant and company administrators.

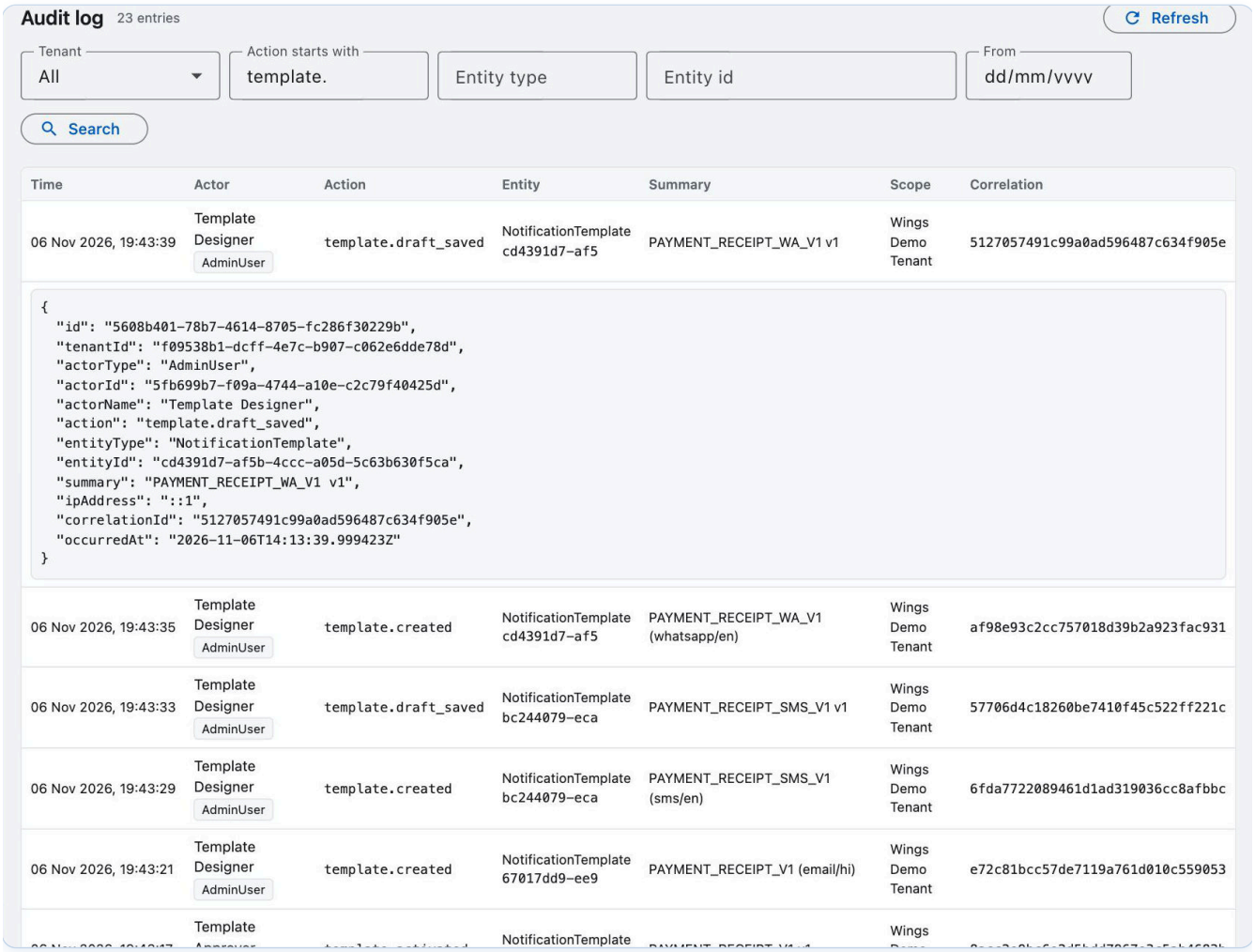


Figure 71: The Audit log, filtered to template actions, with one entry open.

1. Open **Audit log**.
2. Filter with **Action starts with**, such as `template.`, **Entity type**, such as `NotificationTemplate` or `ProviderAccount`, **Entity id** and **From**, plus **Tenant** if you see more than one.
3. Select **Search**.
4. Select a row to see the full entry.

Each row shows the **Time**, **Actor**, **Action**, **Entity**, **Summary**, **Scope** and **Correlation**. Action names read like `template.draft_saved`, `credential.created`, `company.suspended` and `credits.top_up`. Addresses are masked in summaries.

Tip If someone quotes a correlation id from an error banner, it links their error to the actions around it.

REFERENCE

Appendices

Statuses, template syntax, role permissions, troubleshooting, and pointers for developers and installers.

A	Delivery status reference	78
B	Template syntax reference	79
C	Role and permission matrix	80
D	Troubleshooting	81
E	For developers and installers	83
F	Glossary	84

APPENDIX A

Delivery status reference

A delivery's status always moves forward. Not every channel reaches every status.

Status	What it means	What to do
Draft	Not yet released for sending.	Nothing.
Scheduled	Waiting for its send time, or for quiet hours to end.	Nothing.
Queued	Waiting for a worker.	Nothing, unless many stay here.
Processing	A worker is handling it.	Nothing.
Submitted	Handed to the provider.	Nothing.
Provider accepted	The provider took it.	Nothing. Some providers report no further.
Sent	The provider says it left for the recipient.	Nothing.
Delivered	Confirmed as delivered.	Nothing.
Read	The recipient opened it, on channels that report this.	Nothing.
Retry pending	A try failed, and another is scheduled.	Wait, or select **Retry**.
Failed	It could not be delivered.	Read the failure, fix it, select **Retry**.
Expired	Its time limit passed before it could be sent.	**Retry** if still useful.
Cancelled	Stopped before sending, for example with a campaign.	Nothing.
Suppressed	Blocked by consent, preference or the suppression list.	See section 12.4 on page 46.
Unknown	Nobody can tell whether the provider took it.	Select **Resolve**. See section 6.5 on page 19.

Other statuses you will meet

On	Statuses
Templates	Draft · In review · Approved · Active · Retired · Rejected
Campaigns	Draft · Pending approval · Approved · Scheduled · Running · Paused · Completed · Cancelled
Campaign recipients	Pending · Queued · Skipped · Failed
Providers	Healthy · Degraded · Unhealthy · Circuit open · Unknown · Suspended · Disabled
Approvals	Pending · Approved · Rejected · Cancelled
Inbound replies	Pending · Routed · No subscriber · Failed

APPENDIX B

Template syntax reference

Templates use a small, safe language. Output is always escaped, and no code can run.

B.1 Variables

Write	Meaning
`{{CustomerName}}`	A variable. Names ignore case.
`{{Customer.Name}}`	A value inside another value.
`{{CompanyName}}`, `{{TenantName}}`, `{{RecipientName}}`, `{{UnsubscribeUrl}}`, `{{CurrentDate}}`	Built-in: the hub supplies these when sending.

B.2 Filters

Filters follow a `|`, can be chained, and take one argument after a `:`.

Filter	Example	Result
`upper`	`{{CustomerName | upper}}`	UPPER CASE
`lower`	`{{Email | lower}}`	lower case
`title`	`{{CustomerName | title}}`	Title Case
`trim`	`{{Comment | trim}}`	Removes surrounding spaces
`default:text`	`{{Nickname | default:Customer}}`	Fallback when empty
`truncate:n`	`{{Comment | truncate:80}}`	Cuts to n characters (…)
`number:decimals`	`{{Quantity | number:2}}`	Grouped number
`currency:code`	`{{InvoiceAmount | currency:INR}}`	Currency amount, such as ₹1,84,500.00
`date:format`	`{{DueDate | date:dd-MMM-yyyy}}`	Date, such as 12-Nov-2026
`url`	`{{PaymentLink | url}}`	Safe http(s) link only

B.3 Blocks

Write	Meaning
`{{#if Discount}}Save {{Discount}}{{else}}No offer{{/if}}`	Conditional section
`{{#unless Paid}}Payment pending{{/unless}}`	Inverse conditional
`{{#each Items}}{{@index}}. {{this.Name}}{{/each}}`	Repeat for each list item
`{{! internal note }}`	Comment, not rendered

⚠ **Warning** Every `{{#if}}`, `{{#unless}}` and `{{#each}}` needs its closing tag. While a block is unclosed, **Live preview** says so, for example `Unclosed '#if' block (line 1, column 4)`.

APPENDIX C

Role and permission matrix

What each role can do within its scope. Platform and tenant administrators hold every permission; the differences for them are noted below the table.

Permission	Platform	Tenant	Company	Application	Designer	Approver	Campaign	Operator	Auditor
View dashboard	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
View notifications (explorer)	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	✓ Yes	✓ Yes	✓ Yes
Read message content and full addresses	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	No	✓ Yes	No
Retry and resolve deliveries	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	No	✓ Yes	No
Send notifications, real tests, schedules	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	No	No	No
View templates	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	✓ Yes
Edit templates, notification types	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	No	No
Approve, activate, retire templates	✓ Yes	✓ Yes	✓ Yes	No	No	✓ Yes	No	No	No
View rules	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	No	✓ Yes
Edit rules	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	No	No	No
View campaigns	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	✓ Yes	✓ Yes	✓ Yes
Manage campaigns	✓ Yes	✓ Yes	✓ Yes	No	No	No	✓ Yes	No	No
Approve campaigns	✓ Yes	✓ Yes	✓ Yes	No	No	No	No	No	No
View Inbox	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	No	✓ Yes	✓ Yes
Reply in Inbox	✓ Yes	✓ Yes	✓ Yes	No	No	No	No	✓ Yes	No
View recipients	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	✓ Yes	No	✓ Yes
Manage recipients	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	✓ Yes	No	No
View providers	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	No	✓ Yes	✓ Yes
Manage providers	✓ Yes	✓ Yes	✓ Yes	No	No	No	No	No	No
Approve provider credentials	✓ Yes	✓ Yes	✓ Yes	No	No	No	No	No	No
View usage and reports	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	✓ Yes	No	✓ Yes
Set quotas	✓ Yes	✓ Yes	No	No	No	No	No	No	No
View system monitoring	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	No	✓ Yes	✓ Yes
View audit log	✓ Yes	✓ Yes	✓ Yes	No	No	No	No	No	✓ Yes
View tenants, companies, applications	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Manage tenancy, applications, calendars	✓ Yes	✓ Yes	No	No	No	No	No	No	No
Manage client credentials	✓ Yes	✓ Yes	No	✓ Yes	No	No	No	No	No
Manage webhooks	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	No	No	No
Acknowledge alerts	✓ Yes	✓ Yes	✓ Yes	✓ Yes	No	No	No	✓ Yes	No
Emergency suspension	✓ Yes	✓ Yes	✓ Yes	No	No	No	No	No	No
Manage users and roles	✓ Yes	✓ Yes	No	No	No	No	No	No	No

Platform administrator only: create tenants, create shared providers, top up credits, change billing and cost rates, suspend tenants, grant the Platform administrator role, create platform-wide calendars and routing rules, and see **Simulate reply**.

Requeue dead letters: platform administrators, tenant administrators and notification operators. It needs a grant that covers a whole tenant, because queues are shared.

APPENDIX D

Troubleshooting

D.1 A message wasn't delivered: what do I check?

1. **Find it.** In the **Notification explorer**, enter the full address in **Recipient (exact email or phone)**. Clear any date filters.
2. **Nothing found?** The request never reached the hub, or the application got an error. Ask the application's owner. Check **Emergency suspension** and **Quotas**, which stop sends before they are recorded.
3. Suppressed ? Read the reason. See section 12.4 on page 46. A retry will not help until consent or the suppression changes.
4. Failed ? Read **Failure** and the **Attempts** tab. A bad address must be corrected. A provider fault can be retried once it recovers.
5. Unknown ? Check with the provider, then **Resolve**.
6. Queued or Scheduled for a long time? Check the **Attempts** line on the delivery, which shows when the next try is due, quiet hours on the company, and queues and workers on **System monitoring**.
7. Delivered , but the person says it didn't arrive? Ask them to check spam or blocked senders. For email, check **Sender domain guidance**. Quote the provider message id to the provider.

D.2 Other common problems

Problem	Check
A button in this manual is missing	Your role does not include it. See Appendix C on page 80.
A link returns you to the Dashboard	Your role cannot open that screen.
The console shows **Offline**	The live connection dropped. It keeps retrying. Refresh the page if it stays offline.
A WhatsApp message is always Suppressed	The person has no **Channel opt-in**. Record it, or set an SMS fallback on the rule.
A promotional message is delayed	Quiet hours on the company or rule. It waits for the window to end.
A rule never fires	Simulate it. Check **Enabled**, the event type, and whether an earlier rule has **Stop evaluating later rules...** ticked.
A template change is not going out	Saved drafts do not send. It must be approved and show Active .
Campaign recipients all show Skipped	Promotional class with no marketing consent. Read the **Error** column.
Campaign recipients all show Failed	The template needs a variable the campaign does not supply. See the warning in section 11.1 on page 40.
A provider shows Unhealthy	Select **Test** and read the result. Check credentials and network access.
A webhook shows Disabled	Read **Attempts**, fix the receiver, select **Enable**.
A provider's receipts never arrive	The provider needs the account's **Callback URL**. SMTP has no receipts.

D.3 Common error messages

Errors appear in a red banner, with the code and a correlation id.

Message	What it means
`Invalid email or password.`	Sign-in failed. Check both, or ask your administrator.
`You do not have access to this resource.`	Your role cannot do this. Ask an administrator.
`The notification could not be rendered.`	A required template variable is missing.

Message	What it means
`None of the requested channels could be addressed for the recipient(s).`	The recipient has no address for the chosen channels.
`The template is invalid. body: Template HTML contains a disallowed construct`	Remove the script, event handler, frame or `javascript:` link.
`The rule is invalid.`	The message lists what to complete.
`Invalid cron expression`	Check the five fields of the schedule.
`Passwords must be at least 12 characters.`	Enter a longer password.
`A user with this email already exists.`	Select **Grant** on the person's row to add a role.
`… schedule(s) use this calendar.`	Choose another calendar for those schedules, then delete it.
`The provider account does not support template submission.`	Choose a real WhatsApp account, not a Mock one.
`An unexpected error occurred.`	Something failed inside the hub. Quote the correlation id to support.

Errors your application sees

Code	Meaning
`quota_exceeded`	A monthly quota is used up (section 19.2 on page 68).
`insufficient_credits`	The prepaid balance is too low (section 19.3 on page 69).
`company_suspended`, `tenant_suspended`, `application_suspended`	See Chapter 22 on page 75.
`no_deliverable_channels`	No channel could be used for the recipient.
`rate_limited`	The application is sending too fast for its rate limit.

APPENDIX E

For developers and installers

Everything in the console is also available through a REST API at `/api/v1`. A full, generated reference is linked as **API reference** at the foot of the menu.

Sending a notification

```
# 1. Get a token with a client credential from Applications
curl -X POST https://hub.example.com/connect/token \
  -d grant_type=client_credentials -d client_id=$ID -d client_secret=$SECRET

# 2. Send
curl -X POST https://hub.example.com/api/v1/notifications/send \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -H "Idempotency-Key: invoice-10045-created" \
  -d '{ "companyId": "<company id>",
        "notificationType": "INVOICE_CREATED",
        "recipient": { "customerId": "CUST-1001" },
        "channels": ["email", "whatsapp"],
        "variables": { "CustomerName": "ABC Traders", "InvoiceNumber": "INV-10045",
                       "InvoiceAmount": "18400", "InvoiceDate": "2026-11-06" } }'
# 202 Accepted: { "notificationId": "…", "status": "accepted", "deliveries": [ … ] }
```

Publishing a business event for rules

```
curl -X POST https://hub.example.com/api/v1/events \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{ "companyId": "<company id>", "eventType": "payment.overdue",
        "payload": { "customer": { "customerId": "CUST-1001" }, "DaysOverdue": 12 } }'
```

Where to go next

Document	Covers
`docs/api.md`	Authentication, scopes, idempotency, errors, every resource, webhooks
`docs/providers.md`	Each provider type, its settings, secrets and callbacks; WhatsApp and DLT
`docs/wings-integration.md`	Wings MCP Server and AI agents, Wings Code, SDKs
`docs/architecture.md`	How the hub is built
`docs/operations.md`	Topologies, configuration, backup and retention
`deploy/iis/README.md`	Hosting on Windows Server with IIS

APPENDIX F

Glossary

Application
A system that sends through the hub.

Approval
A second person's decision on a change. See maker-checker.

Attempt
One try at handing a delivery to a provider.

Campaign
One message sent to many recipients at a controlled pace.

Channel
Email, SMS, WhatsApp, in-app, push, web push or webhook.

Circuit breaker
Automatic rest for a provider that keeps failing.

Client credential
The id and secret an application uses to sign in.

Company
A business inside a tenant.

Consent
A recorded agreement to receive messages for a channel and purpose.

Correlation id
A reference linking an error or request to its records.

Cron expression
Five fields that describe a repeating time.

Dead letter
An item that used up its retries in a queue.

Delivery
One message to one address on one channel.

DLT
India's telecom registry for SMS senders and templates.

E.164
The full international phone format, such as +919845012345.

Escalation
A follow-up message when the first is not acted on in time.

Event
Something an application reports, such as `payment.overdue`.

Fallback
A second channel or provider used when the first fails.

Hard bounce
A permanent email delivery failure.

Locale
A language code, such as `en` or `hi`.

Maker-checker
The rule that the person who asks cannot also approve.

Message class
Transactional, promotional or critical.

Notification
One request to tell someone something.

Notification type
A named kind of message with its defaults.

Payload
The data sent with an event.

Provider
The outside service that carries a message.

Quiet hours
Times when chosen messages wait, in the recipient's local time.

Quota
A monthly limit on messages.

Retention
How long message bodies and records are kept.

Rule
Turns an event into notifications.

Sandbox
An application mode that never reaches real providers.

Scope
On a role, what it covers. On a credential, what it allows.

Segment
A filter that selects contacts for a campaign.

Sender identity
The address or number a message goes out under.

Suppression
A block on an address.

Template
The wording of a message, versioned and approved.

Tenant
A customer organisation on the hub.

Throttle
The most messages a campaign sends per minute.

Webhook
A signed call from the hub to an application.

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